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Maths Olympiad

Unleash The Maths Olympian In You!

Advanced (Pr 5 & 6, suitable for 11-13 years old)

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- **Q.E.D. (Quite Easily Done!)**
  Adopt and apply elegant and efficient solutions to complex MO-type problems
- **Step-by-step worked solutions**
  Implicit and alternative answers to further learning and understanding

Terry Chew

Maths Olympiad

Unleash The Maths Olympian In You!

Advanced (Pr 5 & 6, suitable for 11-13 years old)

Terry Chew

Master trainer at Terry Chew Academy
and Co-Founder/Academic Director of
SEAMO

Maths Olympiad – Unleash The Maths Olympian In You! (Advanced)

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A Word by the Author

Russian mathematicians have suggested that ***mathematics is the gymnasium of the brains***. Likewise, in China, Mathematics Olympiad is ferociously practised by the large student population, all for a coveted place in one of the elite high schools.

Children who exhibit certain traits and penchant for numbers at the age of 5 or 6 years old, or even earlier, have great potential to be the mathematical olympians among their peers – provided they are groomed via a systematic, rigorous and routinized training.

Singapore was ranked 3rd in Mathematics in a recent TIMSS survey, after Hong Kong and Taiwan. Notably, China was not among the list of countries surveyed.

The most prestigious competition locally is **RIPMWC** (Raffles Institution Primary Mathematics World Competition). Meant for primary 6 students or younger, the top 50 to 60 or so participants are selected from Round 1 to compete in Round 2. Thereafter, 6 top participants emerge to take part in the world competition for primary school mathematics in Hong Kong. Another popular competition, also meant for primary 6 students, is **APMOPS** (Asia Pacific Mathematical Olympiad for Primary Schools), which is organized by Hwa Chong Institution since 1989. The following awards are being given at the end of two rounds of competition: ***Platinum, Gold, Silver and Bronze.***

At primary 5 level, the yearly **NMOS** (National Mathematical Olympiad of Singapore) competition has also captured the attention of parents since 2006, who eye NUS High as their most preferred high school.

The first series of books ***Maths Olympiad: Unleash the Maths Olympian in You!*** published in 2007 and 2008, has served as an ideal companion to students looking to establish a strong foundation in Mathematics – be it for PSLE preparation or in hope that they might one day take part in the various local and international competitions. The books are, therefore, also first-choice materials for parents of primary 3 students looking for quality content in gifted programme training.

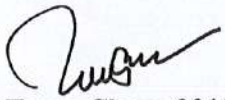
In this new edition you will find the following additions:

- Write Equations
- Geometric Problem
- Ratio
- Complex Fraction
- Speed 2
- Approximation

Area and Perimeter of Circles has also been largely enhanced.

The objective is to cater to increasingly smarter children who have been exposed to a wide variety of topics. Some of these topics, which overlap the local mathematics syllabus, have also been adopted by schools here for students to practice on.

I feel extremely privileged and honoured to be able to continue serving students in this field. My latest series |***Conquer*| *Exam-Standard Mathematics Problem Sums with Terry Chew*** and ***Mathematical Olympiad Training Book*** are currently out on shelves.



Terry Chew, 2019

For related courses and workshops, please visit
terrychew.com.sg.

SEAMO

SOUTHEAST ASIAN MATHEMATICAL OLYMPIAD

Southeast Asian Mathematical Olympiad (**SEAMO**) is a unique assessment platform originating from Singapore and presided over by Mr Terry Chew, the author of the celebrated book series 'Unleash the Maths Olympian in You!'. The Singapore Maths curriculum places strong emphasis on Critical Thinking Skills as part of its 21st Century Core Competencies. Students must be able to make connections and think outside the box in order to be successful in school and at the workplace. With this vision in mind, **SEAMO** has grown to become the most highly acclaimed Maths Olympiad competition in the region.

Over the years, **SEAMO** has established a wide network of Maths Olympians across S. E. Asia, which knowledge, skills and culture are shared. Hundreds of thousands of eager young minds have received encouragement and recognition as they grow their passion for mathematics.

With competition papers suitable for all levels, **SEAMO** aims to stretch the capabilities of its candidates one bite-sized step at a time and in tandem with the school mathematics curriculum.

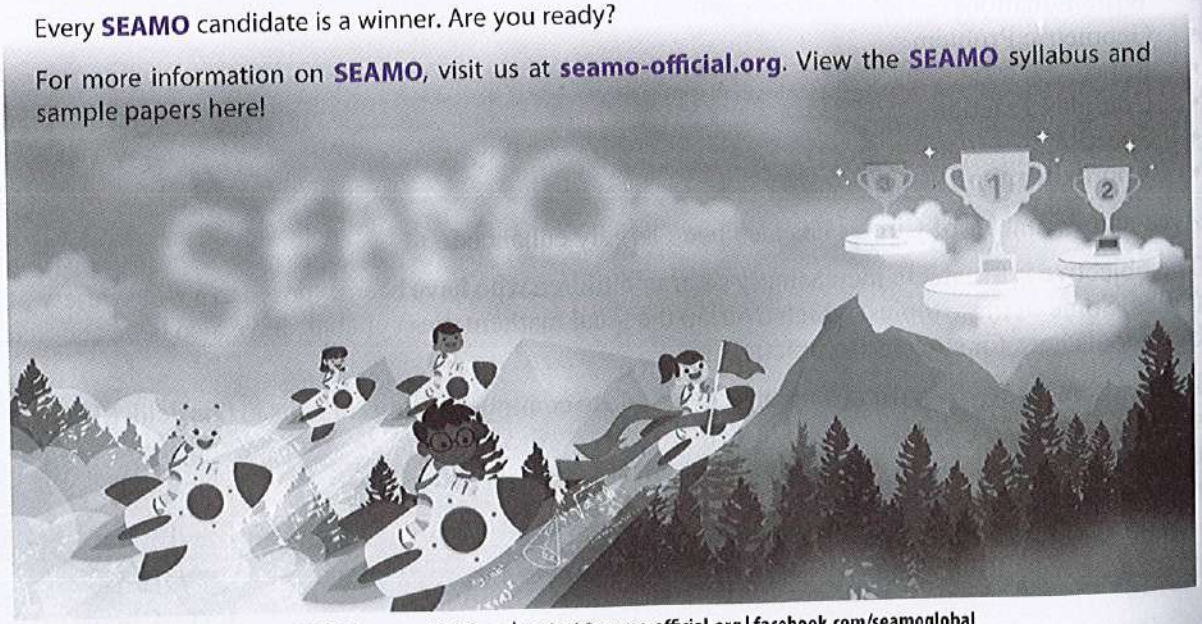
A reward-driven awarding structure also serves to motivate candidates and eliminate the belief that 'only geniuses can win'. As stated on every certificate,

> **"This certificate recognises your outstanding attitude towards accepting mathematical challenges; for representing your country among thousands from the region and beyond; and for your everlasting passion for mathematics."**

Active and dedicated participants will consequently emerge with confidence and a robust mindset towards problem-solving, which will in turn drive them to conquer more difficult challenges ahead.

Every **SEAMO** candidate is a winner. Are you ready?

For more information on **SEAMO**, visit us at **seamo-official.org**. View the **SEAMO** syllabus and sample papers here!



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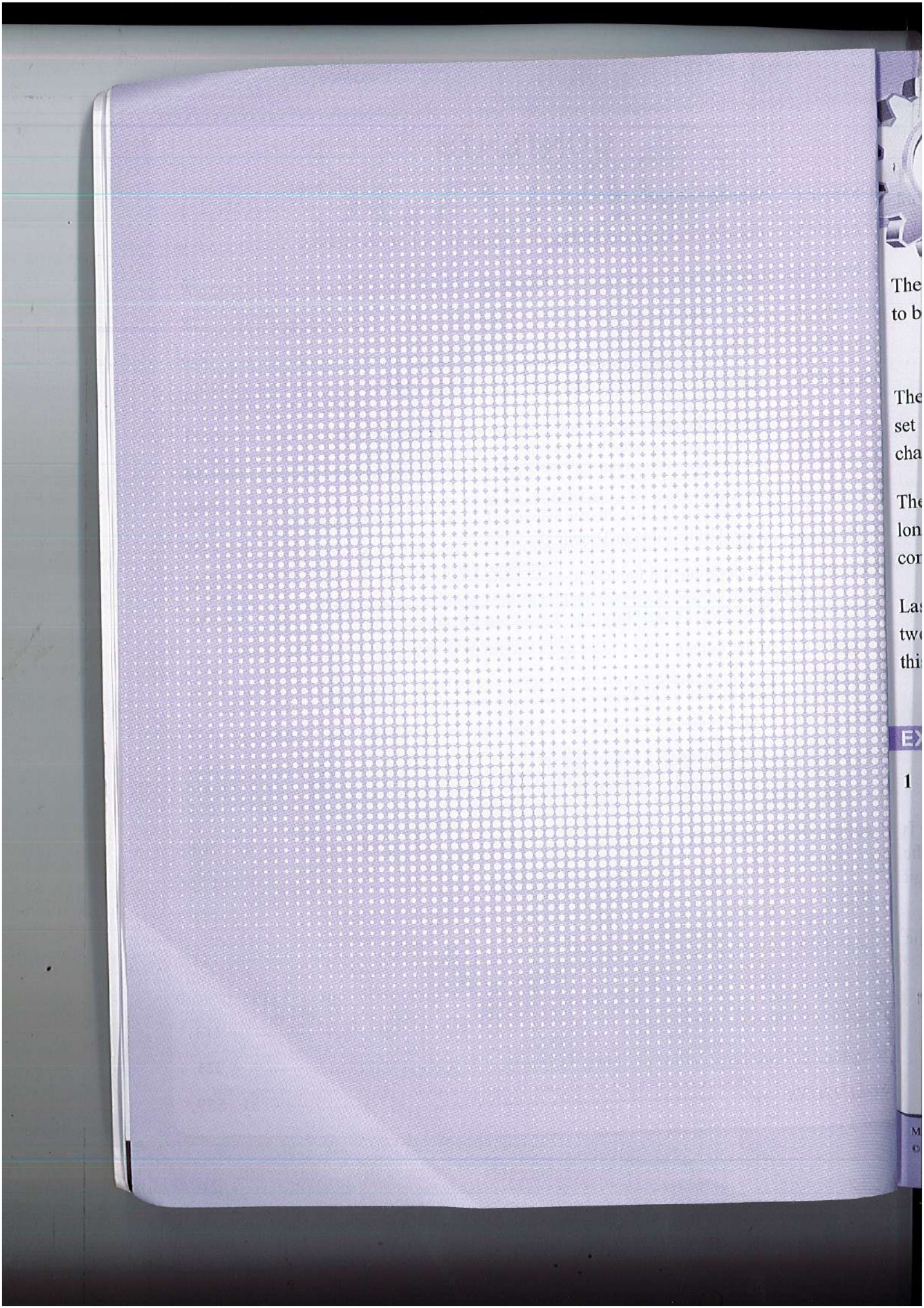

Whole Numbers

The most important and sophisticated technique that we will learn from this chapter is to be able to express, say, *abcd*, which denotes a 4-digit number, in the form of

$$1000a + 100b + 10c + d.$$

The usefulness of this simple expression is to help us solve and appreciate a unique set of mathematical problems such as Example 3, Example 4 and Question 10 in this chapter.

The other category of mathematical problems, namely multiplication of two extremely long string of numbers, uses a simple concept ($10 - 1 = 9$) for problem-solving. This concept is demonstrated in Example 2.

Last but not least, we will learn to simplify the computation of the sum or difference of two sets of products through skilful factorisation. Example 1 and Question 7 illustrate this technique.

EXAMPLES

1 $999\,999 \times 222\,222 + 333\,333 \times 333\,334 = ?$

***Solution*:** $999\,999 \times 222\,222 + 333\,333 \times 333\,334$

$= 333\,333 \times 3 \times 222\,222 + 333\,333 \times 333\,334$

$= 333\,333 \times 666\,666 + 333\,333 \times 333\,334$

$= 333\,333 \times (666\,666 + 333\,334)$

$= 333\,333 \times 1\,000\,000$

$= \mathbf{333\,333\,000\,000}$

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2 Find the sum of all the digits of $\underbrace{333\ldots333}_{2008\ 3s} \times \underbrace{666\ldots666}_{2008\ 3s}$.

Analysis: It is not possible to multiply two numbers of such magnitude. The whole trick to this question lies in a simple relationship:
$10 - 1 = 9$.

Solution:

$$\underbrace{333\ldots333}_{2008\ 3s} \times \underbrace{666\ldots666}_{2008\ 3s}$$

$$= \underbrace{333\ldots333}_{2008\ 3s} \times 3 \times \underbrace{222\ldots222}_{2008\ 2s}$$

$$= \underbrace{999\ldots999}_{2008\ 9s} \times \underbrace{222\ldots222}_{2008\ 2s}$$

$$= (\underbrace{1000\ldots000}_{2008\ 0s} - 1) \times \underbrace{222\ldots222}_{2008\ 2s}$$

$$= \underbrace{222\ldots222}_{2008\ 2s}\ \underbrace{000\ldots000}_{2008\ 0s} - \underbrace{222\ldots222}_{2008\ 2s}$$

$$= \underbrace{222\ldots222}_{2007\ 2s}\ 1\ \underbrace{777\ldots778}_{2007\ 7s}$$

$2 + 7 = 9$ (there are 2007 pairs of 9)
$1 + 8 = 9$ (there is one more pair of 9)
$2008 \times 9 = 18\,072$
The sum of all the digits of $\underbrace{333\ldots333}_{2008\ 3s} \times \underbrace{666\ldots666}_{2008\ 6s}$ is **18 072**.

3 The sum of all the digits of a three-digit number is 21. The digit in the ones place is greater than the digit in the tens place. A new number, which is 198 more than the original one, is formed by interchanging the digit in the ones place with the digit in the hundreds place. What is the original number?

Analysis: It will not take us long to figure out that $876 - 678 = 198$. Hence, the answer is 678. The question lies with whether this is the only answer.

Solution:

$100a + 10b + c = abc$ — (1)
Interchange the digits in the ones and hundreds places, it will become
$100c + 10b + a$ — (2)
(2) – (1)

$$100c + 10b + a - abc = 198$$
$$100c + 10b + a - 100a - 10b - c = 198$$
$$99c - 99a = 198$$
$$99(c - a) = 198$$
$$c - a = 198 \div 99 = 2$$

Since the digit in the ones place is greater than the digit in the hundreds place by 2, we try 597. However, 9 is greater than 7.

Next, we try 759.

$957 - 759 = 198$

Hence, the original number can be **678** or **957**.

4 Miss Cussler was born on the 1st of January many years ago. In 2002, her age was the sum of all the four digits of the year that she was born in. How old was Miss Cussler in 2002?

***Analysis*:** Again, we express the year that she was born in as $1000 + 100a + 10b + c$, if we assume it is $1abc$.

***Solution*:**

$$2002 - (1000 + 100a + 10b + c) = 1 + a + b + c$$
$$1002 - 100a - 10b - c = 1 + a + b + c$$
$$1001 = 101a + 11b + 2c$$

Take $a = 9$, so that it becomes $19bc$.

$$1001 = 101 \times 9 + 11b + 2c$$
$$1001 - 909 = 11b + 2c$$
$$92 = 11b + 2c$$
$$b = \frac{92 - 2c}{11}$$

When $c = 2$,

$$b = \frac{92 - 2 \times 2}{11}$$
$$= \frac{88}{11}$$
$$= 8$$

Miss Cussler was born in 1982.

$1 + 9 + 8 + 2 = 20$

as $1982 + 20 = 2002$.

Miss Cussler was **20** years old in 2002.

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PRACTICE

1 For each question below, do only the first three multiplication problems. Write out the next three products based on your conjecture.

(a) $3 \times 4 =$
$33 \times 34 =$
$333 \times 334 =$
$3333 \times 3334 =$
$33\,333 \times 33\,334 =$
$333\,333 \times 333\,334 =$

(b) $6 \times 7 =$
$66 \times 67 =$
$666 \times 667 =$
$6666 \times 6667 =$
$66\,666 \times 66\,667 =$
$666\,666 \times 666\,667 =$

(c) $5 \times 9 =$
$55 \times 99 =$
$555 \times 999 =$
$5555 \times 9999 =$
$55\,555 \times 99\,999 =$
$555\,555 \times 999\,999 =$

(d) $8 \times 9 =$
$88 \times 99 =$
$888 \times 999 =$
$8888 \times 9999 =$
$88\,888 \times 99\,999 =$
$888\,888 \times 999\,999 =$

2 Find the value of $1\,111\,111\,122\,222\,222 \div 33\,333\,334$.

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roblems. Write out

3 Find the value of $\underbrace{666\ldots666}_{\text{2008 6s}} \times \underbrace{666\ldots667}_{\text{2007 6s}}$.

4 Compute 11 111 111 × 99 999 999.

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5 Find the value of $\underbrace{888\ldots888}_{20\ 8s} \times \underbrace{333\ldots334}_{19\ 3s} + \underbrace{666\ldots666}_{20\ 6s} \times \underbrace{888\ldots888}_{20\ 8s}$.

6 Compute $\underbrace{333\ldots333}_{2008\ 3s} \times \underbrace{888\ldots888}_{2008\ 8s} \div \underbrace{666\ldots666}_{2008\ 6s}$.

… 888.
20 8s

7 Compute 2007 2008 × 2008 2007 – 2007 2007 × 2008 2008.

8 Find the value of $\underbrace{999\ldots999}_{2008\ 9s} \times \underbrace{999\ldots999}_{2008\ 9s} + 1\ \underbrace{999\ldots999}_{2008\ 9s}$.

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9 The sum of the digits of a two-digit number is multiplied by 8. The result is 8 more than the number. Find the two-digit number.

10 James' age in 2008 was the sum of all the digits of his year of birth. How old was James in 2008?

Solve by Comparison and Replacement

The techniques of comparison and replacement, also known as substitution, have been studied in detail in Maths Olympiad – Beginner.

In the first case, we will write two statements about the problems. We then bring the statements to a comparable stage when they cannot be compared directly in the first place.

In the second case, we treat two items or events as if they are the same. In other words, we substitute one item for another. Adjustments need to be made to facilitate the solving of such problems.

EXAMPLES

1 5 similar tables and 18 similar chairs cost $594.
The cost of one such table is the same as the cost of 3 such chairs.
How much does each table cost?
How much does each chair cost?

Solution:

'Change' all items to chairs.

1 table → 3 chairs

5 tables → 15 chairs

15 + 18 = 33 chairs

$594 ÷ 33 = $18

Each chair costs **$18**.

3 × $18 = $54

Each table costs **$54**.

2 4 basketballs and 6 footballs cost \$460.
A basketball is \$15 more expensive than a football.
How much is each basketball?
How much is each football?

***Solution*:**

Method 1

'Change' all items to footballs.

$\$460 - 4 \times \$15 = \$400$

$4 + 6 = 10$

Cost of each football $= \$400 \div 10 = \$40$

Cost of each basketball $= \$40 + \$15 = \$55$

Method 2

'Change' all items to basketballs.

$\$460 + 6 \times \$15 = \$550$

$4 + 6 = 10$

Cost of each basketball $= \$550 \div 10 = \$55$

Cost of each football $= \$55 - \$15 = \$40$

Each basketball costs **\$55**. Each football costs **\$40**.

3 5 kg of sugar and 6 kg of rice cost \$84.50.
2 kg of sugar and 3 kg of rice cost \$41.
How much is 1 kg of sugar?
How much is 1 kg of rice?

***Solution*:**

2 kg of sugar + 3 kg of rice $\rightarrow$ \$41

4 kg of sugar + 6 kg of rice $\rightarrow 2 \times \$41 = \$82$

5 kg of sugar + 6 kg of rice $\rightarrow$ \$84.50

1 kg of sugar $\rightarrow \$84.50 - \$82 = \$2.50$

3 kg of rice $\rightarrow \$41 - 2 \times \$2.50 = \$36$

1 kg of rice $\rightarrow \$36 \div 3 = \$12$

1 kg of sugar costs **\$2.50**. 1 kg of rice costs **\$12**.

4 If 10 rabbits can be exchanged for 2 goats, 9 goats be exchanged for 3 cows, and 8 cows be exchanged for 2 horses, how many rabbits can 5 horses be exchanged for?

Solution:

10 rabbits → 2 goats
9 goats → 3 cows
8 cows → 2 horses

(5 ÷ 2) × 8 cows ← 5 horses
20 cows ← 5 horses
(3 × 20) goats ← 20 cows
60 goats ← 20 cows
(30 × 10) rabbits ← 60 goats
300 rabbits ← 60 goats

5 horses can be exchanged for **300** rabbits.

5 It takes Mark a total of 25 minutes to run a distance of 1000 m and then walk another 2000 m. It also takes him 25 minutes to run a distance of 4000 m and then walk another 500 m. How fast can Mark run?

Solution:

Let r and w represent Mark's speeds for running and walking respectively.

1000r + 2000w → 25 min
4000r + 500w → 25 min
16 000r + 2000w → 100 min
15 000r → 75 min
15 000 ÷ 75 = 200 m/min

Mark can run at **200 m per minute**.

PRACTICE

1 $\square = \Delta + \Delta + \Delta$

$\square + \square + \bigcirc = \bigcirc + \bigcirc$

$\bigcirc = ? \Delta$

2 5 raisin rolls and 6 donuts cost $14.70.
Each raisin roll costs 3 times as much as each donut.
How much is a raisin roll?
How much is a donut?

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3 5 similar tables and 12 similar chairs cost $720.
Each table costs $25 more than each chair.
What is the cost of a table?

4 On 1st of March some years ago, one US dollar could be exchanged for 1.45 Singapore dollar. On the same day, 1000 Thai Baht could buy 50 Singapore dollars. How much US dollars could 29 000 Thai Baht buy on that day?

5 4 footballs and 4 volleyballs cost $240.
3 footballs and 5 volleyballs cost $234.
What is the cost of a volleyball?
What is the cost of a football?

6 In an ancient village, 10 rabbits can be exchanged for 2 turkeys.
6 turkeys can be exchanged for 2 goats.
10 goats can be exchanged for 2 cows.
How many rabbits can 5 cows be exchanged for?

7 2 peaches and 12 apples cost \$14.60.
6 peaches and 4 apples cost \$18.20.
How much is an apple?
How much is a peach?

8 There are 160 white and blue balls, 180 blue and yellow balls, and 170 white and yellow balls. Find the number of each colour of balls.

9 3 packets of French fries, 2 cheese burgers and 1 cup of soft drink cost \$10.95.
1 packet of French fries, 2 cheese burgers and 3 cups of soft drink cost \$11.25.
Find the price of a set.

10 3 pieces of item A, 7 pieces of item B and 1 piece of item C in a toy set co
\$31.50.
4 pieces of item A, 10 pieces of item B and 1 piece of item C in the same set co
\$42.
Find the price of item A + item B + item C.

cost $10.95.
cost $11.25.

in a toy set cos

the same set cos

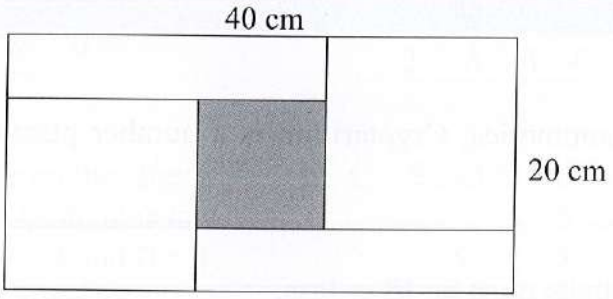
11 5 cows and 6 goats feed on 27 kg of grass every day.
6 cows and 5 goats feed on 28 kg of grass every day.
How many kilograms of grass does a cow feed on each day?

12 3 basketballs and 5 footballs cost $390.
6 basketballs and 3 footballs cost $465.
What is the cost of a basketball?
What is the cost of a football?

13 Amanda took 30 minutes to cycle 3000 m and then swim for 1000 m.
It took her 48 minutes to cycle 2400 m and swim for 2000 m.
How fast could Amanda cycle?

14 A basket of pears and 2 baskets of oranges weigh 13 kg.
2 baskets of apples and 2 baskets of oranges weigh 22 kg.
3 baskets of pears and 2 baskets of apples weigh 21 kg.
How much does each basket of fruit weigh?

15 A rectangle has a length of 40 cm and a breadth of 20 cm. It is cut into 2 identical squares, 2 identical rectangles and a shaded small square.
Find the area of the shaded square.



3 Cryptarithm

Also known as Alphanumerics, Cryptarithm is a number puzzle whereby a digit is replaced by a letter.

Important notes:

The sum of two digits must be 18 or less.

The product of two digits must be 81 or less.

EXAMPLES

1 What digit does each letter represent?

$$\begin{array}{rcccc} & & A & B \\ & & A & B \\ + & & A & B \\ \hline & C & C & C \\ \hline \end{array}$$

Solution:

If C = **1**,
111 ÷ 3 = 37
A = **3** and B = **7**

If C = **2**,
222 ÷ 3 = 74
A = **7** and B = **4**

2 What digit does each letter represent?

(a)
$$\begin{array}{rcccc} & & A & B & C \\ \times & & & & A \\ \hline & 1 & 0 & A & C \\ \hline \end{array}$$

(b)
$$\begin{array}{rcccc} & A & B & C & D \\ \times & & & & 9 \\ \hline & D & C & B & A \\ \hline \end{array}$$

Solution:

(a) 3 × 3 = 9
10 = 9 + 1
A = **3**
C = **5**

$$\begin{array}{rcccc} & & 3 & B & 5 \\ \times & & & & 3 \\ \hline & 1 & 0 & 3 & 5 \\ \hline \end{array}$$

B = **4**

(b) A = **1** and D = **9**

$$\begin{array}{rcccc} & 1 & B & C & 9 \\ \times & & & & 9 \\ \hline & 9 & C & B & 1 \\ \hline \end{array}$$

B = **0** (otherwise there will be regrouping in the thousands place.)

C = **8**

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3 What digit does each letter represent?

(a)

$$\begin{array}{rcccc} & & C & D \\ & B & C & D \\ + B & A & C & D \\ \hline 1 & 9 & 8 & 9 \end{array}$$

(b)

$$\begin{array}{rcccc} & & & C \\ & & C & A & C \\ + A & C & B & C \\ \hline & 2 & A & A & C \end{array}$$

***Solution*:**

(a) From the above, it is obvious that $D = \mathbf{3}$ and $B = \mathbf{1}$.

$3 \times 6 = 18$

$C = \mathbf{6}$

$9 - 1 - 1 = 7$

$A = \mathbf{7}$

(b) $C = \mathbf{5}$ and $A = \mathbf{1}$

$$\begin{array}{rcccc} & & & 5 \\ & & 5 & 1 & 5 \\ + 1 & 5 & B & 5 \\ \hline & 2 & 1 & 1 & 5 \end{array}$$

$B = \mathbf{9}$

4 Alice's house number is a 4-digit number. When she moves the first digit to the ones place, she notices that the new 4-digit number is bigger than her house number by 4707. What is her house number?

***Solution*:**

Let her house number be *abcd* and the new number be *bcda*.

$$bcda - abcd = 4707$$

$$1000b + 100c + 10d + a - 1000a - 100b - 10c - d = 4707$$

$$900b - 999a + 9d + 90c = 4707$$

Divide both sides by 9.

$100b - 111a + d + 10c = 523$

When $b = 6$ and $a = 1$,

$489 + d + 10c = 523$

$d = 4$ and $c = 3$

Her house number is **1634**.

PRACTICE

1 Find at least 5 possible values of the digit that represents each letter.

$$\begin{array}{r} A\ B \\ +\ B\ A \\ \hline C\ C \end{array}$$

2 What digit does each letter represent?

(a)
$$\begin{array}{r} A\ B\ C \\ \times\quad\ A \\ \hline 2\ C\ 9\ 0 \end{array}$$

(b)
$$\begin{array}{r} A\ B \\ \times\ A\ B \\ \hline A\ C\ D \end{array}$$

3 What digit does each letter represent?

(a)
$$\begin{array}{r} A\ B\ C \\ \times\quad D \\ \hline D\ 9\ B \end{array}$$

(b)
$$\begin{array}{r} B\ C \\ \times\ B\ C \\ \hline A\ B\ C \end{array}$$

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4 Find the value of the digit represented by each letter.

(a)

$$\begin{array}{rcccc} & & A & B & C \\ \times & & & & A \\ \hline & 1 & 8 & D & A \\ \hline \end{array}$$

(b)

$$\begin{array}{rcccc} & & & A & B \\ \times & & & A & B \\ \hline & & B & 7 & B \\ \hline \end{array}$$

5 Find the value of each digit to make the addition work.

$$\begin{array}{rcccc} & D & C & B & A \\ + & F & C & F & A \\ \hline F & A & B & D & A \\ \hline \end{array}$$

6 Find the value of each digit to make the addition work.

$$\begin{array}{rccccc} & A & B & C & D & E \\ + & 4 & 1 & 9 & 7 & 6 \\ \hline E & D & C & B & A \\ \hline \end{array}$$

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7 Find the value of each letter.

$$\begin{array}{r} \quad C\ D \\ \quad A\ C\ D \\ +\ A\ B\ C\ D \\ \hline 2\ 0\ 0\ 5 \end{array}$$

8 What digit does each letter represent?

$$\begin{array}{r} A\ B\ C\ D\ E\ F \\ \times \qquad\qquad\ F \\ \hline G\ G\ G\ G\ G\ G \end{array}$$

9 Find the value of each digit represented by different letters.

$$\begin{array}{r} A\ B\ C \\ +\ B\ A\ C \\ \hline D\ A\ D\ C \end{array}$$

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10 Find the values of the digits represented by A and B.

$$\begin{array}{rcccc} & & & A & B \\ \times & & & B & A \\ \hline & & 1 & 0 & B \\ + & 1 & 7 & B & \\ \hline & 1 & 8 & B & B \end{array}$$

11 What digit does each letter represent?

$$\begin{array}{rcccccc} & 1 & A & B & C & D & E \\ \times & & & & & & 3 \\ \hline A & B & C & D & E & 1 \end{array}$$

12 Find out what digit each letter represents.

$$\begin{array}{rccc} & & C & D \\ \times & & D & C \\ \hline & 1 & 1 & 7 \\ + & C & 5 & 1 \\ \hline & C & 6 & 2 & 7 \end{array}$$

13 Find out what digit each letter represents.

$$\begin{array}{r} \quad\quad\quad A \\ \quad\quad A\ B \\ \quad A\ B\ C \\ +\ A\ B\ C\ D \\ \hline 2\ 7\ 0\ 5 \end{array}$$

14 What digit does each letter represent?

$$\begin{array}{r} B\ A\ A\ B\ B \\ +\ B\ A\ A\ B\ B \\ \hline A\ A\ C\ C\ C \end{array}$$

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15 What digit does each letter represent?

$$\begin{array}{r} B\ \ C \\ B\ \ C \\ B\ \ C \\ +\ B\ \ C \\ \hline A\ \ B \end{array}$$

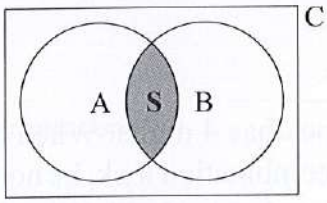
16 Mr Turner's combination lock has 4 digits. When he changes the digit in the ones place to the first digit of the combination lock, he notices that the new 4-digit number is greater than the original number by 5877. What is the original combination lock number?

4 Venn Diagram

There is a string of mathematical problems that can arise when some items or events a counted twice in the process of counting.

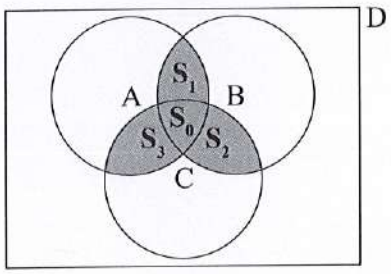
To avoid such mathematical errors in counting, Venn diagrams can be used. A Ven diagram is a special drawing that can clearly show the relationship between two or mo items or events.

Examples of a Venn diagram are shown below. The Venn diagram below shows th relationship between two different items, A and B. The shaded area, S, is a region th is common to both A and B.



Total number of items or events, $C = A + B - S$

The Venn diagram below shows the relationship between three items or events.



The shaded area, S_1, is common to both A and B. The shaded area, S_2, is comm to both B and C. The shaded area, S_3, is common to both A and C. The shaded are S_0, is common to A, B and C.

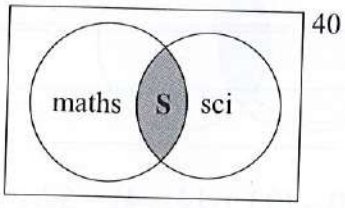
Total number of items or events, $D = A + B + C - S_1 - S_2 - S_3 + S_0$

EXAMPLES

1 30 out of a group of 40 students completed their mathematics homework. 25 students completed their science homework.
How many students completed both homework?

Solution:

Let the number of students who completed both homework be S. Note that S has been counted twice.



$30 + 25 - S = 40$

$55 - S = 40$

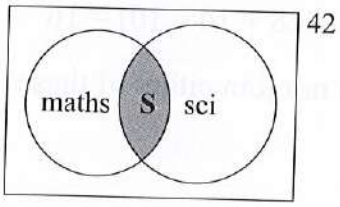
$S = 55 - 40 = 15$

15 students completed both homework.

2 22 students from Class 5F scored 90 marks and above in a mathematics examination. 30 students from the same class scored 90 marks and above in a science examination. If the class size is 42, how many students scored 90 marks and above in both examinations?

Solution:

Let the number of students who scored 90 marks and above in both examinations be S. Note that S has been counted twice.



$22 + 30 - S = 42$

$52 - S = 42$

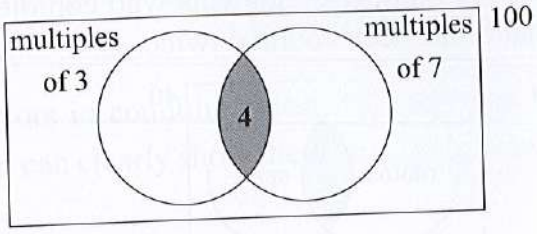
$S = 52 - 42 = 10$

10 students scored 90 marks and above in both examinations.

3 How many numbers ranging from 1 to 100 are not multiples of 3 and not divisibl by 7?

Solution:

Common multiples of 3 and 7: 21, 42, 63, 84.
Number of multiples of 3 ranging from 1 to 100: $100 \div 3 = 33$ R 1
Number of multiples of 7 ranging from 1 to 100: $100 \div 7 = 14$ R 2

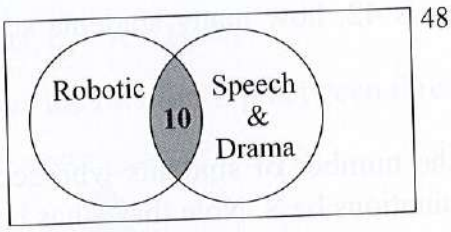


$$100 - (33 + 14 - 4) = 57$$

57 numbers ranging from 1 to 100 are not multiples of 3 and not divisibl by 7.

4 Class 5D has 48 students. 28 of them join the Robotic Club. 14 of them join th Speech and Drama Club. 10 students join both clubs. How many students in Cla 5D do not join either of these clubs?

Solution:



$$48 - (28 + 14 - 10) = 16$$

16 students in Class 5D do not join either of these clubs.

3 and not divisibl

) ÷ 3 = 33 R 1
) ÷ 7 = 14 R 2

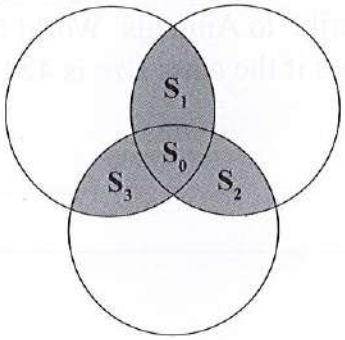
00

f 3 and not divisibl

14 of them join th
ny students in Cla

clubs.

5 3 identical circles each has an area of 157 cm^2. When the circles overlap one another partially, the total area becomes 280 cm^2. If $S_0 = 56$ cm^2, find the value of $S_1 + S_2 + S_3$.



Solution:

$$157 + 157 + 157 - (S_1 + 56) - (S_2 + 56) - (S_3 + 56) + 56$$

$$= 471 - S_1 - S_2 - S_3 - 56 - 56 - 56 + 56$$

$$= 471 - 112 - (S_1 + S_2 + S_3)$$

$$= 359 - (S_1 + S_2 + S_3)$$

$$359 - (S_1 + S_2 + S_3) = 280 \text{ cm}^2$$

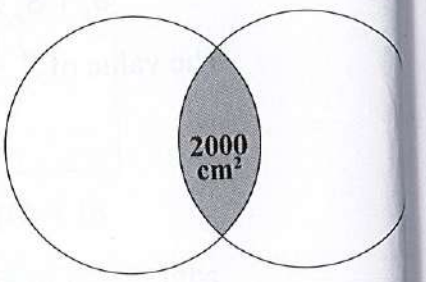
$$S_1 + S_2 + S_3 = 79 \text{ cm}^2$$

The value of $S_1 + S_2 + S_3$ is **79 cm^2**.

PRACTICE

1 34 students from Class 4E subscribe to Today's Scientists Magazine. 22 student from the same class subscribe to Animals' World Magazine. How many student subscribe to both magazines if the class size is 42?

2 Two pieces of table cloth with areas of 8000 cm^2 and 6000 cm^2 are overlapped shown below. There is an overlap of 2000 cm^2. Find the total area of the two piec of table cloth.



azine. 22 student
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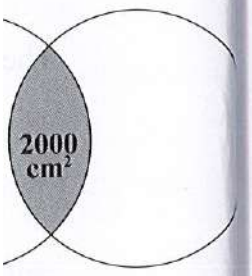
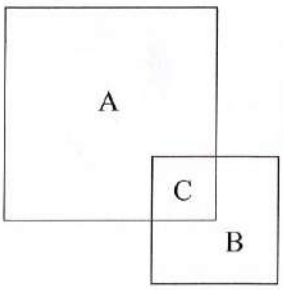
² are overlapped
ea of the two piec



3 Among 100 people who are being interviewed, 68 like music and 75 like sports. How many people like both music and sports?

4 The figure below is made up of A, B and C, which are squares of sides 10 cm, 6 cm and 3 cm respectively. What is the area of the figure?



5 How many numbers ranging from 1 to 100 are not multiples of 3 and not divisible by 11?

6 How many numbers ranging from 1 to 200 are multiples of 5 and divisible by 3?

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'3 and not divisib

7 Among the Primary 5 students who took part in the final examination, 45 of them scored distinction for mathematics. 78 students scored distinction for English, 38 of them scored distinction for both subjects and 64 students did not get distinction for either subject. How big was the cohort of Primary 5 students?

ınd divisible by 3

8 Class 5J has 42 students. 18 of them enjoy swimming, 14 of them enjoy cycling and 6 students like both activities. How many students do not like either activity?

9 Of all the Primary 5 students in a school, 26 took part in a mathematical competitio and 32 took part in a science competition. 20 students took part in both competitions and 64 of them did not take part in either competition. How many Primary 5 student were there?

10 38 students from Class 6D took part in a mathematical competition. 28 of them got the first question correct and 23 of them got the second question correct. If 1 students answered both questions correctly, how many students did not get eithe question correct?

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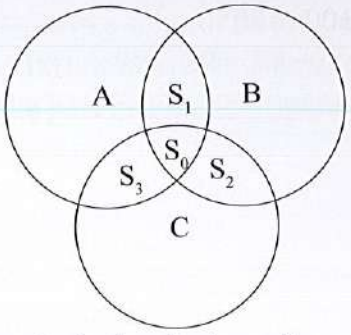
11 Given $1 + 2 + 3 + \ldots + 99 + 100 = 5050$,
find the value of $1 + 2 + 3 + \ldots + 97 - 98 - 99 - 100$.

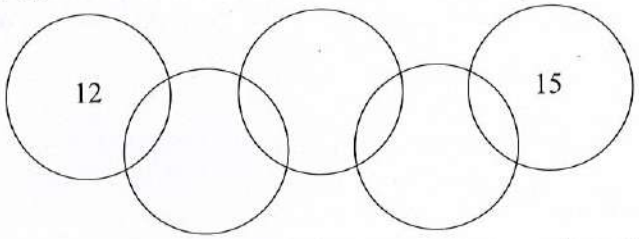
12 Among 100 club members, 56 like to play tennis and 74 like to swim. At most how many of these club members like both sports? At least how many of them like both sports?

13



The figure above shows 3 circles that overlap one another. The areas of A, B an C are 20 cm^2, 26 cm^2 and 18 cm^2 respectively. The following areas in the Ven diagram are $S_1 = 6\ cm^2$, $S_2 = 8\ cm^2$ and $S_3 = 10\ cm^2$. Find the area of S_0 if the tot area of the figure is 45 cm^2.

14 A total of 9 regions are formed by 5 circles. Two such regions are assigned th numbers 12 and 15.



Use the numbers 3, 4, 5, 6, 8, 9 and 11 only once to assign the rest of the regio so that the sum of numbers in each circle is 20.

15 Among the Primary 6 students in a school, 60 like mathematics, 50 like English and 42 like sports. 30 of them like both mathematics and English, 20 of them like both English and sports and 35 of them like both mathematics and sports. If 18 students like all the three subjects, how many students have exactly one favourite subject?

16 How many numbers from 1 to 80 are divisible by 3, 4 and 5?

Fibonacci Numbers

Born in the city-state of Pisa in Italy, Leonardo of Pisa, also known as Fibonacci, mad a significant contribution in the field of mathematics. A statue of Fibonacci was erecte in his honour and it can be found along the Arno River.

Fibonacci was one of the greatest Western mathematicians of the Middle Ages. The was not much record about his personal life, except he was born in roughly 1170 ar passed away around 1250.

His greatest contribution in the field of mathematics included the introduction of th Hindu-Arabic numerals 0 to 9 to Western Europe. His book, Liber Abaci (or Book Calculations), documented the Hindu-Arabic numerals and how they could be useful business transactions.

He also generated the famous Fibonacci numbers, whereby each term after the first tw terms is the sum of the two preceding terms.

1, 1, 2, 3, 5, 8, 13, 21, ...

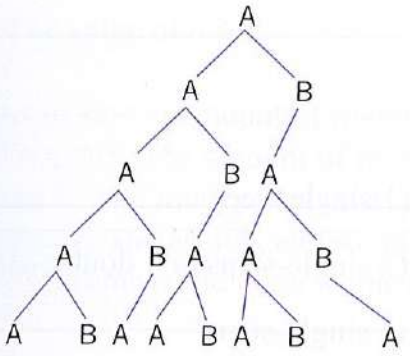
The questions in this chapter will demonstrate the application of Fibonacci numbers as to eliminate laborious mathematical computations.

EXAMPLES

1 A pair of rabbits can reproduce a pair of baby rabbits every month. The baby rabbits take a month to mature before they can reproduce another pair of baby rabbits. How many pairs of rabbits will there be by the end of one year?

***Solution*:**

Let A denote a pair of adult rabbits and B denote a pair of baby rabbits.



→ 1 pair at the end of first month

→ 2 pairs at the end of second month

→ 3 pairs at the end of third month

→ 5 pairs at the end of fourth month

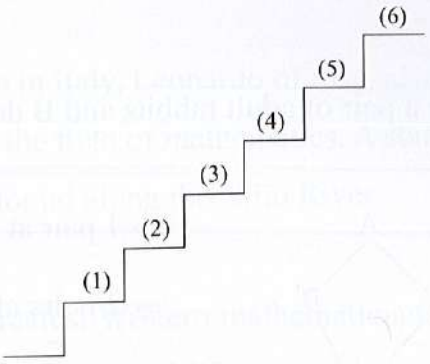
→ 8 pairs at the end of fifth month

1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, ...

There will be **233** pairs of rabbits by the end of one year.

2 There is a flight of 6 steps in a staircase. Dave can take single or double steps each time when he climbs the staircase. In how many ways can he complete the climbing?

Solution:



To reach step (1) : 1 way (1 single-step)

To reach step (2) : 2 ways (2 single-steps or 1 double-step)

To reach step (3) : 3 ways (3 single-steps,
1 double-step followed by 1 single-step,
1 single-step followed by 1 double-step)

To reach step (4) : 5 ways (4 single-steps,
2 single-steps followed by 1 double-step,
1 single-step followed by 1 double-step and 1 single-step,
1 double-step followed by 2 single-steps and 1 single-step,
2 double-steps)

We have shown that this is a set of Fibonacci numbers.

1, 2, 3, 5, 8, 13, ...

He can complete the climbing in **13** ways.

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A sequence follows the pattern of Fibonacci numbers. Find the value of a if the value of the 6th term equals 50 in

$$5, a, 5 + a, \ldots$$

Solution:

The sequence is

$5, a, 5 + a, 5 + 2a, 10 + 3a, 15 + 5a, \ldots$

$15 + 5a = 50$

$5a = 35$

$a = 7$

The value of a is **7**.

Jolene plans to save a amount of money in January, and increase to b amount of money in February. The amount of money she plans to save in March will be the sum of a and b. The money she saves in April will be the amount of money she saves in February and March, and so on. Find the values of a and b if Jolene saves \$75 in June. Assume a and b are whole numbers.

Solution:

Jan : a

Feb : b

Mar : $a + b$

Apr : $a + 2b$

May : $2a + 3b$

Jun : $3a + 5b$

$3a + 5b = 75$

$5b = 75 - 3a$

$b = \dfrac{75 - 3a}{5}$

When $a = 5$,

$$b = \frac{75 - 3 \times 5}{5} = 12$$

The values of a and b are **\$5** and **\$12** respectively.

PRACTICE

1 Find the missing numbers in each of the following.

(a) 3, 6, ( ), ( ), 24, ( ), …

(b) 10, 15, ( ), ( ), 65, ( ), …

(c) 2, 2, 4, ( ), ( ), ( ), …

(d) 3, 5, ( ), ( ), 21, ( ), …

2 Find the missing numbers in each of the following.

(a) ( ), ( ), 12, 20, ( ), ( ), …

(b) ( ), ( ), 17, 29, ( ), ( ), …

(c) ( ), 20, ( ), 60, ( ), ( ), …

(d) ( ), 18, ( ), 53, ( ), ( ), …

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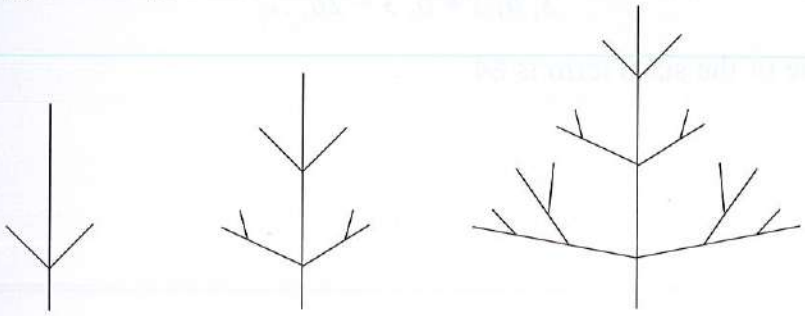
3 The three drawings below depict the initial stages of growth of a tree. Draw the 4th and 5th stages of the growth.



1st 2nd 3rd

4 Find the value of a in the following sequence. The value of the sixth term equals 60.

$a, 3, a + 3, a + 6, \ldots$

5 Find the value of b in

$$3, b, 3 + b, 3 + 2b, \ldots$$

if the value of the sixth term is 84.

7

6 It is assumed that a pair of adult guinea pigs can reproduce another pair of bal 8
guinea pigs every month. The baby guinea pigs take one month to grow before th
can reproduce another pair of baby guinea pigs. In which month will there be
pairs of guinea pigs? How many pairs of guinea pigs will there be in a year?

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7 There are 12 matchsticks on a table. In how many ways can you draw the matchsticks if you can draw one or two matchsticks each time?

nother pair of ba
to grow before th
ıth will there be
: be in a year?

8 There is a flight of 8 steps in a staircase. A person can either take one or two steps up each time. In how many ways can the person reach the eighth step?

9 Vanessa bought a pair of baby rabbits in January. The baby rabbits grew up to adu
rabbits in the second month and became productive in the third month. How mar
pairs of rabbits would there be in 12 months?

10 The second term of a Fibonacci number is 12. Its eighth term is 180. Find the val
of its first term.

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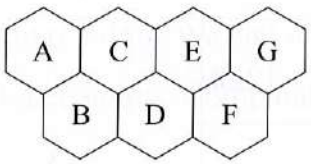
its grew up to adu
month. How ma

180. Find the val

11 Tanya saved a dollars in the first month and b dollars in the second month. The money she saved in the third month was the sum of the money she saved in the first and second months. The money she saved in the fourth month was the sum of the money she saved in the second and third months, and so on. If Tanya saved \$280 in the ninth month, what were the values of a and b given that they were whole numbers?

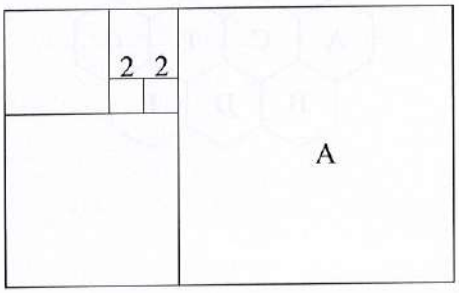
12 In how many ways can a bee reach cell G if it starts its journey from either cell A or cell B?



13 Find the Fibonacci numbers in the Pascal Triangle shown below.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
... ...

14 The figure below is made from squares of different sizes. Find the area of A if t
side of each of the two identical smallest squares is 2 cm.



6 Permutation and Combination

Permutation is a form of arrangement that chooses **r** items from a total of **n** items and then arranging them according to a specific requirement. Thus, it is important that the order chosen or position arranged is taken into consideration.

For example, we write ${}^5P_3$ to represent 3 items chosen from 5 items for arrangement.

The symbol for permutation is ${}^nP_r$.

Combination, on the other hand, is an arrangement of items regardless of order or position.

The symbol for combination is ${}^nC_r$.

For example, there are 10 points on a circle, and we want to know how many straight lines can be drawn between any two points. We can write ${}^{10}C_2$. This means 2 points are needed for a line and there are 10 such points. Computation of ${}^nC_r$ is found in Example 1.

EXAMPLES

1 Evaluate

(a) ${}^5P_3$

(b) ${}^{30}P_4$

(c) ${}^5C_3$

(d) ${}^{10}P_4$

(e) ${}^6C_4$

(f) $3 \times {}^4P_1 - {}^3P_2$

(g) ${}^{10}C_3 \times {}^6C_2$

(h) ${}^8P_{4\times}{}^4P_4$

Solution:

(a) $^5P_3 = 5 \times 4 \times 3$
$= \mathbf{60}$

(b) $^{30}P_4 = 30 \times 29 \times 28 \times 27$
$= \mathbf{657\,720}$

(c) $^5C_3 = \dfrac{5 \times 4 \times 3}{3 \times 2 \times 1}$
$= \mathbf{10}$

(d) $^{10}P_4 = 10 \times 9 \times 8 \times 7$
$= \mathbf{5040}$

(e) $^6C_4 = \dfrac{6 \times 5 \times 4 \times 3}{4 \times 3 \times 2 \times 1}$
$= \mathbf{15}$

(f) $^4P_1 = 4$
$^3P_2 = 3 \times 2 = 6$
$3 \times 4 - 6 = \mathbf{6}$

(g) $^{10}C_3 \times {}^6C_2$
$^{10}C_3 = \dfrac{10 \times 9 \times 8}{3 \times 2 \times 1}$
$= 120$
$^6C_2 = \dfrac{6 \times 5}{2 \times 1}$
$= 15$
$120 \times 15 = \mathbf{1800}$

(h) $^8P_4 \times {}^4P_4$
$= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
$= \mathbf{40\,320}$

2 There are 12 players in a volleyball team. However, only 6 players are needed for volleyball match. How many ways are there to choose 6 players from 12 players

Analysis: The order of the players is not a concern in this case, therefore we choo 6 players from 12 players, $^{12}C_6$.

Solution:

$$^{12}C_6 = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$
$$= 924$$

There are **924** ways to choose 6 players from 12 players.

3 In how many ways can you form a three-digit number using the digits 1, 2, 3, and 5 if all the digits in the number are different? How many of these numbers a odd?

Solution:

There are 5 ways to choose the hundreds digit, 4 ways to choose the te digit and 3 ways to choose the ones digit.

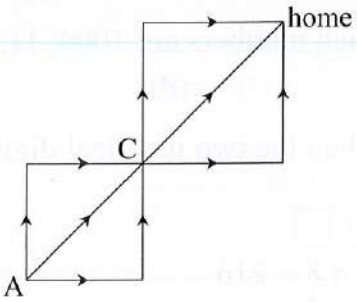
$$^5P_3 = 5 \times 4 \times 3 = 60$$

I can form a three-digit number using the digits 1, 2, 3, 4 and 5 in **60** wa There are 3 ways to form the ones digit using either the digits 1, 3 o so that the number is odd. There are 4 numbers left to form the hundre digit and 3 numbers left to form the tens digit.

$$3 \times 4 \times 3 = 36$$

36 of these numbers are odd.

4 In how many ways can an ant return home from Point A by following the directions of the arrows?



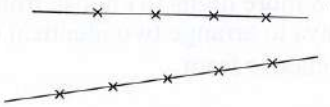
Solution:

There are 3 ways to move from Point A to Point C. There are 3 ways to move from Point C to home.

$$3 \times 3 = 9$$

An ant can return home from Point A in **9** ways.

5 Line **a** has 5 points on it and line **b** has 4 points. How many triangles can be formed using these points as vertices? How many quadrilaterals can be formed?



Analysis: For a triangle, we choose any two points on, say, line **a** to form the base. The remaining vertex must then be a point on line **b**.

Solution:

Scenario 1: When the base is on line **a**, the combination is ${}^5C_2$. Since the remaining vertex is on line **b**, the combination is ${}^4C_1$.

$${}^5C_2 \times {}^4C_1 = \frac{5 \times 4}{2 \times 1} \times 4 = 40$$

Scenario 2: When the base is on line **b** and the third vertex is on line **a**.

$${}^4C_2 \times {}^5C_1 = \frac{4 \times 3}{2 \times 1} \times 5 = 30$$

$$40 + 30 = 70$$

70 triangles can be formed using these points as vertices.

To form a quadrilateral, we can choose any 2 points from line **a** and another 2 points from line **b** (the order of which is not important).

$${}^5C_2 \times {}^4C_2 = \frac{5 \times 4}{2 \times 1} \times \frac{4 \times 3}{2 \times 1}$$

$$= 10 \times 6$$

$$= 60$$

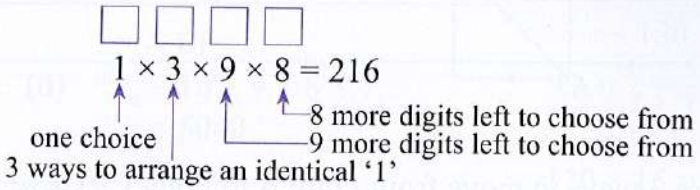
60 quadrilaterals can be formed.

6 How many four-digit numbers that begin with the digit 1 have two identical digits including 1?

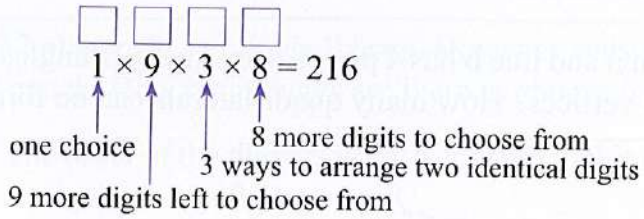
***Analysis*:** Examples of such numbers are 1004, 1153, 1353, ...

***Solution*:**

Scenario 1: When the two identical digits are 1,



Scenario 2: When the two identical digits are other than 1,



$$216 + 216 = 432$$

432 four-digit numbers that begin with the digit 1 have two identical digits, including 1.

PRACTICE

1 Evaluate

(a) ${}^{5}P_{4}$

(b) ${}^{18}P_{3}$

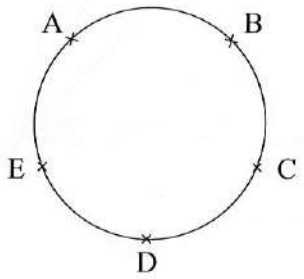
(c) ${}^{10}C_{4}$

(d) ${}^{8}C_{3}$

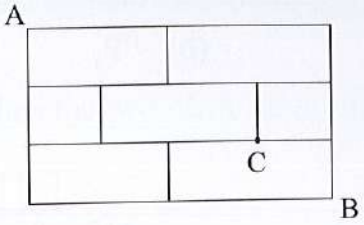
(e) $2 \times {}^{5}P_{2} - {}^{3}P_{2}$

(f) ${}^{7}C_{3} \times {}^{6}C_{2}$

2 There are 5 points, A, B, C, D and E, on the circumference of a circle. How many triangles can be formed if at least one of the points, A and/or B, is/are used as a vertex/vertices?



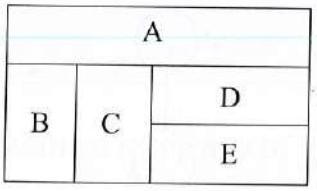
3 A worker wants to walk from site A to site B using the routes as shown below. I site C is out of bounds, how many ways are there for the worker to reach site B Only → and ↓ directions are allowed.



4 There are ten points on the circumference of a circle. How many triangles can b formed by using any three points as their vertices?

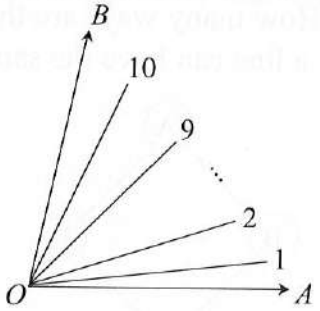
is shown below. I
er to reach site B

5 In how many ways can each region of the figure be coloured using 4 different colours so that no adjacent regions will have the same colour?

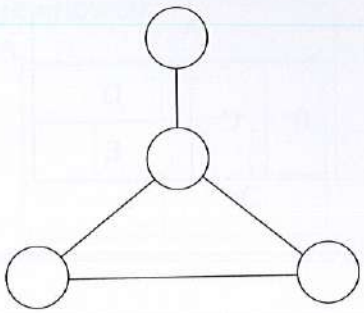


y triangles can

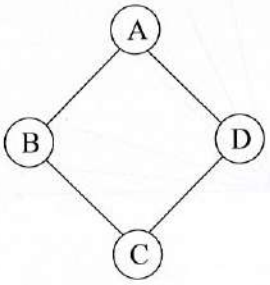
6 How many acute angles can be formed from the figure if $\angle BOA$ is less than 90°?



7 How many ways are there to colour the circles using 3 different colours so that no two circles joined by a line have the same colour?



8 Four circles shown below are connected by lines. Each circle is to be coloured using either yellow, black or red. How many ways are there to colour the four circles so that no two circles connected by a line can have the same colour?



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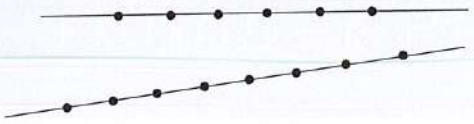
nt colours so that n

s to be coloured usi
our the four circles

9 The two lines shown below have 8 points and 6 points on them.



How many triangles can be formed using any of these 3 points as their vertices?

10 How many three-digit odd numbers can be formed using the digits 1, 3, 6, 7 and 8? Each digit can only be used once in every number.

11 A staircase has 10 steps. If Peter can climb either 2 or 3 steps each time, in ho many ways can he reach the tenth step?

12 Mrs Reed forgets the number combination to her safe. The combination h 4 numbers, each a 1-digit number. The 4 digits add up to 10 and there is no ze How many times at the most does Mrs Reed have to try to open her safe?

Number Pattern

"A great discovery solves a great problem, but there is a grain of discovery in the solution of any problem. Your problem may be modest, but if it challenges your curiosity and brings into play your inventive faculties, and if you solve it by your own means, you may experience the tension and enjoy the triumph of discovery."

~ George Polya ~

George Polya (1887 – 1985) was one of the greatest contemporary mathematicians. Born in Hungary, he migrated to the United States of America in 1940. He was a professor of mathematics in Stanford University.

His greatest contribution is his effort to characterize the various methods used in solving mathematical problems. In 1945, his first work, *How to Solve It*, was published. This book became a household name very quickly and was later translated into many languages.

In this book, he elaborated on the four principles of problem solving:

1. Understand the Problem

- Do you understand all the words used in stating the problem?
- What are you asked to find or show?
- Can you think of a picture or diagram that may help you understand the problem?
- Is there enough information to find the solution?
- What information, if any, is missing?

2. Devise a Plan

What will be the best approach to address the problem?

Listed below are some strategies that can help.

- Make a systematic list / table
- Write an equation
- Consider special cases
- Use direct reasoning
- Use indirect reasoning
- Look for a pattern
- Draw a picture
- Solve a simpler problem
- Use a model
- Work backwards

3. Carry Out the Plan

- Solve the problem with great care and patience
- Discard the plan if it does not work and devise a new one
- Record what you have done to avoid repetitive work

4. Looking Back or Checking

- Have you addressed the problem?
- Is your answer reasonable?
- Can the method be applied to other similar problems?

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PRACTICE

1 Observe the following number pattern. What is the first number in the 15^{th} row?

					1					
				3	5	7				
			9	11	13	15	17			
		19	21	23	25	27	29	31		
	...	...	...	...	...	...	...	...	...	
...	...	...	...	...	...	...	...	...	...	...

2 Find the sum of the number pattern below.

1	2	3	4	...	...	20
2	3	4	5	...	...	21
3	4	5	6	...	...	22
⋮	⋮	⋮	⋮	...	...	⋮
⋮	⋮	⋮	⋮			⋮
20	21	22	23	...	...	39

3 In which row and column will the number 2008 appear?

A	B	C	D	E
	8	6	4	2
10	12	14	16	
	24	22	20	18
26	28	30	32	
	...	...	...	...
...	...	...	...	

4 The first three numbers of a series are 1, 9 and 8. The fourth number is the remaind when the sum of the first three numbers is divided by 3. The fifth number is th remainder when the sum of the second, third and fourth numbers is divided by and so on. What is the 2008th number?

5 1, 2, 4, 7, 11, 16, 22, 29, 37, 46, 56, …
Find the remainder when the 2008th number in the above pattern is divided by 5.

6 In which row and column will the number 2008 be in the number pattern shown below?

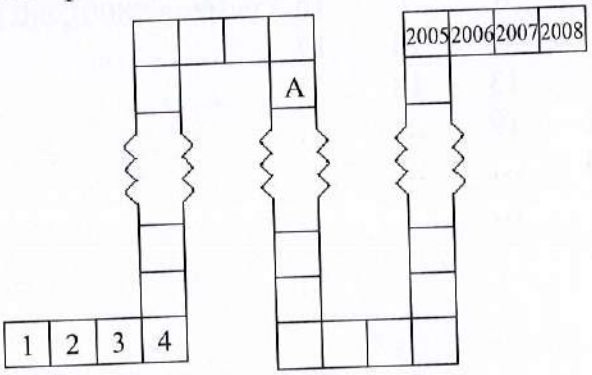
1	2	6	7	15	16	…	…
3	5	8	14	17	…	…	
4	9	13	18	…	…		
10	12	19	…	…			
11	20	…	…				
21	…	…					
…	…						
…							

7

$$
\begin{array}{ccccccccc}
26 & \rightarrow & \cdots & \cdots & \cdots & \cdots & \cdots & & \\
\uparrow & & & & & & & & \\
25 & & 10 & \rightarrow & 11 & \rightarrow & 12 & \rightarrow & 13 \\
\uparrow & & \uparrow & & & & & & \downarrow \\
24 & & 9 & & 2 & \rightarrow & 3 & & 14 \\
\uparrow & & \uparrow & & \uparrow & & \downarrow & & \downarrow \\
23 & & 8 & & 1 & & 4 & & 15 \\
\uparrow & & \uparrow & & & & \downarrow & & \downarrow \\
22 & & 7 & \leftarrow & 6 & \leftarrow & 5 & & 16 \\
\uparrow & & & & & & & & \downarrow \\
21 & \leftarrow & 20 & \leftarrow & 19 & \leftarrow & 18 & \leftarrow & 17
\end{array}
$$

In the spiral number pattern, the number at the first corner is 2. The number at th second corner is 3 and so on. Find the number at the 22nd corner.

8 The house numbers in a residential area run from 1, 2, 3, ..., 2007, 200 Help the postman to figure out the house number of the house marked with "A" that he can deliver his parcel.



9

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24

The sum of the five numbers in the shape of a cross is 55. If the sum of the five numbers in a similar cross is 625, what are the five numbers?

10 In the number pattern from 1 to 2000,

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
…	…	…	…	…	…	…
…	…	…	…	…	…	…

find a 3-by-3 number block where the sum of the nine numbers add up to 12 573.

Speed 1

This chapter aims to combine various mathematical problem-solving techniques suc as writing equations, using ratio and using logical reasoning to solve dynamic spee problems besides the use of conventional formulae mentioned below.

$$\text{Speed} = \text{Distance} \div \text{Time}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Time} = \text{Distance} \div \text{Speed}$$

To equip students with skills to solve problems of this nature, it is noteworthy t understand that the ratio of speed of two moving objects is the reverse of the ratio o their time. The ratio of distance travelled by the two objects, on the other hand, is th same as the ratio of their speeds.

This knowledge is particularly useful when we attempt to solve Example 7, an subsequently Questions 10 and 12.

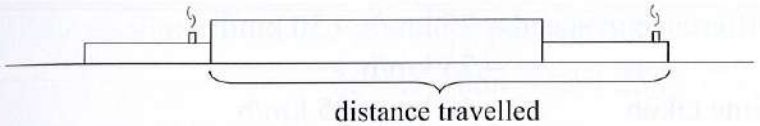
In Example 4 and Question 5, where the speed of an escalator is studied, we are give the opportunity to appreciate the nature of one moving object on another moving objec The problems in Example 3 and Question 4 are solved using simple reasoning skills.

The problems on average speed in Example 2 and Question 3 are attempted by tw methods, namely using a hypothetical distance, which is a common multiple, and b writing equations.

EXAMPLES

1 It took 65 seconds for a train to pass a bridge that is 1200 m long. It took the same train, travelling at the same speed, 85 seconds to pass a tunnel that is 1600 m long. How long was the train? At what speed was the train travelling?

Analysis: The distance travelled is the length of the train and the length of the bridge (or tunnel).



Solution:

Difference in distance covered = 1600 m – 1200 m = 400 m
Difference in time taken = 85 s – 65 s = 20 s
The train can travel 400 m in 20 s.

$$400 \text{ m} \div 20 \text{ s} = 20 \text{ m/s}$$

The train was travelling at **20 m/s**.

$20 \times 65 - 1200 = 100$ m or $20 \times 85 - 1600 = 100$ m
The train was **100 m** long.

2 Mr Brown drove at a speed of 96 km/h to his office every morning. He returned home from work at a speed of 72 km/h. What was his average speed for the round-trip?

Solution:

Method 1: Writing Equations

Let the distance be d.

$$\text{Total time} = \frac{d}{96} + \frac{d}{72} = \frac{3d}{288} + \frac{4d}{288} = \frac{7d}{288}$$

Total distance covered = $2d$

$$\text{Average speed} = 2d \div \frac{7d}{288} = 2d \times \frac{288}{7d} = 82\frac{2}{7} \text{ km/h}$$

Method 2: Solve by Reasoning

Assume the distance to be 96 km.

$$\text{Time taken for the round-trip} = 96 \div 96 + 96 \div 72 = 1 + 1\frac{1}{3} = 2\frac{1}{3}\text{h}$$

$$\text{Average speed} = 96 \times 2 \div 2\frac{1}{3} = 192 \times \frac{3}{7} = \frac{576}{7} = 82\frac{2}{7} \text{ km/h}$$

His average speed for the round-trip was $\mathbf{82\frac{2}{7}}$ **km/h**.

3 Professor Greenberg travelled by scooter from his house to a town that was som distance away. If he travelled at a constant speed of 50 km/h, he would have arrive at 11 am. If he travelled at 75 km/h, he would have arrived at 10 am. At what spee should he be riding if he wanted to arrive at 10.30 am?

Solution:

Method 1: Solve by Reasoning

At 10 am, he still had 50 km × 1 h = 50 km to go if he travelled at 50 km/h

Difference in speed = 75 km/h – 50 km/h
= 25 km/h

Time taken = 50 km ÷ 25 km/h
= 2 h

It took 2 hours to be 50 km behind.

It would take him 2 hours to arrive at the destination if he travelled a 75 km/h. It would take him 3 hours to arrive at the destination if h travelled at 50 km/h.

50 km/h × 3 h = 150 km

or

75 km/h × 2 h = 150 km

The distance between his house and the town was 150 km.

150 km ÷ 2.5 h = 60 km/h

Method 2: Writing Equations

Let n be the number of hours needed for him to arrive at the destinatio at the speed of 50 km/h.

Distance = 50 km/h × n or 75 km/h × $(n - 1)$
= $50n$ or $75(n - 1)$

$$50n = 75(n - 1)$$
$$50n = 75n - 75$$
$$75n - 50n = 75$$
$$25n = 75$$
$$n = 75 \div 25$$
$$= 3$$

50 km/h × 3 h = 150 km

150 km ÷ 2.5 h = 60 km/h

He should be riding at a speed of **60 km/h** if he wanted to arrive at 10.30 a

4 Frank and Ronnie embarked on a moving escalator to the second floor of a shopping mall. Frank walked up the escalator at a speed of 2 steps every second. At the same time, Ronnie walked up the same escalator at a speed of 3 steps every 2 seconds. Frank arrived at the second floor in 25 seconds. It took Ronnie 30 seconds to reach the same floor. How many steps had the escalator when it was stationary?

Analysis: First, we need to find the speed of the escalator.

Solution:

If the escalator travelled at n steps per second,

$$25 \times 2 + 25n = 50 + 25n$$

Frank would have travelled $50 + 25n$ steps.

$$30 \times \frac{3}{2} + 30n = 45 + 30n$$

Ronnie would have travelled $45 + 30n$ steps.

$$50 + 25n = 45 + 30n$$

$$5n = 5$$

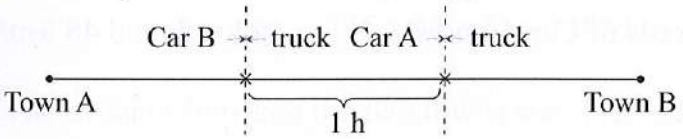
$$n = 1 \text{ step every second}$$

$50 + 25 \times 1 = 75$ or $45 + 30 \times 1 = 75$

The escalator had **75** steps when it was stationary.

5 Two cars, travelling at 62 km/h and 54 km/h respectively, left Town A for Town B at the same time. The faster car passed an oncoming truck 10 hours later. An hour after that, the slower car and the truck also passed each other. Find the speed of the truck.

Analysis: The clue lies in the two meeting points.



Solution:

When the faster Car A met the truck,

the slower Car B was $(62 - 54) \times 10 = 80$ km behind Car A.

Together, Car B and the truck travelled 80 km in an hour.

80 km/h – 54 km/h = 26 km/h

The speed of the truck was **26 km/h**.

6 Town A and Town B are 420 km apart. A car leaves Town A for Town B a another car leaves Town B for Town A. If Car A has a head start of 2.5 hours, t two cars will pass each other 2.5 hours after Car B starts its journey. If Car B h a head start of 2 hours, the two cars will pass each other 3 hours after Car A sta its journey. Find their respective speeds.

***Solution*:**

Scenario 1:
The two cars pass each other 5 hours after Car starts its journey a 2.5 hours after Car B starts its journey.

Scenario 2:
They meet 5 hours after Car B starts its journey and 3 hours after Ca starts its journey.

Let a and b be the speeds of cars A and B respectively.

$5a + 2.5b = 420$ — (1)
$3a + 5b = 420$ — (2)

(1) × 2
$10a + 5b = 840$ — (3)

(3) – (2)
$7a = 420$
$a = 420 \div 7 = 60$ km/h

$10 \times 60 + 5b = 840$
$5b = 840 - 600 = 240$
$b = 240 \div 5 = 48$ km/h

The speeds of Car A and Car B are **60 km/h** and **48 km/h** respectiv

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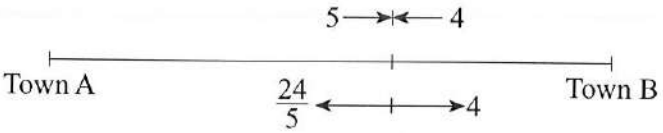
A car left Town A for Town B. Another car left Town B for Town A at the same time. The ratio of their speeds was 5 : 4. The first car reduced its speed by $\frac{1}{5}$ and the second car increased its speed by 20% after they had passed each other. When the first car reached its destination, the second car was still 25 km away from Town A. Find the distance between the two towns.

Analysis: First car's reduced speed $= 5 \times (1 - \frac{1}{5}) = 4$ units

Second car's increased speed $= 4 \times (1 + \frac{1}{5}) = \frac{24}{5}$ units

Solution:



Time taken for first car to reach Town B after they passed each other $= 4 \div 4 = 1$ h

Distance covered by second car after the meeting point $= \frac{24}{5} \times 1$

$= \frac{24}{5}$ units

Distance that the second car need to travel $= 5 - \frac{24}{5}$

$= \frac{1}{5}$ unit

$25 \div \frac{1}{5} = 125$ units

$125 \times (4 + 5) = 1125$ km

The distance between the two towns was **1125 km**.

PRACTICE

1 It took a train 50 seconds to pass a tunnel that was 1000 m long. It took the same train, travelling at the same speed, 75 seconds to pass a bridge th was 1625 m long. How long was the train? At what speed was it travelling?

2 Two trains travelled towards each other on different tracks. Train A travelled a speed of 54 km/h and Train B travelled at 72 km/h. A commuter aboard Train recorded that Train B took 8 seconds to pass his window seat completely. How lo was Train B?

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long.
ds to pass a bridge th
was it travelling?

s. Train A travelled a
mmuter aboard Train
at completely. How lo

3 Mr Welch drives at a constant speed of 108 km/h to his office every morning. He drives at a constant speed of 72 km/h for his return trip. What is his average speed for the round-trip?

4 Mr Cartland is driving from Red Deer to Okonagan for a business meeting. If he drives at a speed of 90 km/h, he will arrive at 4 pm. If he drives at a speed of 120 km/h, he will arrive 2 hours earlier. At what speed should Mr Cartland be driving if he wants to arrive at 3.12 pm?

5 Henry and Robert embarked on a moving escalator. Impatient with the spee the escalator, Henry walked at a speed of 3 steps per second while Robert wa at a speed of 2 steps per second. How many steps had the escalator when it stationary? It was given Henry took 20 seconds to arrive at the second floor Robert took 25 seconds to arrive at the same floor.

6 A backpacker was walking at a speed of 5 km/h towards Mississauga that 40 km away. He got a ride from a motorist who drove at 25 km/h in the s direction. They arrived at Mississauga 2.5 hours later. How far had the backpa walked before he took a ride from the motorist?

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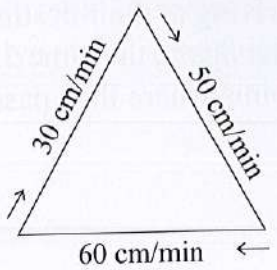
7 Car A departed from Town A and travelled towards Town B. At the same time, Car B started its journey in the direction of Town A from Town B. The two cars passed each other 64 km away from Town B and carried on with their journeys. They made U-turns upon arriving at their destinations so as to return to their own towns. They passed each other again, this time 52 km away from Town A. Find the distance between the two points where they passed each other.

8 Barton, Carl and Dan were having a 100-m race. When Carl dashed across the finishing line, Barton was 10 m away and Dan was 15 m away. How far was Dan away from the finishing line when Barton completed the race?

9 A beetle is climbing along the three sides of an equilateral triangle. The speeds which it travels along the three sides are 30 cm/min, 50 cm/min and 60 cm/m What is the average speed for the beetle to complete the whole journey?



10 A car left Town A for Town B. Another car left Town B for Town A at the sa time.
The ratio of the speeds of the two cars was 6 : 5 initially.
After the two cars passed each other, Car A's speed was reduced by $\frac{1}{6}$ and Car speed was reduced by 25%. When Car A arrived at Town B, Car B was still 54 away from Town A. Find the distance between Town A and Town B.

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11 In school, Adam and Bernard run along a 400-m track in opposite directions. Their speeds are 6 m/s and 5 m/s respectively. How many times do they pass each other in 15 minutes?

12 A car travels at a constant speed towards a town. If it increases its speed by 15 km/h, the time required is in a ratio of 6 : 5. If it reduces its speed by 15 km/h, it needs another 105 minutes to arrive at the destination. Find the distance travelled by the car.

Divisibility

Properties of Divisibility:

(a) If a is divisible by c,

$$c \mid a$$

and b is divisible by c,

$$c \mid b$$

then $(a + b)$ is divisible by c,

$$c \mid (a + b)$$

and $(a - b)$ is divisible by c,

$$c \mid (a - b).$$

(b) If a is divisible by a product, $b \times c$,

$$b \times c \mid a$$

then a is divisible by b,

$$b \mid a$$

and a is divisible by c,

$$c \mid a.$$

(c) If a is divisible by b,

$$b \mid a$$

and b is divisible by c,

$$c \mid b$$

then a is divisible by c,

$$c \mid a.$$

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EXAMPLES

1 The sum of 2008 and a number n is divisible by 13 and 19.
Find the smallest value of number n.

Analysis: If $(2008 + n)$ is divisible by 13 and 19, it will then be divisible by the product of 13 and 19, which is 247.
We, therefore, only need to find a multiple of 247 that is greater than 2008.

Solution:

$247 \times 8 = 1976$ (smaller than 2008)
$247 \times 9 = 2223$ (greater than 2008)

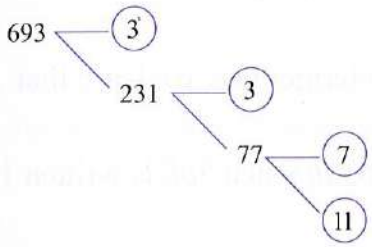
$2223 - 2008 = 215$
$n = 215$

The smallest value of n is **215**.

2 The product of 693 and a number n is a square number.
What is the smallest possible value of n?

Analysis: The prime factors of 693 and n must make pairs!

Solution:



$693 = 3 \times 3 \times 7 \times 11$

To be a square number,

$693 \times n = 3 \times 3 \times 7 \times 11 \times n$
$n = 7 \times 11$
$= 77$

The smallest possible value of n is **77**.

3 Find the possible values of a 6-digit number 568*abc* so that it can be div 3, 4 and 5.

***Analysis*:** (1) $5 + 6 + 8 + a + b + c$ must be divisible by 3.
(2) bc must be divisible by 4.
(3) c must either be 0 or 5.

***Solution*:**

When $c = 0$,
b must be one of the digits 2, 4, 6, 8 or 0.

When $b = 0$, 5 6 8 a 0 0, $a = 2, 5, 8$
$b = 2$, 5 6 8 a 2 0, $a = 0, 3, 6, 9$
$b = 4$, 5 6 8 a 4 0, $a = 1, 4, 7$
$b = 6$, 5 6 8 a 6 0, $a = 2, 5, 8$
$b = 8$, 5 6 8 a 8 0, $a = 0, 3, 6, 9$

The possible values of the 6-digit number are, therefore,

568200	**568020**	**568140**	**568260**	**568080**
568500	**568320**	**568440**	**568560**	**568380**
568800	**568620**	**568740**	**568860**	**568680**
	568920			**568980**

4 If 3*ab* is written 100 times in this way, $\underbrace{3ab3ab \ldots \ldots 3ab}_{100\ 3abs}$, it will become a of 91.
Find the values of a and b.

***Analysis*:** In Prime Factorisation from Intermediate, we learn that $1001 = 7 \times 11 \times 13$.

Now, there are 50 pairs of 3*ab*3*ab* when 3*ab* is written 100 tim

***Solution*:** $3ab \times 1001 = 3ab \times 11 \times 91$

Therefore, for any value of a or b, this condition will always be s and it will be a multiple of 91.

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PRACTICE

1 Find the value of $a + b + c$ if $173a$ is divisible by 9, $173b$ is divisible by 11 and $173c$ is divisible by 6.

2 The sum of 1997 and a number is divisible by 23 and 19.
Find the smallest possible value of this number.

3 A class monitor helped his teacher to collect some money from 72 students fo movie outing. The record of money collected was accidentally smeared with wa and only 3 digits of the total cash amount, $\square$71.5$\square$, were legible. Help the cl monitor to figure out the correct amount of money he had collected.

4 The product of 2860 and m is a square number. Find the smallest possible value m.

7 How many numbers from 1 to 2008 are divisible by either 2, 5 or 7?

8 Find the values of m and n if $m1998n$ is divisible by 9 and 11.

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, 5 or 7?

9 A series of new electronic parts numbered from 1 to 2008 are to be tested. Starting from the right, the 1st to the 5th electronic parts skip the testing and the 6th electronic part undergoes the testing. The 7th to the 11th electronic parts skip the testing and the 12th electronic part undergoes the testing, and so on. This process continues until there are 9 electronic parts left in the testing process. What is the original number of the 3rd electronic part from the left?

11.

10 A 6-digit number, 15*abcd*, is divisible by 36. Find the values of a, b and c so that the number has the least value of quotient when it is divided by 36.

11 If $7ab$ is written 2008 times in this manner, $\underbrace{7ab7ab \ldots \ldots 7ab}_{2008\ 7ab\text{s}}$, it will becom
a multiple of 143.
Find the values of a and b.

10 Solve Using Table or Drawing

Deployment of different strategies for different mathematical problems helps to alleviate the laborious process of working through them. Solving a mathematical problem using a table or drawing is another useful resource to be tapped into.

Tables and drawings are simple yet powerful tools for solving mathematical problems such as logic, working backwards, speed, etc.

The beauty of solving mathematical problems using table or drawing is manifold.

A well-sketched drawing sometimes will present the solution by itself almost instantly. A table, on the other hand, makes problem-solving systematic and allows us to do checking.

Furthermore, the tools of drawing or table help us translate a question into visual diagrams or images. This is especially true when we work on the speed problems.

Drawing is also a useful tool when we work on mathematical problems related to working backwards.

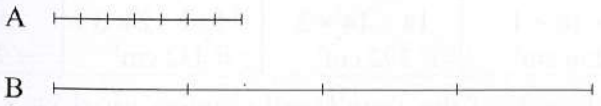
EXAMPLES

1 $\frac{5}{7}$ of number A is the same as $\frac{1}{4}$ of number B.

What is the minimum sum of numbers A and B?

Solution:

Method 1: Solve Using Drawing



1 part of B → 5 parts of A

$4 \times 5 + 7 = 27$

Method 2: Solve by Algebra

$$\frac{5}{7}A = \frac{1}{4}B$$

$$A = \frac{1}{4}B \times \frac{7}{5}$$

$$= \frac{7}{20}B$$

$$A + B = \frac{7}{20}B + B$$

$$= \frac{27}{20}B$$

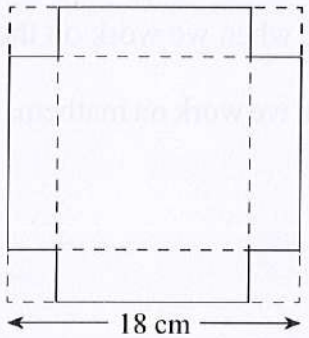
When B = 20,

$$\frac{27}{20} \times 20 = 27$$

$$A + B = 27$$

The minimum sum of numbers A and B is **27**.

2 The four corners of a square piece of paper of side 18 cm are cut off as show below. The piece of paper is then folded along the dotted lines into a tray. Wh must be the length of each of the four small squares so that the volume of the tr is the maximum? (Consider only whole numbers.)



***Analysis*:** Make a table that will allow us to check systematically.

***Solution*:**

side	1 cm	2 cm	3 cm	4 cm
base	16 cm	14 cm	12 cm	10 cm
volume	$16 \times 16 \times 1$ $= 256\ cm^2$	$14 \times 14 \times 2$ $= 392\ cm^3$	$12 \times 12 \times 3$ $= 432\ cm^3$	$10 \times 10 \times 4$ $= 400\ cm^3$

The length of each of the four small squares must be **3 cm** so that t volume of the tray is the maximum.

3 Three friends happened to wash their cars on 24th of June, a Sunday.

Alan: I wash my car every other day.
Brian: I wash my car every three days.
Charles: I wash my car every four days.
All of them will not wash their cars on Fridays.
They will do it on Saturdays instead.
Which day of the week will it be the next time they wash their cars on the same day?

Analysis: Making a table will allow us to check systematically.

Solution:

	Sun	M	T	W	Th	F	S	Sun	M	T	W	Th	F	S	Sun	M
Alan	✓		✓		✓	×	✓		✓		✓		×	✓		
Brian	✓			✓		×	✓			✓			×	✓		
Charles	✓				✓	×			✓				×	✓		

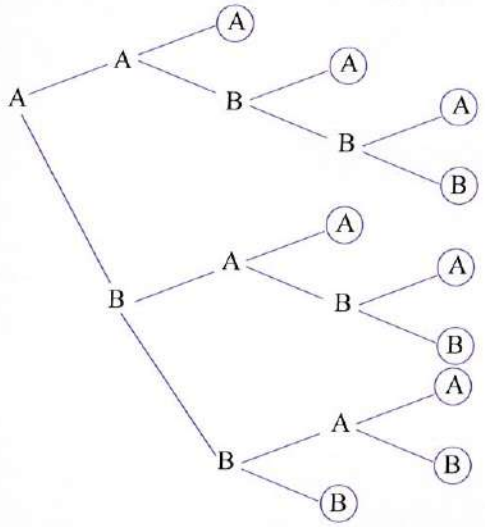
It will be a **Saturday** the next time they wash their cars on the same day.

4 The two finalists in a Snooker League championship will continue to play until one player wins a total of three games. How many ways are there for any finalist to win the championship?

Analysis: There are two scenarios: Finalist A takes the first game, or Finalist B takes the first game.

Solution:

If Finalist A takes the first game,



There are 10 possible results if A takes the 1st game.

$10 \times 2 = 20$ ways

There are **20** ways for any finalist to win the championship.

PRACTICE

1 There are three groups of red and blue marbles.
Each group has the same number of marbles.
The number of red marbles in the first group is the same as the number of marbles in the second group.
The number of red marbles in the third group is $\frac{2}{5}$ of the total number of red mar in all the three groups.
What is the fraction of blue marbles in all the three groups of marbles?

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2 156 marbles were to be given to A, B and C.
At first, A got the most number of marbles and C got the least number of marbles.
A then gave some marbles to B and C that doubled the number of marbles they had plus another 4 marbles each.
B also gave some marbles to A and C that doubled the number of marbles they had plus another 4 marbles each.
Finally, C gave A and B some marbles that were 4 more than doubled the number of marbles they had. Now A, B and C had the same number of marbles.
How many marbles did each of them get at first?

3 Jonathan and Samuel saved \$712 altogether.

$\frac{1}{4}$ of Jonathan's savings was \$48 more than $\frac{1}{6}$ of Samuel's savings.

How much did each boy save?

4 Whole numbers 1 to 10 are randomly arranged in the following manner:

10 3 7 4 6 5 8 9 1 2

How many times will you need to adjust the numbers so that it will become

1 2 3 4 5 6 7 8 9 10?

(Assume each number can only cross one number from its right at one time.)

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5 Suppose a coach left Calgary for Toronto every 4 hours. Another coach left Toronto for Calgary every 4 hours at the same time. It took the coaches 48 hours to arrive at their destinations. How many coaches would an eastbound coach come across before its arrival at the destination?

6 Machines A and B were supposed to produce 560 spare parts altogether. In the end, the two machines jointly produced 656 spare parts. If Machine A produced $\frac{1}{5}$ more spare parts than scheduled and Machine B produced $\frac{3}{20}$ more spare parts than scheduled, how many spare parts was each machine supposed to produce at first?

7 Three football teams played exactly one match against one another in the North Zone Football League. The table below shows the results of each team after the matches.

Team	Win	Loss	Draw	Goals Scored	Goals Conceded
A	2	0	0	6	1
B	0	1	1	2	4
C	0	1	1	3	6

What was the result of each match?

8 A cow produces a calf at the beginning of every year. The young cow will become productive and conceive in the fourth year and give birth to another calf the beginning of every year, starting from the following year. How many cows will there be in the eighth year?

9 When Jeremy divided his marbles into 3 groups of the same number, the remainder was 2. When he again divided his marbles from 2 of the groups into 3 groups of the same number, the remainder was still 2. Lastly, he arranged 2 groups of marbles from the previous division into 3 equal groups of the same number, and the remainder was still 2. How many marbles did Jeremy have at first?

Observation and Induction

In mathematics, inductive reasoning is a process whereby a general conclusion is drawn upon observing a pattern. However, this general conclusion may not stay true for all situations and it eventually leads to an analysis on the hypothesis if it is true, false or somewhere in between. Once a counterexample is cited, we can hypothesize it as false.

Inductive reasoning is particularly useful in patterns or number patterns. Its four basic steps are:

1. **Observation:** At this preliminary stage, the problem solver carries out careful observation on the pattern or number pattern.
2. **Conjecture of a Pattern:** The problem solver forms an opinion on the trend of the pattern or number pattern based on his observation.
3. **Testing of the Pattern:** The problem solver continues to come up with a few more patterns or number patterns to confirm what he conjectures.
4. **Write a Formula:** At this stage, it is possible to write a formula that will work for all cases of the pattern or number pattern that the problem solver is working on.

Example:

What is the eighth term in 1, 3, 8, 15, …?

In the number pattern above, one can observe that the numbers are not increasing at the same rate.

We then look for a similar number pattern whereby every term in the number pattern is the square of the term number.

$$1, 4, 9, 16, \ldots$$

Back to our original number pattern, we observe that, from the second term onwards, each term in the number pattern is the result of subtracting 1 from the square of the term number.

$$2^2 - 1 = 3$$
$$3^2 - 1 = 8$$
$$4^2 - 1 = 15$$

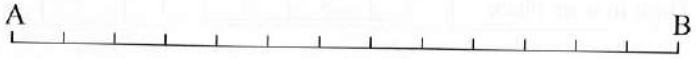
Hence, the formula will be $n^2 - 1$, where n is the term number.

$$\text{eighth term} = 8 \times 8 - 1 = 63$$

The formula $n^2 - 1$, indeed, works for this number pattern.

EXAMPLES

1 There are 12 points along Line AB, excluding A and B.
How many line segments can be counted altogether?



***Analysis*:** Let's observe the pattern carefully.

***Solution*:**

1 point between A and B = 1 + 2 = 3 line segments
2 points between A and B = 1 + 2 + 3 = 6 line segments
3 points between A and B = 1 + 2 + 3 + 4 = 10 line segments
⋮
12 points between A and B = 1 + 2 + 3 + … + 12 + 13
= (1 + 13) × 13 ÷ 2
= 91
91 line segments can be counted altogether.

2 Compute the sum of all the odd numbers smaller than 100.

***Solution*:**

Method 1: Solve by Reasoning

$$1 + 3 = 4 = 2 \times 2$$
$$1 + 3 + 5 = 9 = 3 \times 3$$
$$1 + 3 + 5 + 7 = 16 = 4 \times 4$$
$$\vdots$$
$$1 + 3 + 5 + \ldots + 97 + 99 = 50 \times 50 = 2500$$

Method 2: Solve by Commutative Law

1 + 3 + 5 + … + 97 + 99
= (1 + 99) + (3 + 97) + (5 + 95) + … + (49 + 51)
= 100 × 25
= 2500

Method 3: Solve by Formula

(first term + last term) × number of terms ÷ 2
= (1 + 99) × 50 ÷ 2
= 100 × 25
= 2500
The sum of all the odd numbers smaller than 100 is **2500**.

3 What is the digit in the ones place for 2007^{2007}?

Analysis: $7^1 = 7, 7^2 = 7 \times 7, 7^3 = 7 \times 7 \times 7, \ldots$

Power	7^1	7^2	7^3	7^4	7^5	7^6	7^7	7^8	...
Digit in ones place	7	9	3	1	7	9	3	1	...
					↑ R_1	↑ R_2	↑ R_3	↑ R_4	

Solution:

The digits in the ones place recurs at 7, 9, 3, 1, ...

$2007 \div 4 = 501$ R 3

The digit in the ones place for 2007^{2007} is **3**.

4 Evaluate $1 + 3 + 3^2 + 3^3 + \ldots + 3^8$.

Analysis: We do not want to end up with laborious computation for questions of this nature. We must find a method to get to the answers efficiently.

Solution:

If $A = 1 + 3 + 3^2 + 3^3 + \ldots + 3^8$ —— (1)

We can also write $3A = 3 + 3^2 + 3^3 + 3^4 \ldots + 3^9$ —— (2)

(2) – (1)

$3A - A = 3^9 - 1$

$2A = 19\,683 - 1$

$2A = 19\,682$

$A = 19\,682 \div 2 = 9841$

$1 + 3 + 3^2 + 3^3 + \ldots + 3^8 = \mathbf{9841}$

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PRACTICE

1 How many small triangles has Figure 8?

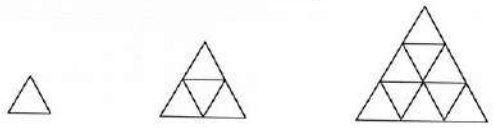

Figure 1 Figure 2 Figure 3 ... Figure 8

2 How many odd digits are there in the product of $\underbrace{999\ldots999}_{2008\ 9s}$ and $\underbrace{999\ldots999}_{2008\ 9s}$?

3 Evaluate $A = 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \ldots + 12^3$.

4 Study the number pattern below.

$\frac{1}{1}$,

$\frac{2}{1}, \frac{1}{2}$,

$\frac{3}{1}, \frac{2}{2}, \frac{1}{3}$,

$\frac{4}{1}, \frac{3}{2}, \frac{2}{3}, \frac{1}{4}$,

$\frac{5}{1}, \frac{4}{2}, \frac{3}{3}, \frac{2}{4}, \frac{1}{5}$,

$\vdots$

In which row and what term number from the left is $\frac{2008}{1998}$?

5 20 lines are drawn on a piece of paper.
How many intersections at the most can these lines form? Assume no line will lie directly over another.

6 A rectangle divides a plane into 2 regions at the most.
How many regions at the most can 5 such identical rectangles divide the plane into?

7 What is the sum of the first 1953 terms of the series,
$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}, \ldots$?

8 In a sequence, 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, 16, 21, …, state the law that governs this series. Write down the 16^{th} and the 20^{th} terms of the series.

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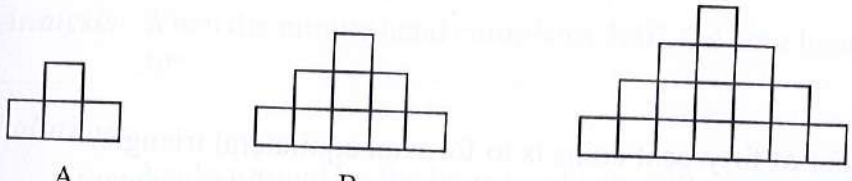
9 Find the 1008th term in the series, $\frac{1}{2}, \frac{1}{5}, \frac{1}{10}, \frac{1}{17}, \frac{1}{26}, \cdots$

10



A B C

Figure A, formed by 4 squares, has a perimeter of 30 cm.
What is the perimeter of the figure formed by 64 squares?

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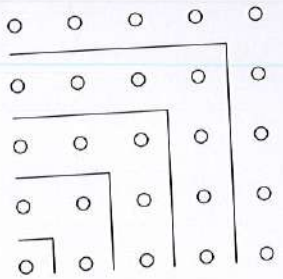
11 There is a formula that governs the dot-pattern shown below.



(a) Write down the pattern, in terms of n, which represents the n^{th} pattern.

(b) Evaluate $2008^2 - 2007^2 + 2006^2 - 2005^2 + \ldots + 2^2 - 1^2$ based on your findings in part (a).

12 A certain number of fifty-cent coins is to form an equilateral triangle. The same number of fifty-cent coins can also be used to form a square. The number of fifty-cent coins on each side of the square is 6 fewer than the number of fifty-cent coins on each side of the equilateral triangle. How many fifty-cent coins are there altogether?

12 Mathematics of Time

How many times does the hour hand of a clock lie directly over the minute hand in 24 hours?

What action must you take when your watch is slower or faster than the actual time?

Have you ever wondered how many degrees about the centre of the clock will the hour hand have turned when the minute hand turns one complete round?

EXAMPLES

1 Find the angle formed by the minute hand and hour hand of a clock at 7.10 pm.

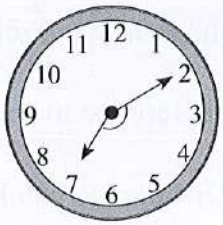
***Analysis*:** When the minute hand completes 360°, the hour hand will have travelled 30°.

***Solution*:**

Angle formed by the hour hand using 12 as the beginning point

$= 210° + 30° \times \frac{10}{60}$

$= 215°$



Angle formed by the minute hand

$= 360° \times \frac{10}{60}$

$= 60°$

$215° - 60° = 155°$

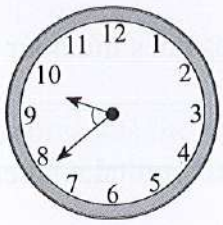
The angle formed by the minute hand and hour hand of the clock at 7.10 pm is **155°**.

2 Find the angle formed by the hour hand and minute hand of a clock at 9.38 pm.

***Analysis*:** When the minute hand completes 360°, the hour hand will have travelled 30°.

***Solution*:**

$$270° + 30° \times \frac{38}{60} = 289°$$



The hour hand travelled 289°.

$$360° \times \frac{38}{60} = 228°$$

The minute hand travelled 228°.

$$289° - 228° = 61°$$

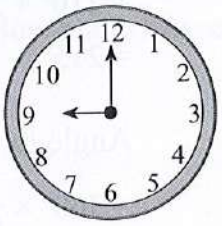
The angle formed by the hour hand and minute hand of the clock at 9.38 pm is **61°**.

3 If it is 9 o'clock now, how long will it take for the minute hand of a clock to lie directly over the hour hand?

***Analysis*:** Very much like catching up in speed, we ask when the minute hand catches up with the hour hand.

***Solution*:**

When the minute hand completes 1 cycle,



the hour hand will complete $\frac{1}{12}$ of a cycle.

Difference in speed $= 1 - \frac{1}{12} = \frac{11}{12}$

Distance to catch up = 45 scales out of 60

Time taken $= 45 \div \frac{11}{12} = 45 \times \frac{12}{11} = 49\frac{1}{11}$ min

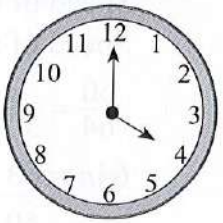
It will take $\mathbf{49\frac{1}{11}}$ **min** for the minute hand of the clock to lie directly over the hour hand.

4 At what time do the minute hand and hour hand of a clock form a right angle between 4 and 5 o'clock?

Analysis: There are two scenarios to this problem. One is when the hour hand is ahead of the minute hand. Another is when the minute hand is ahead of the hour hand.

Solution:

Let us start the problem from 4 o'clock as shown in the figure.
In order to form a right angle, there must be 15 scales between the two hands.



Difference in speed $= 1 - \frac{1}{12} = \frac{11}{12}$

1st scenario: Distance to catch up $= 20 - 15 = 5$ scales

$$\text{Time taken} = 5 \div \frac{11}{12} = 5 \times \frac{12}{11} = 5\frac{5}{11} \text{ min}$$

2nd scenario: Distance to catch up $= 20 + 15 = 35$ scales

$$\text{Time taken} = 35 \div \frac{11}{12} = 35 \times \frac{12}{11} = 38\frac{2}{11} \text{ min}$$

The 1st right angle occurs at **4 o'clock and $5\frac{5}{11}$ min**, the 2nd one at **4 o'clock and $38\frac{2}{11}$ min**.

5 Jonathan has a daily frustration. His watch is 4 minutes faster than the actual time in every hour. He sets his watch right at 7 every morning.
What is the actual time when his watch shows 12 noon?

***Analysis*:** The problem is best approached using the concept of speed.

***Solution*:**

Method 1: Think Algebra

Suppose the watch travels $5 \times 60 = 300$ min
Actual time travels n min.

Speed of the watch = 64 min in an hour
Speed of actual time: 60 min in an hour

$$\frac{60}{64} = \frac{n}{300}$$

$$64n = 60 \times 300$$

$$n = \frac{60 \times 300}{64} = \frac{1125}{4}$$

$$= 281\frac{1}{4} \text{ min or } 4 \text{ h } 41\frac{1}{4} \text{ min}$$

Method 2: Solve by Reasoning

The ratio of the two speeds is 64 : 60 = 16 : 15.

$$4 \text{ min} \div \frac{16}{15} = 4 \times \frac{15}{16} = 3\frac{3}{4}$$

$3\frac{3}{4}$ is the amount of time the actual time falls behind his watch in an hour.

$$5 \times 3\frac{3}{4} = 5 \times \frac{15}{4}$$

$$= \frac{75}{4} = 18\frac{3}{4} \text{ min slower in 5 h}$$

$$12 \text{ o'clock} - 18\frac{3}{4} \text{ min} = 11 \text{ o'clock } 41\frac{1}{4} \text{ min}$$

The actual time is **11 o'clock $41\frac{1}{4}$ minutes**.

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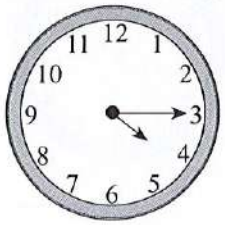
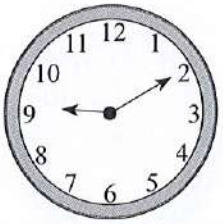
PRACTICE

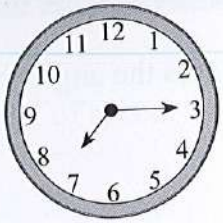
1 What is the angle formed by the minute hand and the hour hand at 4.15 pm?



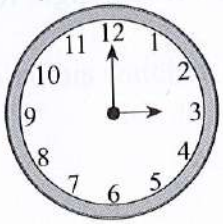
2 What is the angle formed by the minute hand and the hour hand at 9.10 am?



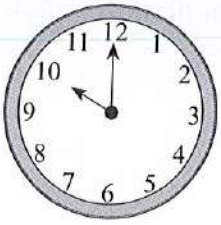
3 Find the angle formed by the minute hand and the hour hand at 7.15 am.



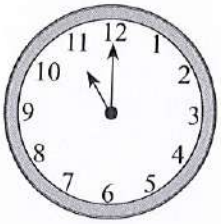
4 At what time, after 3 pm, will the hour hand lie directly over the minute hand?



5 What time will it be after the minute hand completes 2008 cycles, starting from 10 o'clock in the morning?

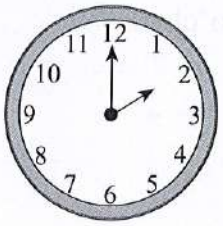


6 At what time do the minute hand and hour hand form a right angle between 11 and 12 o'clock?



7 What is the angle formed by the minute hand and the hour hand at 5.10 pm?

8 At what time will the hour hand and the minute hand make an angle of 150° between 2 o'clock and 3 o'clock?



9 Mr Smith adjusted his watch to the actual time, which was 9 at night. He arrived at his office at 9 the next morning and realised that he was late by 10 minutes. By how much was his watch slower than the actual time in an hour?

10 A clock is slower than the actual time by 1 minute every hour. What is the actual time if the clock indicates 6 am in the morning given that it has been adjusted to the actual time at 12 midnight?

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11 Samuel has a watch that is faster than his alarm clock at home by 20 seconds in an hour. The alarm clock is, in turn, slower than the actual time by 30 seconds every hour. By how many seconds does his watch deviate from the actual time in a day?

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Write Equations

In the Beginner book, we deal with two types of problems:

Chicken and Rabbit, and Solve by Replacement. In both cases, we can write 2 equations with 2 unknowns. Then, we can proceed to bring the 2 equations to a comparable state, in order to eliminate one unknown. For example, if we say "There are 100 chickens and rabbits, and a total of 320 legs. How many chickens are there? How many rabbits are there?", we can write, almost by mathematical intuition or more common sense:

$$c + r = 100 \quad \text{—— (1)}$$
$$2c + 4r = 320 \quad \text{—— (2)}$$

where c and r denote the number of chickens and rabbits respectively. When we multiply equation (1) by 2, we can see $2r = 120$. Therefore, there are 60 rabbits. Another example would be, "8 basketballs and 5 volleyballs cost \$635, while 6 basketballs and 9 volleyballs cost \$765. What is the cost of a basketball? What is the cost of a volleyball?" Again,

$$8b + 5v = 635 \quad \text{—— (1)}$$
$$6b + 9v = 765 \quad \text{—— (2)}$$

Adding (1) and (2), we have $14b + 14v = 1400$, $1b + 1v = 100$, and so on. In the Intermediate book, this skill of writing equations has been extended to writing 1 equation with 2 unknowns. As you might have experienced, there is quite a bit of guessing work in the second part of the solution. We learn then to write 2 equations with 3 unknowns, too. The trick is to eliminate one unknown. Later, by comparing via multiplication and subtraction, we end up with one equation.

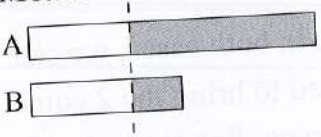
This chapter revises all these and enriches students with more.

EXAMPLES

1 Shop A and Shop B had 250 muffins altogether. Shop A sold 97 muffins and Shop B sold 25 muffins. The number of muffins the two shops had left became the same. How many muffins did each shop have at first?

Solution:

Method 1: Use Model



$2\text{ u} \rightarrow 250 - 97 - 25 = 128$

$\text{u} \rightarrow 64$

$\text{A} \rightarrow 64 + 97 = 161$

$\text{B} \rightarrow 64 + 25 = 89$

Method 2: Write Equation

Suppose Shop A has x muffins at first and Shop B has $(250 - x)$.

A was left with $(x - 97)$ and B was left with $(250 - x) - 25$.

The amount left was the same.

$$x - 97 = (250 - x) - 25$$
$$2x = 250 - 25 + 97$$
$$= 322$$
$$x = 161$$
$$250 - 161 = 89$$

Shop A had **89 muffins** and Shop B had **161 muffins** at first.

2 Bottle A contains 10% syrup. Bottle B contains 30% syrup. When the 2 bottles of syrup are mixed together, the syrup content is 25% and the total volume is 500 mℓ. How much syrup is in each bottle at first?

Solution:

Let m be the volume of A,

n be the volume of B,

$$m + n = 500 \quad \text{—— (1)}$$
$$10\% \times m + 30\% \times n = 500 \times 25\%$$
$$0.1m + 0.3n = 125 \quad \text{—— (2)}$$
$$(2) \times 10, \quad m + 3n = 1250 \quad \text{—— (3)}$$
$$(3) - (1), \quad 2n = 750$$
$$n = 375$$
$$m = 500 - 375$$
$$= 125$$

Bottle A has **125 mℓ** of syrup and Bottle B has **375 mℓ** of syrup at first.

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3 Sam walks to school at a speed of 80 m/min. He returns at a speed of 60 m/min. What is his average speed for the whole trip?

Solution:

Method 1

The LCM of 80 and 60 is 240 m. Let the school be 240 m away.

Time taken to go $= \frac{240}{80} = 3$ min

Time taken to return $= \frac{240}{60} = 4$ min

Average speed $= \frac{2 \times 240}{3 + 4} = \frac{480}{7}$ m/min

Method 2

Let the distance be d.

Time taken to go $= \frac{d}{80}$

Time taken to return $= \frac{d}{60}$

Total time taken $= \frac{d}{80} + \frac{d}{60} = \frac{7d}{240}$

Total distance $= 2d$

Average speed $= 2d \div \frac{7d}{240} = 2d \times \frac{240}{7d} = \frac{480}{7}$ m/min

His average speed for the whole trip is $\mathbf{\frac{480}{7}}$ **m/min.**

4 There is a 3-digit number, abc. The tens digit is larger than the hundreds digit by 5. The ones digit is smaller than the tens digit by 2. Half the sum of the 3 digits is the same as the tens digit. What is abc?

Solution:

We can write 3 simple equations.

$b - 5 = a$ —— (1)

$c + 2 = b$ —— (2)

$\frac{a + b + c}{2} = b$

$a + b + c = 2b$

$a + c = b$ —— (3)

Substitute b in (1) with $(a + c)$

$a + c - 5 = a$

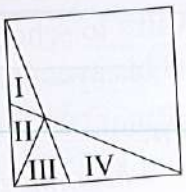
$c = 5$

From (2), $b = 7$

From (1), $a = 2$

The number is **275**.

5 In the square, area of I equals area of II.
Area of IV is twice area of III.
The total area of I, II and III is 8 cm^2.
The total area of I, III and IV is 12 cm^2.
Find the total area of I, II, III and IV.



Solution:

Let the area of regions I, II, III and IV be a, b, c and d respectively.

We know $a = b$.

$a + b + c = 8$

$2a + c = 8$ — (1)

$d = 2c$ — (2)

$a + c + d = 12$ — (3)

From (1), $a = \frac{8 - c}{2}$ — (4)

Substitute (2) and (4) into (3).

$\frac{8 - c}{2} + c + 2c = 12$

$8 - c + 2c + 4c = 24$

$5c = 16$

$c = \frac{16}{5}$

$d = \frac{32}{5}$

$a = 12 - \frac{16}{5} - \frac{32}{5} = \frac{60 - 16 - 32}{5} = \frac{12}{5}$

$2 \times \frac{12}{5} + \frac{16}{5} + \frac{32}{5} = \frac{72}{5}$ cm^2

The total area is $\mathbf{\frac{72}{5}\ cm^2}$.

spectively.

PRACTICE

1 Find the values of x, y and z in

$$2x + 6y + 2z = 60 \quad \text{—— (1)}$$

$$x : y : z = 1 : 2 : 3 \quad \text{—— (2)}$$

2 It is given:

$$\frac{A + C + E}{3} = 40 \quad \text{—— (1)}$$

$$\frac{A + B + D}{3} = 28 \quad \text{—— (2)}$$

$$\frac{B + C + D + E}{4} = 33 \quad \text{—— (3)}$$

$$A = ?$$

3 The sum of 11 consecutive numbers is 187.
Find the sum of the next 15 consecutive numbers.

4 A tub of saline water weighing 50 kg contains 30% salt. The content of salt is 40% when the saline water is left to evaporate. What is the weight of the saline water now?

5 A mathematical competition has 30 questions. 8 marks are awarded for a correctly worked out answer. 5 marks are deducted for a wrong answer. An unattempted question gets 0 marks. A participant scores 49 marks.
Find his number of unattempted questions.

6 There are 120 guests in a party. Every 3 ladies consume a bottle of wine. Every 2 gentlemen consume a bottle of wine. 48 bottles of wine are consumed in all. How many female guests are there?

7 A civil engineer figures out that if the efficiency of a road work can be increased by 40%, then the number of working days can be reduced by 20 days. How many days was the road work scheduled to take in the original plan?

8 A cricket has 6 legs. A spider has 8 legs. A box contains a certain number of spiders and crickets. There is a total of 52 legs. How many crickets are there?

9 Peter, Mary and Nick are playing a card game. The game has 3 cards, each showing a whole number. In each round, each player will get a card, then record down the number. After 3 rounds, the points accumulated are 18, 25 and 23.
What is the number on each of the 3 cards?

10 A paid $700 for two identical fans, one printer and one bookshelf. B paid $800 for one fan, two identical printers and one bookshelf. C paid $900 for one fan, one printer and two identical bookshelves. Find the cost of each item.

11 A bag contains some red balls and black balls. If a red ball is removed, the number of red balls in the bag is $\frac{1}{3}$ of the remaining number of balls. If two black balls are removed instead, the number of red balls is $\frac{2}{5}$ of the remaining number of balls. How many red balls are there?

12 An escalator is moving up at a constant speed. Alan walks up the escalator at a certain speed and reaches the 2nd floor in 55 steps. Bryan also walks up the escalator at twice Alan's speed and reaches the 2nd floor in 60 steps. How many steps has the escalator?

Geometric Problem

Geometry, which means "Each Measuring" in Latin, is a branch of Mathematics that deals with problems of shapes, angle, area, volume and so on. In ancient Greece, the study of geometry originated from its practical use.

In 300 BC, Euclid, the great Greek mathematician, formalized the study of Geometry in his book *Elements*. Using logical reasoning, the book contained proofs of over a few hundred theorems related to the domain.

The study of Angles and Triangles has been dealt with extensively in Chapter 24 of the Intermediate book. This chapter continues to expose some questions in Angles. Further, the chapter gives students insight into:

(a) Problem of Equal Area
(b) Problem of Folded Paper

Example 4 and Question 7 are representations of type (a) problems. They require students to think of the figure as a whole, then divide it into a number of parts as prompted by the question. Question 1, on the other hand, requires students to visualize in terms of resulting shape and solve almost by mathematical intuition. Questions 9 and 12 provide opportunities for students to apply the same technique.

Questions 6 and 8 are exemplifications of type (b) problems. Again, the technique is quite dynamic, and students really have to use their creative imagination to formulate plans in order to solve them.

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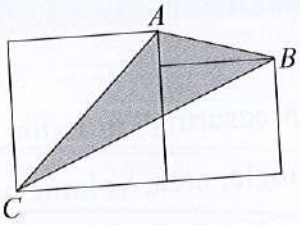
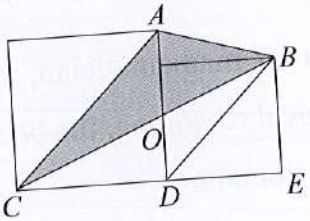
EXAMPLES

1 A square with an area of 60 cm² is placed next to another smaller square. Find the area of the shaded region.



***Solution*:** Draw a line to connect BD as shown.



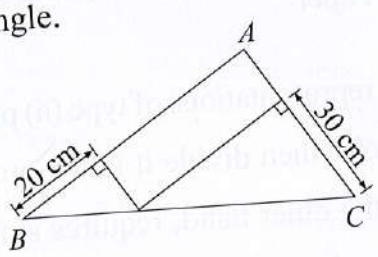
$\angle ACD = \angle BDE = 45°$

Area of ΔADB = Area of ΔBCD as both Δs have same base BD and same height.

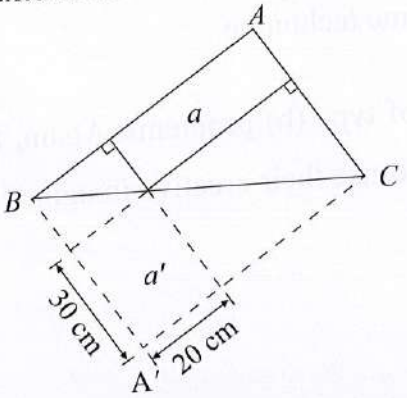
Area of ΔAOB = Area of ΔCOD as ΔBOD is the common area.

Area of shaded region $= \frac{1}{2} \times 60 =$ **30 cm²**

2 Two right-angled triangles and a rectangle form another right-angled triangle, ABC. Find the area of the rectangle.



***Solution*:** Extend the 2 sides of the rectangle.



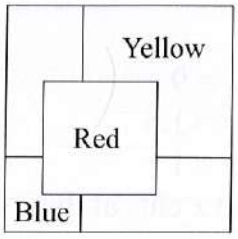
Area of ΔABC = Area of $\Delta A'BC$

Area a = Area a'

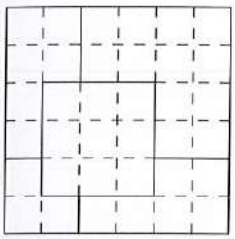
Area of rectangle $= 20 \times 30$

$=$ **600 cm²**

3 Three square pieces of coloured paper are placed at the bottom of a square box. The areas of the yellow paper and the blue paper are 68 cm^2 and 17 cm^2 respectively. One vertex of the red paper rests at the centre of the yellow paper. The other vertex of the red paper rests at the centre of the blue paper.
Find the area of the red paper.

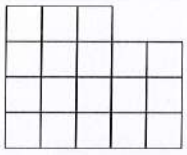


Solution:



$68 = 4 \times 17$
The area of the yellow paper is 4 times the area of the blue paper.
We can sub-divide the area of the box into 24 units.
The area of the yellow paper is 16 units.
16 units $\rightarrow$ 68 cm^2
4 units $\rightarrow$ 17 cm^2
8 units $\rightarrow$ 34 cm^2
The area of the red paper is **34 cm^2**.

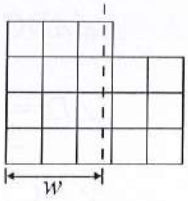
4 The figure is made up of 18 identical squares. Each square has an area of 1 cm^2. Use a straight line to divide the figure into 2 parts of equal area.



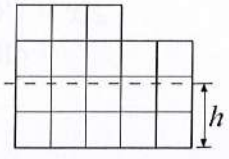
Solution:

Total area = $18 \times 1 = 18$ cm^2
$18 \div 2 = 9$ cm^2 for each part

Method 1
Use a vertical line to cut somewhere.
Height of left-hand-side figure = 4 cm
Its width is w, so $4 \times w = 9$
$w = 2.25$ cm



Method 2
Use a horizontal line to cut somewhere.
Length of the rectangle = 5 cm
Its height is h, so $5 \times h = 9$
$h = 1.8$ cm



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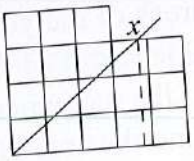
Method 3



Suppose a line cuts the figure as shown.

The area of the upper region

$= 1 \times 3 + \frac{1}{2} \times 3 \times (3 + x)$

$= 3 + 4.5 + 1.5x$

$= 7.5 + 1.5x$

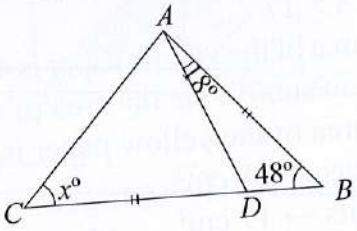
Equating, $7.5 + 1.5x = 9$

$1.5x = 1.5$

$x = 1$

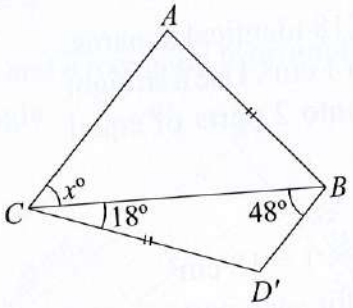
This means the point x cuts at the corner of the square.

5 In ΔABC, $AB = CD$, $\angle ABD = 48°$ and $\angle BAD = 18°$. Find the value represented by x.



Solution:

Since $AB = CD$, we cut along AD and reform the figure.



From the original figure,

$\angle ADC = 18° + 48°$

$= 66°$

$\angle D' = 180° - 48° - 18°$

$= 114°$

$\angle AC \,//\, BD'$

$\angle D' + \angle x + 18° = 180°$

$\angle x = 180° - 18° - 114°$

$= \mathbf{48°}$

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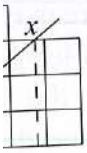
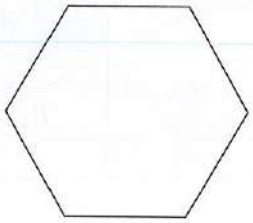


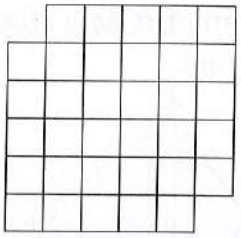
esented by x.

PRACTICE

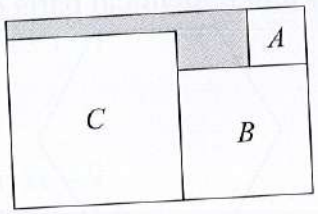
1 Divide the regular hexagon into 8 identical parts of equal area.



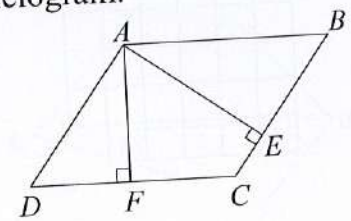
2 Is it possible to cover the following floor area with a certain number of tiles?



3 The area of a rectangle is less than 160 cm^2. A, B and C are 3 squares embedded in it. The ratio of the area of A to the area of B is 4 : 9. Find the area of the shaded region.



4 The perimeter of a parallelogram $ABCD$ is 80 cm. $AE = 9$ cm and $AF = 7$ cm. Find the area of the parallelogram.



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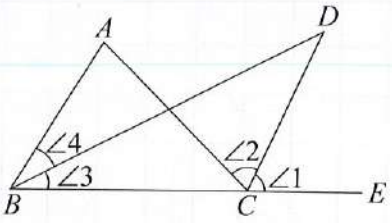
5 Line CD bisects $\angle ACE$ into $\angle 1$ and $\angle 2$. Line BD bisects $\angle ABE$ into $\angle 3$ and $\angle 4$. If $\angle A$ is given as 76°, find $\angle D$.



6 A rectangular piece of paper is folded along a line OP. It is then folded along a symmetrical line. The result is shown in Figure 3. Find the area of Figure 3 if the total area of the shaded regions is given as 10 cm^2 and the area of Figure 3 is $\frac{4}{11}$ of the area of the rectangle.

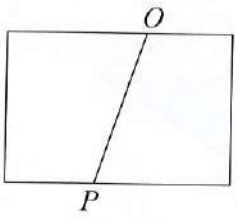


Figure 1

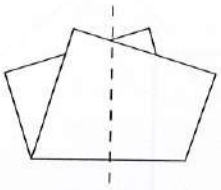


Figure 2

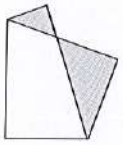
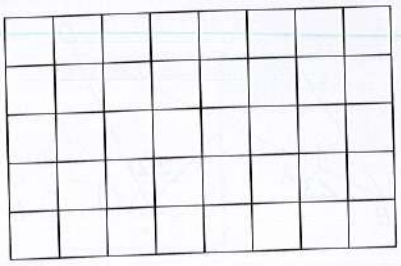
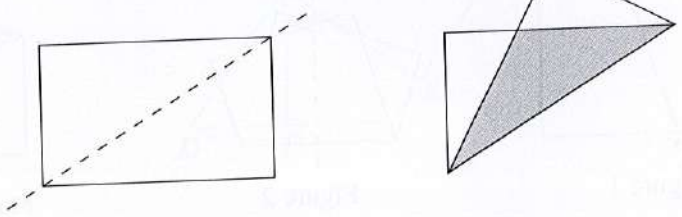


Figure 3

7 Use 2 straight lines to divide the figure into 4 parts, so that the area of the 4 parts has a ratio of 1 : 2 : 3: 4.



8 A rectangular piece of paper is folded along its diagonal line. The length and width of the rectangle are 10 cm and 5 cm respectively. Find the area of the overlapped region.



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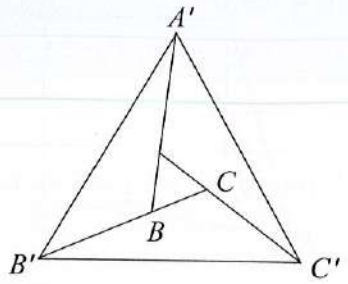
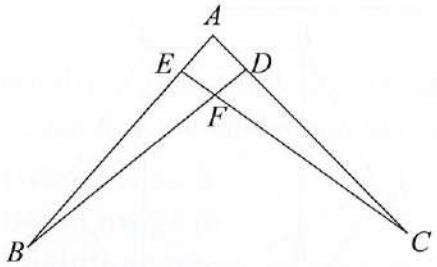
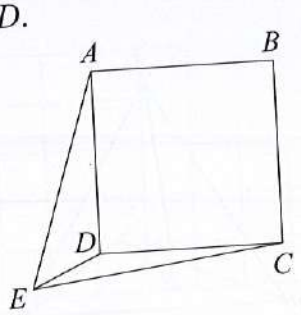
9 $AB = \frac{1}{3}A'B$, $AC = \frac{1}{3}A'C$ and $BC = \frac{1}{3}B'C$. Find the ratio of the area of ΔABC to the area of $\Delta A'B'C'$.



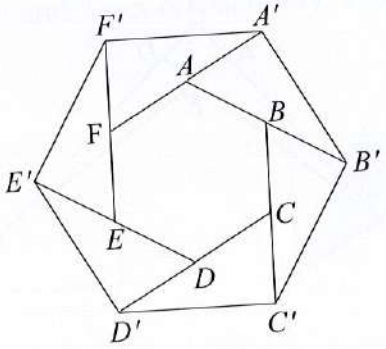
10 In the figure below, $AB = BD$ and $AC = CE$. Find $\angle A$ if $\angle BFC = 105°$.



11 In the figure below, $DE = 4$ cm, area of $\Delta ADE = 18$ cm^2 and area of $\Delta CDE = 10$ cm^2. Find the area of square $ABCD$.



12 In the figure below, $ABCDEF$ is a regular hexagon.
$AA' = AF$, $BB' = AB$, $CC' = BC$, ...
Find the area of hexagon $ABCDE$ if the area of hexagon $A'B'C'D'E'F'$ is given as 9 cm^2.



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15 Logic

Three most commonly deployed techniques in solving logic problems are:

(1) make an assumption
(2) make a list / elimination
(3) draw a table

In technique (1), we assume something to be true at first until it is later proven otherwise by all the given clues.

We make an unrepeated list of all the possibilities in technique (2) and eliminate those that cannot be true.

In technique (3), we create a table or drawing to sort out all the given clues, thus making the reasoning process more systematic. This is especially true in Questions 9 and 11.

EXAMPLES

1 A football flew right through the glass window of the staff room, shattering it. Four probable culprits were called into the Discipline Master's office for investigation.

Alan:	**Dylan broke it.**
Bob:	**Dylan broke it.**
Charles:	**I didn't do it.**
Dylan:	**Bob is lying.**

Only one suspect told the truth.
Who was the culprit?

Solution:

Reasoning:

If Alan was the culprit, then both Alan and Bob lied.

Charles and Dylan told the truth.

If Dylan was the culprit, then Alan, Bob and Charles told the truth.

If Charles was the culprit, then Dylan told the truth and the rest lied.

Charles was the culprit.

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2 The marks scored by students A, B, C, D and E in a mathematics test are above 91.

The average mark of A, B and C is 95.
The average mark of B, C and D is 94.
E's mark of 96 ranks as the second highest among all.
A has the highest mark.
B and D have the same score.

How many marks does D score in the mathematics test?

Solution:

Reasoning: $\frac{A+B+C}{3} = 95$

$A + B + C = 285$ — (1)

$B + C + D = 94 \times 3$

$= 282$ — (2)

(1) – (2)

$A - D = 3$

If $A = 97$, then $D = 94$, $B = 94$ and $C = 94$.

This does not fit the clue that B and D have the same score when C has the same score as both B and D too.

If $A = 98$, then $D = 95$ and $B = 95$ and $C = 92$.

This fits the clue that both B and D have the same score.

D scores **95** marks in the mathematics test.

e above 91.

when C has

3 A science quiz consisted of 6 true-or-false questions. 20 marks were awarded for questions answered correctly. Questions left unanswered would be given 10 marks. No mark was given for a wrong answer.
The answers given by four students are shown below.

Names \ Qns	1	2	3	4	5	6	Score
Melissa	T	F	F	T	T	F	?
Benson	F	T	F	F		F	90
Zachery		T	T	T	T	T	70
Randy	F	F	F	T	F		70

How did Melissa fare?

Solution:

Reasoning:

(1) A full score is 120 marks.
It is clear that Zachery and Randy answered 2 questions wrongly.
Benson has one wrong answer.

(2) Together, Benson and Zachery got questions 3, 4, and 6 wrong.

(3) If Benson had answered question 3 wrongly, then Randy would have answered questions 2, 3 and 4 wrongly, and his marks would then be 50.

(4) If Benson got question 6 wrong, Zachery got questions 3 and 4 wrong and Randy got questions 2 and 4 wrong, then their scores for question 5 show inconsistency.

(5) If Benson got question 4 wrong and Zachery got questions 3 and 6 wrong, then Randy would get questions 2 and 5 wrong.

The right answers: F T F T T F
Melissa's answers: T F F T T F

$4 \times 20 = 80$.

Melissa got **80** marks and she got **questions 1** and **2** wrong.

PRACTICE

1 A diamond is put into one of six identical musical boxes.

Box A is on the left of Box B.
Box B is second on the right of Box C.
Box C is on the right of Box D.
Box D is just next to Box E.
There is a box between Boxes A and E.

The diamond is not in any of the above mentioned boxes. Rank the position of all the musical boxes.

2 Aloysius, Berry, Carl, Dylan and Edward took part in an International Chess Competition. Each contestant must play exactly one game.

Aloysius played 4 games.
Berry played 3 games.
Carl played 2 games.
Dylan played 1 game.

How many games did Edward play so far?

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3 2000 soldiers stand in a row.
Beginning from the first soldier on the left, each calls out a number, 1, 2, 3, 1, 2, 3, ...
Beginning from the first soldier on the right, each calls out a number, 1, 2, 3, 4, 1, 2, 3, 4, ...
How many of these soldiers call out "3" in both scenarios?

4 A jewel was stolen during an exhibition. The Interpol managed to arrest four suspects to assist in the investigation.

David: **Ellen stole it.**
Ellen: **Fred did.**
Fred: **I have an alibi.**
George: **Don't believe anything Ellen said.**

Only one suspect told the truth.
Who was the thief?

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5 Many years ago, there were 4 Thursdays and 5 Wednesdays in the month of May. On which day of the week was 15th of May in that year?

6 In an ancient grave, an archaeologist discovered a tomb that carried a pattern of numbers. Some of these numbers had been defaced.

4	9	9	11	□	12	15	8
□	5	□	3	4	□	9	□
1	4	□	8	2	□	6	□
3	20	14	□	8	35	□	15

Help the archaeologist to decode the number pattern.

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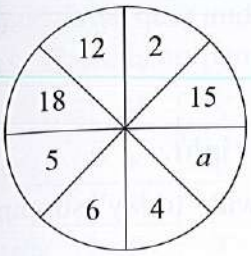
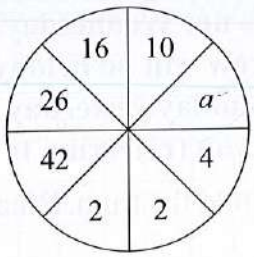
7 What does a represent in each number puzzle?

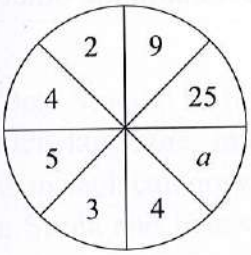
(a)



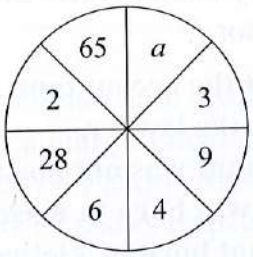
(b)



(c)



(d)



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8 Four students were arguing over what day of the week it should be.

A: Today is not Wednesday.
B: Tomorrow will be Friday.
C: It was Sunday yesterday.
D: Both of you (referring to B and C) are not right.

Only one student told the truth. What day of the week was "today" supposed to be?

9 The countries of birth of Alice, Bryan and Chloe are Korea, Vietnam and Indonesia, but not necessarily in that order. One of them is an accountant, the other two are a lawyer and a doctor.

Bryan is not the accountant.
Chloe is not the lawyer.
The accountant was not born in Indonesia.
The lawyer was born in Korea.
Chloe was not born in Vietnam.

What is Bryan's occupation? Which is his country of birth?

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10 The table below shows the answers provided by four students participating in a true-and-false quiz and their corresponding scores. Each correct answer is given 5 marks. There is no penalty for any wrong answer.

Name \ Qns	1	2	3	4	5	6	7	8	9	10	Score
Jolene	F	F	T	F	F	T	F	T	T	F	35
Leon	F	T	T	T	T	F	F	F	T	T	25
Natalie	F	T	T	T	F	F	F	T	T	T	15
Wilfred	F	F	T	T	F	T	F	T	T	T	?

What is Wilfred's score?

11 The Best Student of the Year award is given based on a student's performance in conduct, language, mathematics, aesthetics and physical education. The highest marks in each category is 5 marks, followed by 3 and 1. Annabelle scores 23 marks while Shola and Ernest score 9 marks and 13 marks respectively. If Shola scores the highest mark in conduct, who scores the least in this category?

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Of Interest, Profit and Loss

(A) Profit and Loss

$$\text{profit} = \text{selling price} - \text{cost price}$$

$$\text{loss} = \text{cost price} - \text{selling price}$$

$$\text{percentage profit} = \frac{\text{profit}}{\text{cost price}} \times 100\%$$

$$\text{percentage loss} = \frac{\text{loss}}{\text{cost price}} \times 100\%$$

(B) Interest is the fee earned when you put money in the bank or the "fee" charged when you borrow money from a bank.

The formula for calculation of interest is:

$$\text{simple interest} = \text{principal} \times \text{rate} \times \text{time}$$

where **principal** is the initial amount of money, **rate** is the percentage earned or charged by the bank, and **time** is the period in order to earn or be charged on the interest, usually on a yearly or quarterly basis.

In the computation of Compound Interest, the interest earned for the first period of the time forms part of the principal for the computation of interest for the second period of the time and so on.

EXAMPLES

1 The selling price of a figurine is \$8000.
If it is sold at \$4800, the profit is 20%.
How much should it be sold for if the profit has to be 75%?

***Solution*:**

$$\text{cost price} = \frac{\$4800}{120\%} \times 100\% = \$4000$$

To make a 75% profit,

$\$4000 \times 175\% = \$7000$

It should be sold for **\$7000**.

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2 A product is marked at 20% above the cost price. It is then sold at 90% of the selling price. The profit is \$120. What is the cost price of this product?

Solution:

If the cost price is \$$n$,

marked price = $n \times (1 + 20\%)$
= $1.2n$

selling price = $1.2n \times 90\%$
= $1.08n$

profit = selling price – cost price
= $1.08n - n$
= $0.08n$

$0.08n$ = \$120

n = \$1500

The cost price of this product is **\$1500**.

3 A shop purchased some shoes at \$65 per pair. It then sold them at \$74 per pair. The shopkeeper calculated that the profit earned was \$440 when there were 5 pairs of shoes left. How many pairs of shoes did the shop purchase initially?

Solution:

Total profit = \$440 + \$74 × 5
= \$810

Profit from each pair = \$74 – \$65 = \$9

Number of pairs of shoes = \$810 ÷ \$9 = 90

The shops purchased **90** pairs of shoes initially.

4 A bank charged 8% annual interest on the money lent to its customers. John borrowed \$20 000 from the bank that was repayable within 6 years.

(a) What was the payable interest?

(b) What was the total amount payable?

(c) What was the monthly repayment?

Solution:

(a) Interest = \$20 000 × 0.08 × 6 = \$9600

The payable interest was **\$9600**.

(b) \$20 000 + \$9600 = \$29 600

The total amount payable was **\$29 600**.

(c) \$29 600 ÷ (6 × 12) = $\$411\frac{1}{9}$ or \$411.11

The monthly repayment was **\$411.11**.

5 Mr Young deposits \$30 000 with a bank that pays an interest of 5%. How much interest will be earned at the end of 3 years if the rate is compounded every year?

***Solution*:**

The interest earned for each year:

End of 1st year = \$30 000 × 0.05 × 1
= \$1500

End of 2nd year = (\$30 000 + \$1500) × 0.05 × 1
= \$1575

End of 3rd year = (\$31 500 + \$1575) × 0.05 × 1
= \$1653.75

\$1500 + \$1575 + \$1653.75 = \$4728.75

\$4728.75 will be earned at the end of 3 years if the rate is compounded every year.

PRACTICE

1 The selling price of a sofa set is $12 000.
It is later sold at $6500 at an exhibition. The profit made is 30%.
What is the percentage of profit if the profit is $5000?

2 The Megamart sold 60% fewer cans of soft drinks than bottles of mineral water on Sunday. The profit of each bottle of mineral water was 70% less than that of each can of soft drink. The profit gained from selling the soft drinks was $1200. What was the profit gained from selling the mineral water?

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3 A shop bought 80 laptops at $4000 each. It then sold 50 of them at a 20% profit in the first month. In the second month, the remaining laptops were sold at 75% of the first month's selling price. How much profit was made altogether?

4 A product was marked at 20% above the cost price. It was eventually sold at 80% of the selling price. The loss was $40. What was the cost price of the product?

5 A shop purchased some tennis rackets at \$150 each. It then sold them at \$175 each. When the shop was left with 8 rackets, the proceeds had covered the initial total cost price plus a profit of \$525. How many tennis rackets did the shop purchase at first?

6 The cost of publishing a children's book for a publishing house has gone up by 10% compared to last year, but the publisher wishes to maintain the selling price of the book. The profit for each book drops by 40% as a result. The number of books purchased, however, increases by 80%. What is the percentage increment in the profit?

7 Two different brands of television sets were sold at the same price of $2040 each. The sale of the first set made a profit of 20% while that of the second set made a loss of 20%. Did the shopowner profit from the sales of the two television sets?

8 Stephen puts $800 in a bank that earns him a simple interest of 8% per annum. What is the amount of simple interest earned in 3 years? When will his interest accumulate to $512?

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9 Maurice deposits $500 with the bank. The interest rate is calculated at 3% on a yearly basis. How much will she get at the end of 2 years?

10 A cricket promised to return $2n$ kg of barley for every n kg of barley borrowed at the end of the first year. This meant that the interest rate was 100%. The greedy ant changed its mind and requested for the repayment with interest compounded every 4 months. Give an explanation for the change of mind.

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17 Ratio

A ratio can be denoted as $a : b$ or $\frac{a}{b}$, where $b \neq 0$.

If $a : b = c : d$, we can express the ratio in fraction:

$$\frac{a}{b} = \frac{c}{d}$$

Cross multiply, we get $a \times d = b \times c$.

At competition level, especially, the presence of ratio is wide and far-reaching. It is, first of all, often associated with fraction. The most sophisticated speed problem is related to ratio, too.

If we say car A is travelling at 90 km/h and car B at 75 km/h, then the ratio of their speeds is 90 : 75 = 6 : 5. The ratio of distance, likewise, is also 6 : 5. The ratio of time will be 5 : 6. Ratio is found to be associated with problems involving area, content of liquid, etc.

EXAMPLES

1 The ratio of the number of yellow balls to the total number of balls in a bag was 5 : 7 at first. 18 more yellow balls were put into the bag. The ratio of the number of yellow balls to the total number of balls became 2 : 1. How many balls were there altogether in the end?

Solution:

Method 1: Solve Using Ratio

At first, yellow : total = 5 : 7

After 18 more yellow balls were put into the bag,

yellow : total

2 : 1

14 : 7

14 – 5 = 9 units

9 units = 18 yellow balls

1 unit = 2 yellow balls

(14 + 7) × 2 = 42 balls

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Method 2: Think Algebra

Let each unit be n, the total number of balls was $(5 + 7)n$ at first.

$$\frac{5n + 18}{12n + 18} = \frac{2}{3}$$

Cross multiply, we have

$3(5n + 18) = 2(12n + 18)$

$15n + 54 = 24n + 36$

$24n - 15n = 54 - 36$

$9n = 18$

$n = 2$

$2 \times (5 + 7) + 18 = 42$

There were **42** balls altogether in the end.

2 A bookshelf had 3 levels and a total of 240 books. The same number of books as the 2nd level were shuffled from the 1st level to the 2nd level. The same number of books as the 3rd level were then shuffled from the 2nd to the 3rd level. Finally, the same number of books as the remaining books on the 1st level were shuffled from the 3rd to the 1st level.
The ratio of the number of books on the 3 levels became 1 : 2 : 3. How many books were on each level at first?

Solution:

$1 + 2 + 3 = 6$

$240 \div 6 = 40$

$2 \times 40 = 80$

$3 \times 40 = 120$

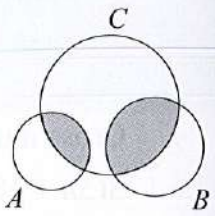
The number of books on each level were 40, 80 and 120 in the end.

Working backwards:

level 1	level 2	level 3
40	80	120
20	80	140
20	150	70
95	75	70

There were **95**, **75** and **70 books** on levels 1, 2 and 3 respectively at first.

3 In the diagram below, not drawn to scale, the sum of the areas of Circle A and Circle B is $\frac{3}{5}$ of the area of Circle C.



$\frac{1}{3}$ of Circle A is shaded.

$\frac{1}{2}$ of Circle B is shaded.

$\frac{1}{4}$ of Circle C is shaded.

Find the ratio of the area of Circle A to the area of Circle B.

Solution:

Method 1: Think Algebra

$$A + B = \frac{3}{5}C \quad \text{—— (1)}$$

$$\frac{1}{3}A + \frac{1}{2}B = \frac{1}{4}C \quad \text{—— (2)}$$

To get rid of the unknown C,

(1) $\times$ 10, $10A + 10B = 6C$ —— (3)

(2) $\times$ 24, $8A + 12B = 6C$ —— (4)

(3) – (4), $2A - 2B = 0$

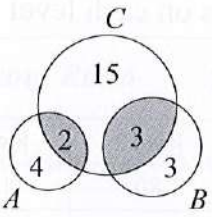
$$2A = 2B$$

$$A = B$$

$$A : B = 1 : 1$$

Method 2: Solve Using Ratio

We assign values to both shaded and unshaded regions of Circles A, B and C according to their ratio until all the conditions stated in the question are satisfied.



$$A : B = 6 : 6$$

$$A : B = 1 : 1$$

The ratio of the area of Circle A to the area of Circle B is **1 : 1**.

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4 A and B set off on a journey at the same time. A has walked $\frac{1}{4}$ more distance than B. B has used $\frac{1}{10}$ more time than A. Find the ratio of the speed of A to B.

Solution:

Ratio of distance,
A : B = (4 + 1) : 4
= 5 : 4

Ratio of time, A : B = 10 : (10 + 1)
= 10 : 11

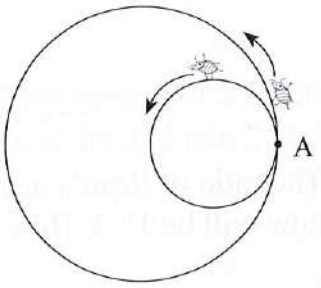
Ratio of speed, $A : B = \frac{5}{10} : \frac{4}{11}$

$= 110 \times \frac{5}{10} : 110 \times \frac{4}{11}$

= 55 : 40
= 11 : 8

The ratio of speed is **11 : 8**.

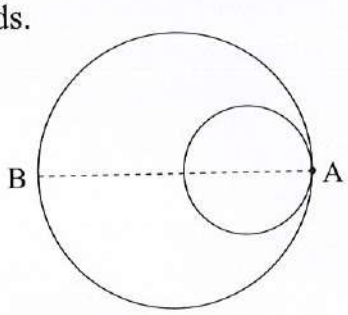
5 2 beetles begin their journey from A in the same direction and at the same speed. One travels along the big circle, the other along the small circle. Given that the radii of the circles are 72 cm and 42 cm, at what point are the 2 beetles furthest apart?



Solution:

72 : 42 = 36 : 21
= 12 : 7

The beetle on the big circle travels 7 rounds, the one on the small circle travels 12 rounds.



They will be furthest apart when one is at A, the other at B.

12 ÷ 2 = 6 rounds

7 ÷ 2 = 3.5 rounds

They are furthest apart when one has travelled 3.5 rounds, the other 6 rounds.

PRACTICE

1 Given $\frac{1}{A} - \frac{1}{B} = \frac{1}{286}$ and A : B = 11 : 13, find A + B.

2 The ratio of Sean's age to Ted's age is 1 : 4. The ratio of their ages 4 years from now will be 1 : 3. How many years ago was the ratio of their ages 1 : 9?

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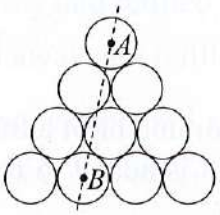
3 The sugar content of a 900 mℓ bottle of rose syrup is 6%. How much sugar must be added so that the sugar content becomes 10%?

4 10 identical circles are put together as shown. A and B are centres of 2 circles. A straight line is drawn through A and B, and the figure is divided into 2 regions. Find the ratio of the areas of the 2 regions.



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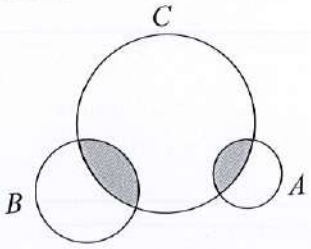
5 The diagram, not drawn to scale, shows circles A, B and C.

$\frac{1}{3}$ of circle A is shaded.

$\frac{2}{5}$ of circle B is shaded.

The total area of circles A and B is $\frac{4}{5}$ of the area of circle C.

Find the ratio of the areas of circles A to B to C.



6 The sum of numerator and denominator of a fraction is 100. The ratio of numerator to denominator is 2 : 3 after 23 is added to the numerator and 32 is added to the denominator. Find the fraction.

7 A conventional football has 32 faces made up of white hexagons and black pentagons.
How many of the faces are white?

8 There was a bottle of red wine.
$\frac{1}{2}$ of the content was poured away and refilled with white wine. $\frac{1}{3}$ of the red-white-wine mixture was then poured away and refilled with white wine.
Finally, $\frac{1}{4}$ of the red-white-wine was poured away and refilled with white wine.
What would be the ratio of the amount of red wine to white wine in the end?

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9 A candle took 3 hours to finish burning.
Another shorter candle, with a bigger diameter, took 5 hours to finish burning. The two candles were lit at the same time and had the same height after 2 hours.
What was the ratio of the heights of the two candles at first?

10 An escalator moves up at a constant rate. Carl takes 90 steps to walk down the escalator. Jane takes 45 steps to walk up the escalator. Given that Carl walks 3 times as fast as Jane, how many steps has the escalator?

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11 The ratio of male members to female members in a club is 3 : 2.
The ratio of members in the golf, tennis and swimming groups is 10 : 8 : 7.
The ratio of male golfers to female golfers is 3 : 1.
The ratio of male tennis players to female tennis players is 5 : 3.
Find the ratio of male swimmers to female swimmers.

12 There are 2 different flavours of sweets in a box – vanilla and orange.
First, Ray removes 36 vanilla and 4 orange sweets.
The remaining sweets are then divided into groups of 11. In each group, there are 8 vanilla and 3 orange sweets.
What can be the greatest number of sweets in the box if at least 80% of the sweets are vanilla flavoured?

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18 Rate

In this chapter, we will be learning how to work on three classic problems related to Rate.

The first type is the '**project**' problems. When referring to the 'whole' project, we often use the number '1'. The keys to problems of this type are to find the working rate of each individual and the combined working rate of two or more persons. From there, the amount of work done or the duration to complete the project will then be computed.

The second type is the '**candle in the wind**' problems. This type of problems touches on two topics, namely Ratio and Rate. We have to find the ratio of the heights of two candles, usually of different diameters, after they have been lit for a period of time.

The last type is the '**green pasture**' problems. It refers to a piece of pasture that is growing at a constant rate. This piece of pasture is consumed usually by a herd of cows or sheep at another constant rate. The key to solving problems of this nature lies in finding the original amount of pasture and the rate of growth of the pasture.

EXAMPLES

1 It takes Randy and Sam 12 hours to complete a science project.
If Randy works alone on the same project for 16 hours and let Sam take over, Sam needs another 7 hours to complete it.
How long does it take Sam to complete the science project alone?

Analysis: It can be taken that Randy and Sam work together for 7 hours and Randy finishes the remaining work in $(16 - 7) = 9$ hours.

Solution:

Combined working rate of the project in one day $= \frac{1}{12}$

Amount of work done in 7 hours $= \frac{1}{12} \times 7 = \frac{7}{12}$

Amount of work left $= 1 - \frac{7}{12} = \frac{5}{12}$

Randy's working rate on the project $= \frac{5}{12} \div 9 = \frac{5}{108}$

Sam's working rate on the project $= \frac{1}{12} - \frac{5}{108} = \frac{1}{27}$

$1 \div \frac{1}{27} = 27$ hours

It takes Sam **27 hours** to complete the science project alone.

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2 Team A takes 42 days to complete a road repairing work alone. If Team A works in collaboration with Team B, it takes them 14 days to finish $\frac{5}{7}$ of the work. Team A then has to move on to other projects. How many more days does Team B need to complete the road repairing work?

Solution:

Team A's working rate $= \frac{1}{42}$

Team A has done $\left(\frac{1}{42} \times 14\right) = \frac{14}{42}$ of $\frac{5}{7}$

Amount of work done by Team B $= \frac{5}{7} - \frac{14}{42} = \frac{16}{42} = \frac{8}{21}$

Team B's working rate $= \frac{8}{21} \div 14 = \frac{8}{21} \times \frac{1}{14} = \frac{4}{147}$

Amount of work left $= 1 - \frac{5}{7} = \frac{2}{7}$

$$\frac{2}{7} \div \frac{4}{147} = \frac{2}{7} \times \frac{147}{4}$$

$$= \frac{21}{2}$$

$$= 10\frac{1}{2}$$

Team B needs $\mathbf{10\frac{1}{2}}$ more days to complete the road repairing work.

3 A candle took 3 hours to finish burning. Another shorter candle, with a bigger diameter, took 5 hours to finish burning. The two candles were lit at the same time and had the same height two hours later.
What was the ratio of the heights of the two candles at first?

Solution:

Let the initial heights of the two candles be m and n respectively.

Amount of candle burnt $= m \times \frac{120}{180} = \frac{2m}{3}$

Amount of candle burnt $= n \times \frac{120}{300} = \frac{2n}{5}$

Remaining length of the two candles: $m - \frac{2m}{3} = \frac{m}{3}$

$$n - \frac{2n}{5} = \frac{3n}{5}$$

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Since the height is the same two hours later,

$$\frac{m}{3} : \frac{3n}{5} = 1 : 1$$

$$\frac{\frac{m}{3}}{\frac{3n}{5}} = \frac{1}{1}$$

Cross multiply,

$$\frac{m}{3} = \frac{3n}{5}$$

$$\frac{m}{n} = \frac{9}{5}$$

$$m : n = 9 : 5$$

The ratio of the heights of the two candles was **9 : 5** at first.

4 Two candles have the same height but different diameters. It takes one candle 3 hours to burn out and the other 4 hours to burn out. At what time must the candles be lit for the ratio of their heights to be 1 : 2 at 3 pm?

Solution:

Let us assume the initial height of both candles to be 1. Let us also assume it takes n mins for the heights to reach a ratio of 1 : 2.

Step 1: Height of 1st candle after n mins $= \frac{n}{180}$

Height of 2nd candle after n mins $= \frac{n}{240}$

Step 2: Height remained: 1st candle $= 1 - \frac{n}{180}$

2nd candle $= 1 - \frac{n}{240}$

Step 3: We seek a ratio of 1 : 2.

$$\left(1 - \frac{n}{180}\right) : \left(1 - \frac{n}{240}\right) = 1 : 2$$

$$1 - \frac{n}{240} = 2 \times \left(1 - \frac{n}{180}\right)$$

$$1 - \frac{n}{240} = 2 - \frac{n}{90}$$

$$\frac{n}{90} - \frac{n}{240} = 2 - 1$$

$$\frac{8n - 3n}{720} = 1$$

$$5n = 720$$

$$n = 144$$

144 min = 2 h 24 min

2 h 24 min before 3 pm is 12.36 pm

The candles must be lit at **12.36 pm**.

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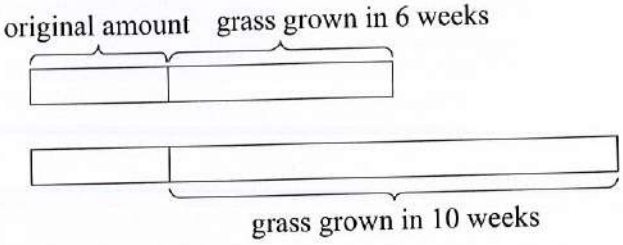
5 The grass on a ranch grows at a constant rate every week.
240 sheep will eat up the grass in 6 weeks.
200 sheep will eat up the grass in 10 weeks.
For how many weeks can the pasture sustain a herd of 190 sheep?

Solution:



$240 \times 6 = 1440$
$200 \times 10 = 2000$

The difference in the total amount
$= 2000 - 1440$
$= 560$ shares if each share can last a sheep for one week.

The growth rate $= 560 \div (10 - 6)$
$= 140$ shares each week

The initial amount $= 1440 - 140 \times 6 = 600$ shares
or
$2000 - 140 \times 10 = 600$ shares
$600 \div (190 - 140) = 12$

The pasture can sustain a herd of 190 sheep for **12 weeks**.

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PRACTICE

1 Mr Kaufman needs a total of 24 hours to harvest his crops. If his son chips in for 6 hours, they will finish the harvesting in 16 hours. How many hours will his son take to harvest the crops all by himself?

2 It takes John and Peter 12 days to complete a project. If John works on it alone for the first 3 days, Peter needs another 8 days to complete $\frac{5}{12}$ of the project. How many days will John take to complete the project all by himself? How long will Peter take to complete the project?

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3 Randy and Steve can finish painting a farmhouse in 6 days. It takes Steve and Tim 9 days to do the same task. Randy and Tim take 12 days to do the same task. How many days will it take if the three boys do the painting together?

4 If Hoses A and B are turned on at the same time, they will take 9 hours to fill a swimming pool. It takes Hoses A and C 12 hours to fill the same pool. Hoses B and C will take 18 hours to do the same job. How many hours will it take to fill the pool if all the three hoses are turned on at the same time?

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5 Paul and Mary are assigned a mathematics project to work on.
If Paul works on it alone for 6 hours and let Mary take over, Mary needs another 12 hours to complete.
If Paul works on it for 8 hours and then hands it over to Mary, she will need another 6 hours.
How many hours does Mary alone need to complete the project if Paul has already worked on it for 3 hours?

6 A candle took 4 hours to finish burning. Another shorter candle, but with a bigger diameter, took 5 hours to finish burning. The candles were lit at the same time and had the same height two hours later. What was the ratio of the heights of the two candles at first?

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7 Two candles have the same length but different diameters. It takes one candle 4 hours and another candle 2 hours to finish burning. How many hours will it take for the ratio of the lengths of the candles to be 1 : 2 if both candles are lit at the same time?

8 A piece of pasture is growing at a constant rate every day.
It is able to sustain a herd of 100 cows for 20 days.
It is able to feed a herd of 150 cows for 10 days.
For how many days is the piece of pasture able to feed 250 cows?

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9 A piece of pasture grows at a constant rate every week.
120 cows will eat up the grass in 12 weeks.
150 cows will eat up the grass in 8 weeks.
How many cows can the pasture sustain for 6 weeks?

19 Complex Fraction

A fraction whereby either the numerator or denominator (or both) consists of

(i) other fractions;

(ii) subsequently the four operations,

is called a complex fraction.

Effectively, the process of solving complex fractions is one of simplification. In this case, the fraction is present in the denominator only, so we can repeat a simple two-step process by:

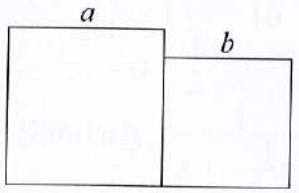
(i) inversing the fractions on the right-hand-side and the left-hand-side;

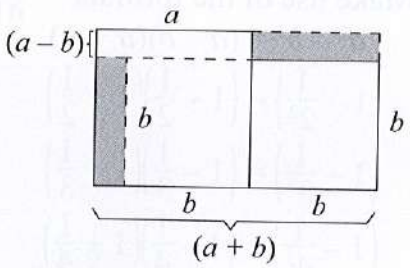
(ii) cancelling the integer part of the fractions.

If the complex fraction is one like Example 5 or Question 11, we realize after four subtractions that the pattern is repeated when we work on the denominator. The problem is then reduced to Remainder Problem.

An important formula is also introduced.

Given two adjacent squares like the ones shown below





whose sides are a and b respectively, the difference in area, which is $(a^2 - b^2)$, can be expressed as $(a - b)(a + b)$. This is because when we move the shaded part to its new position, we actually end up with looking at the area of a rectangle.

This formula will be used in Example 2 and Question 9.

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EXAMPLES

1 Evaluate $\frac{1}{1+2}+\frac{1}{1+2+3}+\frac{1}{1+2+3+4}+\ldots+\frac{1}{1+2+3+\ldots+50}$.

Solution:

Write each denominator as sum $=\frac{\text{(first no. + last no.)} \times \text{how many nos.}}{2}$

We have,

$$\frac{1}{\frac{(1+2)\times 2}{2}}+\frac{1}{\frac{(1+3)\times 3}{2}}+\frac{1}{\frac{(1+4)\times 4}{2}}+\ldots+\frac{1}{\frac{(1+50)\times 50}{2}}$$

$$=\frac{2}{2\times 3}+\frac{2}{3\times 4}+\frac{2}{4\times 5}+\ldots+\frac{2}{50\times 51}$$

$$=2\times\left(\frac{1}{2\times 3}+\frac{1}{3\times 4}+\frac{1}{4\times 5}+\ldots+\frac{1}{50\times 51}\right)$$

We have transformed the question into one of partial fraction.

$$=2\times\left(\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\frac{1}{4}-\frac{1}{5}+\ldots+\frac{1}{50}-\frac{1}{51}\right)$$

$$=2\times\left(\frac{1}{2}-\frac{1}{51}\right)$$

$$=2\times\frac{49}{102}$$

$$=\mathbf{\frac{49}{51}}$$

2 Evaluate $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\left(1-\frac{1}{100^2}\right)$.

Solution:

Make use of the formula

$$a^2-b^2=(a-b)(a+b)$$

$$\left(1-\frac{1}{2^2}\right)=\left(1-\frac{1}{2}\right)\left(1+\frac{1}{2}\right)$$

$$\left(1-\frac{1}{3^2}\right)=\left(1-\frac{1}{3}\right)\left(1+\frac{1}{3}\right)$$

$$\left(1-\frac{1}{4^2}\right)=\left(1-\frac{1}{4}\right)\left(1+\frac{1}{4}\right)$$

$$\vdots$$

$$\left(1-\frac{1}{100^2}\right)=\left(1-\frac{1}{100}\right)\left(1+\frac{1}{100}\right)$$

Let A be the original expression.

$$A=\left(1-\frac{1}{2}\right)\left(1+\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1+\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1+\frac{1}{4}\right)\ldots\left(1-\frac{1}{100}\right)1+\frac{1}{100}$$

$$=\frac{1}{2}\times\frac{3}{2}\times\frac{2}{3}\times\frac{4}{3}\times\frac{3}{4}\times\frac{5}{4}\times\ldots\times\frac{99}{100}\times\frac{101}{100}$$

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Eliminate each pair of inverse fraction accordingly,

$$\frac{3}{2} \times \frac{2}{3} = 1$$

$$\frac{4}{3} \times \frac{3}{4} = 1$$

$$\vdots$$

$$\frac{100}{99} \times \frac{99}{100} = 1$$

$$A = \frac{1}{2} \times \frac{101}{100} = \mathbf{\frac{101}{200}}$$

3 $A = \frac{1000 \times 1003}{1001 \times 1002}$, $B = \frac{1000 \times 1002}{1001 \times 1003}$, $C = \frac{1000 \times 1001}{1002 \times 1003}$.

Which has the greater value?

Solution:

$$A = \frac{1000 \times 1003}{1001 \times 1002} \times \frac{1003}{1003}$$

$$B = \frac{1000 \times 1002}{1001 \times 1003} \times \frac{1002}{1002}$$

$$C = \frac{1000 \times 1001}{1002 \times 1003} \times \frac{1001}{1001}$$

Clearly, $1000 \times 1003^2 > 1000 \times 1002^2 > 1000 \times 1001^2$. $\therefore$ **A > B > C.**

4 Find the value of a in $\dfrac{1}{1+\dfrac{1}{2+\dfrac{1}{a+\frac{1}{2}}}} = \frac{16}{23}$.

Solution:

$$1 + \frac{1}{2+\dfrac{1}{a+\frac{1}{2}}} = \frac{23}{16} = 1 + \frac{7}{16}$$

Similarly, $\dfrac{1}{2+\dfrac{1}{a+\frac{1}{2}}} = \frac{7}{16}$

$$2 + \frac{1}{a+\frac{1}{2}} = \frac{16}{7} = 2 + \frac{2}{7}$$

$$\frac{1}{a+\frac{1}{2}} = \frac{2}{7}$$

$$a + \frac{1}{2} = \frac{7}{2} = 3 + \frac{1}{2}$$

$$a = \mathbf{3}$$

5 Evaluate $1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{\vdots\;\;1-\cfrac{127}{365}}}}$ } 1000 division lines

Solution:

Work out the first few steps.

1^{st}: $1-\dfrac{127}{365}=\dfrac{239}{365}$

2^{nd}: $1-\dfrac{1}{\frac{239}{365}}=1-\dfrac{365}{238}=-\dfrac{127}{238}$

3^{rd}: $1-\dfrac{1}{\left(-\frac{127}{238}\right)}=1+\dfrac{238}{127}=\dfrac{365}{127}$

4^{th}: $1-\dfrac{1}{\frac{365}{127}}=1-\dfrac{127}{365}=\dfrac{238}{365}$

$\vdots$

The pattern recurs at $\dfrac{238}{365}, -\dfrac{127}{238}, \dfrac{365}{127}, \dfrac{238}{365}, \ldots$

$1000 \div 3 = 333 \times 3 + 1$

The expression has a value of $\mathbf{\dfrac{238}{365}}$.

PRACTICE

1 Evaluate $\dfrac{17\frac{2}{3}+21\frac{1}{2}-9\frac{1}{3}}{79\frac{1}{2}-49\frac{2}{3}}$.

2 Evaluate $1\frac{1}{2}+2\frac{1}{6}+3\frac{1}{12}+4\frac{1}{20}+5\frac{1}{42}+6\frac{1}{56}+7\frac{1}{72}+8\frac{1}{90}+9\frac{1}{110}+10\frac{1}{132}+11\frac{1}{156}$.

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3 Evaluate $\dfrac{2000}{123456^2 - 123457 \times 123455}$.

4 Evaluate $\dfrac{1}{1+2} + \dfrac{1}{1+2+3} + \dfrac{1}{1+2+3+4} + \ldots + \dfrac{1}{1+2+3+4+\ldots+100}$.

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5 Evaluate $\frac{1}{3 \times 2 \times 3} + \frac{1}{6 \times 3 \times 4} + \frac{1}{9 \times 4 \times 5} + \ldots + \frac{1}{294 \times 99 \times 100}$.

6 It is given $a * b = \frac{ab}{a+b}$.

Evaluate $\frac{1}{1 * 2} - \frac{1}{2 * 3} + \frac{1}{3 * 4} - \frac{1}{4 * 5} - \ldots - \frac{1}{2000 * 2001}$.

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7 Evaluate $A = \frac{3}{2} - \frac{5}{6} + \frac{7}{12} - \frac{9}{20} + ... + \frac{199}{9900}$.

8 Find the value of b in $\cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{b + \frac{1}{6}}}} = \frac{130}{421}$.

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9 Evaluate $5000 \times \left(1 - \frac{1}{5^2}\right) \times \left(1 - \frac{1}{6^2}\right) \times \cdots \times \left(1 - \frac{1}{1000^2}\right)$.

10 Evaluate $\frac{\frac{1}{2} - \frac{1}{3}}{\frac{1}{3} - \frac{1}{4}} \times \frac{\frac{1}{4} - \frac{1}{5}}{\frac{1}{5} - \frac{1}{6}} \times \ldots \times \frac{\frac{1}{1999} - \frac{1}{2000}}{\frac{1}{2000} - \frac{1}{2001}}$.

11 Evaluate $1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{\vdots\;1-\cfrac{139}{365}}}}$ } 1000 division lines

12 Evaluate $1+\frac{1}{3}+\frac{1}{3^2}+\ldots+\frac{1}{3^{12}}$.

20 Speed 2

This chapter looks into more dynamic questions on speed, so as to let students gain valuable insight. At a more advanced level, we often deploy the help of writing equations to solve these sophisticated problems.

EXAMPLES

1 Car A left Town A for Town B. Car B left Town B for Town A at the same time. The ratio of the speed of Car A to the speed of Car B was 3 : 2. Each continued with its journey after they met and made a U-turn upon arrival at their own destination. When they met the second time, it was 30 km away from where they did the first time. How far was Town B from Town A?

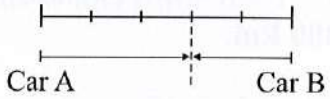
Solution:

Method: Solve by Drawing

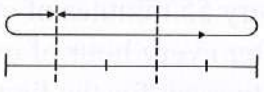
$3 + 2 = 5$

Divide the distance into 5 parts.

In Drawing 1, Car A travelled 3 parts, Car B travelled 2 parts.



In Drawing 2, Car A travelled 6 parts, Car B travelled 4 parts.



$2\text{ u} = 30\text{ km}$

$1\text{ u} = 15\text{ km}$

$5\text{ u} = 75\text{ km}$

Town B was **75 km** away.

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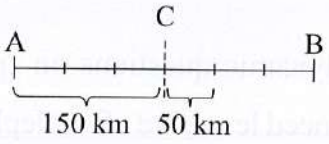
2 Car A begins its journey from Town A towards Town B and Car B begins its journey from Town B towards Town A at the same time. C is a station in the middle of the two towns. Both cars continue their journey after A meets B 50 km past Station C. Car A then makes a U-turn upon reaching Town B and catches up with Car B 150 km past Station C. Find the distance between Town A and Town B.



Solution:

Let the distance be d. Car A has travelled $\left(\frac{d}{2} + 50\right)$ km before meeting Car B.

$$\frac{d}{2} - 50 + \frac{d}{2} + 150 = (d + 100) \text{ km}$$

It travels another $(d + 100)$ km when it catches up with B for the second time.

$$(d + 100) = 2 \times \left(\frac{d}{2} + 50\right)$$

It has since travelled twice the distance.

Car B has travelled $\left(\frac{d}{2} - 50\right)$ km before meeting Car A.

It travels another $(50 + 150)$ km before Car A catches up.

200 km is twice the distance of $\left(\frac{d}{2} - 50\right)$.

$$2 \times \left(\frac{d}{2} - 50\right) = 200$$

$$d - 100 = 200$$

$$d = 300 \text{ km}$$

The distance is **300 km**.

3 The circumference of a circular lake is 30 km. Peter and Jane begin to cycle from the same point around the lake in opposite directions. Peter's speed is 250 m/min, but he rests for 5 minutes after every 55 minutes of cycling. Jane's speed is 200 m/min, but he rests for 10 minutes after every hour of cycling.
Find the time taken for them to meet for the first time.

Solution:

First, we convert the speed to km/h, as the other way round would be quite tardy.

$$250 \text{ m/min} = \frac{250 \times 60}{1000} = 15 \text{ km/h}$$

$$200 \text{ m/min} = \frac{200 \times 60}{1000} = 12 \text{ km/h}$$

Let t be the time taken to meet.

$15 \cdot \frac{55}{60} \cdot t$ is the distance travelled by Peter.

$12 \cdot \left(t + \frac{10}{60}\right)$ is the distance travelled by Jane.

$$15 \cdot t + 12 \cdot \left(t + \frac{10}{60}\right) = 30$$

$$13t + 12t + 2 = 30$$

$$25 \cdot 75t = 28$$

$$t = \frac{28 \times 4}{103} \text{ as } 25 \times 75 = \frac{103}{4}$$

$$= \frac{112}{103}$$

$$= \mathbf{1\frac{9}{103}\ h}$$

4 Peter was driving from his hometown to Town X for business. If he increased his speed by 20%, he would arrive 1 hour ahead of schedule. If he drove at his planned speed for the first 120 km before increasing it by 25%, he would arrive 40 minutes earlier. How far was Town X?

Solution:

In Scenario 1,

original speed : new speed = 100 : 120 = 5 : 6

original time : new time = 6 : 5

6 – 5 = 1 unit

1 unit = 1 hour

Time taken for the whole trip = 6 × 1 = 6 hours

In Scenario 2,

original speed : new speed = 100 : 125 = 4 : 5

original time : new time = 5 : 4

5 – 4 = 1 unit

1 unit = $\frac{2}{3}$ h

5 units = $\frac{10}{3}$ h

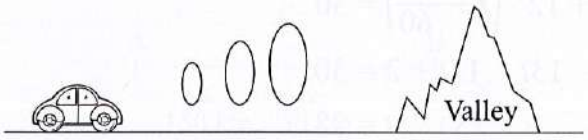
$\left(6 - \frac{10}{3}\right)$ h = $2\frac{2}{3}$ h for the first 120 km.

The speed = $120 \div \frac{8}{3}$ = 45 km/h

The distance = 45 × 6 = **270 km**.

5 I was once driving towards a valley in a foreign land. When I honked the car, I heard the echo 6 seconds later. I wondered how far I was from the valley the moment I honked. The car was moving at a constant speed of 108 km/h. I understand the speed of sound is 340 m/s.

Solution:



Let the distance be d.

$\frac{d}{340}$ was the time taken for sound to travel to the valley.

Now, convert 108 km/h to m/s.

$108 \times \frac{1000}{3600} = 30$ m/s

I was travelling at 30 m/s!

$30 \times 6 = 180$ m

The car would be $\left(d - \frac{d}{340} \times 180\right)$ m away from the valley after 6 seconds.

The car and the sound met $\left(6 - \frac{d}{340}\right)$ seconds later.

$$d - \frac{d}{340} \times 180 = \left(6 - \frac{d}{340}\right) \times (340 + 180)$$

$$\text{LHS} = d - \frac{180d}{340}$$

$$= \frac{340d - 180d}{340}$$

$$= \frac{160d}{340}$$

$$\text{RHS} = \left(6 - \frac{d}{340}\right) \times 520$$

$$= 3120 - \frac{520d}{340}$$

Equate again,

$$3120 = \frac{520d + 160d}{340}$$

$$3120 = \frac{68d}{34}$$

$$d = 3120 \div 2$$

$$= \mathbf{1560\ m}$$

PRACTICE

1 A car and a truck left for Town A, 1000 km away, at different times. The car began its journey when the truck had travelled 200 km. When the car arrived at Town A the truck had 200 km more to go. At which point did the car catch up with the truck?

2 Town A is 300 km away from Town B. Car A leaves Town A for Town B, Car B leaves Town B for Town A at the same time. 2 hours later, Car A is 50 km away from Car B for the first time. Suppose they begin their journey at 0800 h and keep making U-turns after arriving at their destinations, at which point are the two cars 50 km apart for the fifth time?

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3 Mark and Nick were both going to a public library after school. Mark walked at a speed of 80 m/min half of the time. He walked at 70 m/min the other half of the time. Nick walked at 80 m/min for the first half of the journey. He walked at 70 m/min for the remaining journey. Who arrived at the library first?

4 An escalator at a shopping mall ascends at a constant rate.
Lenny walks up the moving escalator at a speed of 40 steps/min.
Marie walks up the moving escalator at a speed of 20 steps/min.
They arrive at the 2nd floor at respective times of 2 and 3 minutes. How many steps has the escalator?

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5 A platoon of soldiers were marching to a camp site from the HQ at a speed of 7.2 km/h. 2 km into the march, the messenger was sent back to deliver a document to the HQ. The messenger rushed back to the HQ at a speed of 10.8 km/h. He caught up with the troop again 2 km before the camp site. What was the distance between the HQ and the camp site?

6 Jessie was walking along a straight road. Mark was jogging along the same road in the same direction. His speed was 3 times that of Jessie's.
A bus caught up with Jessie every 10 minutes.
A bus caught up with Mark every 20 minutes.
Assume buses left the terminal at fixed time intervals, find the time intervals.

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7 Mr Bond was going from Town A to Town B for a mission. He travelled by helicopter for the first half of the journey and drove for the second half of the journey. On his return journey, he flew half of the time and drove the other half of the time. As a result, he saved half an hour. What was the distance between the two towns if the speed of the helicopter and the car were given as 200 km/h and 100 km/h respectively?

8 Mary brisk-walks at a speed of 9 km/h along a small road. The road is parallel to a Light Rail Transit track. She realizes a train passes her by from behind every 8 minutes. An oncoming train also passes her every 5 minutes. Assume the trains travel at a constant speed and leave their terminals at fixed intervals. What is the speed of the trains?

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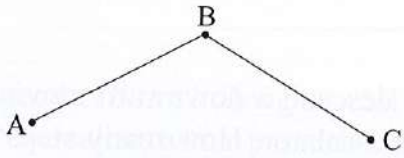
9 Mary, Nikki and Polly were running on a 600 m running track. Their speeds were 8 m/s, 6 m/s and 4 m/s respectively. They started at the same place, at the same time. Polly ran in the opposite direction. How long would it take the three of them to meet for the first time?

10 It takes Sam 24 steps to descend a downward moving escalator. It takes Ted 48 steps to ascend the same escalator. How many steps has the escalator when it is stationary? Assume Ted can walk 4 times as fast as Sam.

11 Mark, Nick and Peter cycled from Town A to Town B. Nick left Town A 10 minutes later than Peter, but caught up with him 30 minutes later. Mark left 30 minutes later than Nick and caught up with Peter 1 hour and 40 minutes later.
How long did it take Mark to catch up with Nick?

12 The journey from Town A to Town C includes an uphill slope towards point B and a downhill slope after point B.



Cycling speeds for uphill and downhill are 200 m/min and 500 m/min respectively. Mark takes 11.5 minutes to reach Town C from Town A. The return trip takes 13 minutes. How far is point B to Town C?

Approximation

In Approximation, we are not involved in solving for the exact value of an answer. Rather, we find an integer closest to it. The signs "<", meaning "smaller than", and ">", meaning "greater than", are often used.

The techniques we deploy to solve problems of this nature are diversified.

They are:

(A) For comparison of values of two or more fractions, we can use one or more of the following:

(i) Inverse the fractions

(ii) Use a formula

(iii) Subtract a common value or from a common value

Example 4, Questions 9 and 11 belong to this group.

(B) For evaluating the integer of a series of fractions or complex fractions:

(i) We break the series into smaller groups of fractions like in Question 2.

(ii) Set the minimum and maximum value like in Example 2, Questions 3, 5 and 7.

(iii) Prime factorize, like in Example 1, Question 4.

(C) Others, like using Partial Fraction (Example 5) and insertion of another series (Question 12).

EXAMPLES

1 Find the integer part of $S = \left(\frac{1}{2}+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+\frac{1}{11}+\frac{1}{13}\right) \times 231$.

Solution:

$$231 = 3 \times 7 \times 11$$

$$S = \frac{231}{2} + 77 + \frac{231}{5} + 33 + 21 + \frac{231}{13}$$

$$= 115\frac{1}{2} + 46\frac{1}{5} + 17\frac{10}{13} + 131$$

Now consider the fractions alone, $\frac{1}{2}+\frac{1}{5}+\frac{10}{13} = \frac{65}{130}+\frac{26}{130}+\frac{100}{130}$.

Clearly, $1 < \frac{1}{2}+\frac{1}{5}+\frac{10}{13} < 2$.

The integer part of the expression

$131 + 115 + 46 + 17 + 1 = \mathbf{310}$

2 Find the integer part of $\dfrac{1}{\frac{1}{80}+\frac{1}{81}+\frac{1}{82}+\frac{1}{83}+\frac{1}{84}}$.

Solution:

Step 1: Compare each fraction with the highest value $\frac{1}{80}$.

$$\frac{1}{81} < \frac{1}{80}, \frac{1}{82} < \frac{1}{80}, \frac{1}{83} < \frac{1}{80}, \frac{1}{84} < \frac{1}{80}$$

$$\frac{1}{80} + \frac{1}{81} + \ldots + \frac{1}{84} < \frac{1}{80} \times 5 = \frac{5}{80}$$

Step 2: Compare each fraction with the lowest value $\frac{1}{84}$.

$$\frac{1}{84} < \frac{1}{80}, \frac{1}{84} < \frac{1}{81}, \frac{1}{84} < \frac{1}{82}, \frac{1}{84} < \frac{1}{83}$$

$$\frac{1}{84} \times 5 < \frac{1}{80} + \frac{1}{81} + \ldots + \frac{1}{84}$$

$$\frac{5}{84} < \frac{1}{80} + \frac{1}{81} + \ldots + \frac{1}{84}$$

$$\frac{5}{84} < \frac{1}{80} + \frac{1}{81} + \ldots + \frac{1}{84} < \frac{5}{80}$$

$$\frac{84}{5} > \frac{1}{\frac{1}{80}+\frac{1}{81}+\ldots+\frac{1}{84}} > \frac{80}{5}$$

$$168.8 > S > 16$$

The integer part of the expression is **16**.

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3 If $a = 1\frac{10}{100} + 2\frac{10}{101} + 3\frac{10}{103} + 4\frac{10}{104} + \ldots + 1\frac{110}{110}$, find the integer part of a.

Solution:

There are 2 parts to the solution. First,

$$1 + 2 + 3 + 4 + \cdots + 11 = \frac{(1 + 11) \times 11}{2}$$
$$= 6 \times 11$$
$$= 66$$

Next, find the maximum and minimum possible values, or estimates, of $\frac{10}{100} + \frac{10}{101} + \frac{10}{102} + \ldots + \frac{10}{110}$.

Possible maximum value,

$\because \frac{10}{101} < \frac{10}{100}, \frac{10}{102} < \frac{10}{100}, \ldots, \frac{10}{100} < \frac{10}{100}$

$$\because \frac{10}{100} + \frac{10}{101} + \frac{10}{102} + \ldots + \frac{10}{110} < \underbrace{\frac{10}{100} + \frac{10}{100} + \ldots + \frac{10}{100}}_{11 \times \frac{10}{100}}$$

$$\frac{10}{100} \times 11 = 1.1 < 2$$

Possible minimum value,

$\because \frac{10}{100} > \frac{10}{110}, \frac{10}{101} > \frac{10}{110}, \ldots, \frac{10}{109} > \frac{10}{110}$

$$\because \frac{10}{100} + \frac{10}{101} + \frac{10}{102} + \ldots + \frac{10}{110} > \underbrace{\frac{10}{100} + \frac{10}{100} + \ldots + \frac{10}{100}}_{11 \times \frac{10}{100}}$$

$$11 \times \frac{10}{110} = \frac{10}{110} \times 11 = 1$$

The integer part of $\frac{10}{100} + \frac{10}{101} + \ldots + \frac{10}{110}$ is 1.

The integer part of a is $(66 + 1) = \mathbf{67}$.

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4 Show that $\frac{555\,553}{555\,555} > \frac{333\,331}{333\,333}$.

Solution:

Method 1: Find the inverse.

$$\frac{555\,553}{555\,555} \to 1\frac{2}{555\,553}$$

$$\frac{333\,331}{333\,333} \to 1\frac{2}{333\,331}$$

Clearly, $1\frac{2}{333\,331} > 1\frac{2}{555\,553}$.

But we have inversed the fraction,

$$\therefore \quad \frac{555\,553}{555\,555} > \frac{333\,331}{333\,333}$$

Method 2: Subtract from a common value.

$$1 - \frac{555\,553}{555\,555} = \frac{2}{555\,555}$$

$$1 - \frac{333\,331}{333\,333} = \frac{2}{333\,333}$$

Clearly, $\frac{2}{333\,333} > \frac{2}{555\,555}$.

$$\therefore \quad \frac{555\,553}{555\,555} > \frac{333\,331}{333\,333}$$

Method 3: Use the $\frac{A+n}{B+n} > \frac{A}{B}$ concept.

$$\frac{555\,553}{555\,555} = \frac{333\,331 + 222\,222}{333\,333 + 222\,222}$$

$A = 333\,331$

$B = 333\,333$

$C = 222\,222$

$$\frac{555\,553}{555\,555} > \frac{333\,331}{333\,333}$$

5 Given $A = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots + \frac{1}{11^2} + \frac{1}{12^2}$, show that $A < 2$.

Solution:

Since $\frac{1}{1 \times 2} > \frac{1}{2^2}, \frac{1}{2 \times 3} > \frac{1}{3^2}, \frac{1}{3 \times 4} > \frac{1}{4^2}, \ldots, \frac{1}{11 \times 12} > \frac{1}{12^2}$,

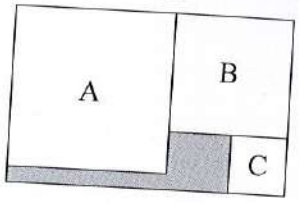
an approximation of A would be

$$1 + 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \ldots + \frac{1}{11} - \frac{1}{12} = 2 - \frac{1}{12}$$

$$= 1\frac{11}{12} < 2$$

PRACTICE

1 Three squares of different sizes are embedded in a rectangle. The area of the rectangle is 228 cm^2. The side of B is $\frac{8}{19}$ of the rectangle's length. The side of C is $\frac{1}{3}$ of its width. Find the area of the shaded region.



2 Find the integer part of $\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\frac{1}{9}+\frac{1}{10}$.

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3 Suppose $A = 9.8 + 9.98 + 9.998 + \cdots + \underbrace{9.99\cdots998}_{7\ 9\text{s}}$, find the integer part of A.

4 Given $A = \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13}\right) \times 1001$, find the integer part of A.

of A.

5 Given $m = \frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \cdots + \frac{\overbrace{999...999}^{20\ 9s}}{\underbrace{1000...000}_{20\ 0s}}$, find the integer part of m.

6 There are 9 rods. The 1st one is 1 cm in length, the 2nd one is 2 cm, the 3rd one is 3 cm, $\cdots$, the 9th rod is 9 cm long. Certain number of rods are selected each time to make a square. How many ways are there to do so?

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7 Find the integer part of $\dfrac{1}{\frac{1}{72}+\frac{1}{73}+\frac{1}{74}+\cdots+\frac{1}{80}}$.

8 Suppose $A = 1-\frac{1}{2}-\frac{1}{4}-\frac{1}{8}-\frac{1}{10}-\cdots-\frac{1}{512}$, show that $A < 0.002$.

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9 Show that $\frac{41245}{123456} > \frac{78282}{234567}$.

10 Find the value of n, so the inequality is satisfied.

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n \times (n+1)} > \frac{1949}{2000}$$

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11 Show that

(a) $\frac{113}{342} > \frac{251}{755}$ and

(b) $\frac{1784}{3456} > \frac{673}{1234}$ are not true.

12 Suppose $A = \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \frac{7}{8} \times \cdots \times \frac{99}{100}$, show that $A < 0.1$.

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22 Bases Other Than Ten

In the Arabic numeral system, also known as decimal or base 10, there are 10 digits, namely 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. We notice that the concept of 10 is represented by two existing digits, 0 and 1.

In the Babylonian numeral system, where 60 is used as the base, we call it the sexagesimal system. Little surprise that the last number of this system is 59, very much like 9 is the last digit of base 10.

Some ancient tribes have used quinary, or base 5, whereby they could count conveniently using one hand. The numbers in the base 5 system are 0, 1, 2, 3 and 4, where the concept of 5 represents 10 because of its carry-over.

The storage and computational function of a computer uses the binary system. Only two numbers, 0 and 1, are present in this system. The concept of 2 represents 10 because of its carry-over.

As a general guide, base 2 is denoted by $(\quad)_2$. Base 3 is denoted by $(\quad)_3$ and so on. If we are in a system of base n, there is a carry-over of 1 whenever we reach n. Likewise, if we reach $n + 1$, the number becomes 11. We look at some examples shown below.

$$(101)_2 + (111)_2 = (1100)_2$$
$$(223)_4 + (323)_4 = (1212)_4$$
$$(345)_6 + (555)_6 = (1344)_6$$

Any guess for 30 of base 10 to be written in base 8?

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EXAMPLES

1 Compute the following.

(a) $(1001)_2 + (1100)_2 = \mathbf{(10101)_2}$

$$\begin{array}{r} 1\ 0\ 0\ 1 \\ +\ 1\ 1\ 0\ 0 \\ \hline 1\ 0\ 1\ 0\ 1 \end{array}$$

Notice that in base 2, (1 + 1) is expressed as 10.

(b) $(2341)_5 + (143)_5 = \mathbf{(3034)_5}$

$$\begin{array}{r} 2\ 3\ 4\ 1 \\ +\ \ \ 1\ 4\ 3 \\ \hline 3\ 0\ 3\ 4 \end{array}$$

Notice that in base 5, (4 + 4) is expressed as 13.

2 Compute the following.

(a) $(1101)_2 \times (101)_2 = \mathbf{(1000001)_2}$

$$\begin{array}{r} 1\ 1\ 0\ 1 \\ \times\ \ 1\ 0\ 1 \\ \hline 1\ 1\ 0\ 1 \\ +\ 1\ 1\ 0\ 1 \\ \hline 1\ 0\ 0\ 0\ 0\ 0\ 1 \end{array}$$

(b) $(10011)_2 \times (110)_2 = \mathbf{(1110010)_2}$

$$\begin{array}{r} 1\ 0\ 0\ 1\ 1 \\ \times\ \ \ 1\ 1\ 0 \\ \hline 1\ 0\ 0\ 1\ 1\ 0 \\ +\ 1\ 0\ 0\ 1\ 1 \\ \hline 1\ 1\ 1\ 0\ 0\ 1\ 0 \end{array}$$

Note that (1 + 1) is expressed as 10 in both (a) and (b).

3 Rewrite the numbers in base 10 and other bases in their expanded form.

(a) $(632)_{10} = \mathbf{6 \times 10^2 + 3 \times 10^1 + 2 \times 10^0}$ $(10^0 = 1)$

(b) $(1864)_{10} = \mathbf{1 \times 10^3 + 8 \times 10^2 + 6 \times 10^1 + 4 \times 10^0}$ $(10^0 = 1)$

(c) $(7453)_8 = \mathbf{7 \times 8^3 + 4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0}$ $(8^0 = 1)$

(d) $(5241)_6 = \mathbf{5 \times 6^3 + 2 \times 6^2 + 4 \times 6^1 + 1 \times 6^0}$ $(6^0 = 1)$

(e) $(1233)_4 = \mathbf{1 \times 4^3 + 2 \times 4^2 + 3 \times 4^1 + 3 \times 4^0}$ $(4^0 = 1)$

(f) $(111001)_2 = \mathbf{1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^0}$ $(2^0 = 1)$

4 Convert these numbers from base 10 to the base indicated below.

(a) $(8540)_{10} = (\quad)_5$

(b) $(72)_{10} = (\quad)_2$

Solution:

(a)

$5 \mid 8\ 5\ 4\ 0$

$5 \mid 1\ 7\ 0\ 8$ — R0

$5 \mid 3\ 4\ 1$ — R3

$5 \mid 6\ 8$ — R1

$5 \mid 1\ 3$ — R3

2 → R3

$(8540)_{10} = \mathbf{(233130)_5}$

(b)

$2 \mid 7\ 2$

$2 \mid 3\ 6$ — R0

$2 \mid 1\ 8$ — R0

$2 \mid 9$ — R0

$2 \mid 4$ — R1

$2 \mid 2$ — R0

1 → R0

$(72)_{10} = \mathbf{(1001000)_2}$

5 Convert these numbers from other bases to base 10.

(a) $(1234)_5 = (\quad)_{10}$

(b) $(7746)_8 = (\quad)_{10}$

Solution:

(a) $(1234)_5 = 1 \times 5^3 + 2 \times 5^2 + 3 \times 5^1 + 4 \times 5^0$

$= 125 + 50 + 15 + 4$

$= \mathbf{(194)_{10}}$

(b) $(7746)_8 = 7 \times 8^3 + 7 \times 8^2 + 4 \times 8^1 + 6 \times 8^0$

$= 3584 + 448 + 32 + 6$

$= \mathbf{(4070)_{10}}$

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PRACTICE

1 Compute the following.

(a) $(1011)_2 + (1111)_2$

(b) $(11011)_2 + (10011)_2$

(c) $(111111)_2 + (11011)_2$

(d) $(11111)_2 - (10011)_2$

2 Convert these numbers from base 10 to the bases indicated below.

(a) $(1237)_{10} = (\qquad)_4$

(b) $(9653)_{10} = (\qquad)_8$

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3 Rewrite these numbers in base 10 or other bases in expanded form.

(a) $(894)_{10}$

(b) $(17653)_{10}$

(c) $(4321)_{5}$

(d) $(6544)_{7}$

(e) $(8888)_{9}$

(f) $(7546)_{8}$

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4 Convert these numbers in other bases to base 10.

(a) $(10111)_2 = (\qquad\qquad)_{10}$

(b) $(100100)_2 = (\qquad\qquad)_{10}$

(c) $(123123)_4 = (\qquad\qquad)_{10}$

(d) $(443322)_5 = (\qquad\qquad)_{10}$

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5 Complete the table below.

base 10	2	( )	6	( )	10
base 3	( )	11	( )	22	( )

6 Convert these number in other bases to base 10.

(a) $(1202)_3 = (\quad)_{10}$

(b) $(4321)_5 = (\quad)_{10}$

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7 Complete the base 6 multiplication table.

	1	2	3	4	5
1	1	2	3	4	5
2		4	10	12	14
3			13	20	23
4				24	32
5					41

8 Find the number base in which

$$\begin{array}{cccccc} & A & B & C & D \\ + & B & D & C & A \\ \hline D & B & 0 & 0 & 0 \end{array}$$

is computed.

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9 Complete the base 8 multiplication table

	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2		4	6	10	12	14	16
3			11	14	17	22	25
4				20	24	30	34
5					31	36	43
6						44	52
7							61

10 Given $(122)_4 = (abc)_5$, find the number represented by abc.

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11 Given $(724)_8 = (abc)_9$, find the value represented by abc.

12 A number in base 10 can be expressed as $(abc)_3$.
It can also be expressed as $(cba)_4$.
Find the number.

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Pigeonhole Principle

It is believed that the Pigeonhole Principle, also known as Dirichlet's Box Principle, was introduced by Dirichlet in 1834. It was known as Drawer's Principle or Shelf's Principle when it was first stated.

The Pigeonhole Principle states that if there are m pigeonholes and n pigeons where $n > m$, then at least one pigeonhole must contain more than one pigeon.

For example, if 6 letters are delivered to 5 mailboxes, then at least one mailbox will have more than one letter.

Let us look at another example. How do we show that at least two in a group of 13 people are born in the same month? There are only 12 months in a year, thus we have only 12 pigeonholes. Consider the worst case scenario: if each person was born in a different month, then the 13th person, no matter which month he or she was born in, would be born in the same month as one of the other 12 people.

At an advanced level, we say that if there are $(mn + 1)$ objects, but only n pigeonholes, then at least one of the pigeonholes will contain at least $(m + 1)$ objects.

For example, if 19 tennis balls are to be put back into 9 tubes, then at least one tube will have more than 2 tennis balls. This is because $19 \div 9 = 2$ R 1. This means that each tube will have 2 tennis balls in the worst case scenario. The remaining tennis ball, no matter which tube it is put back into, will result in that tube having $2 + 1 = 3$ tennis balls.

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EXAMPLES

1 A bag contains 10 red, 9 white, 8 yellow and 2 blue marbles. How many marbles must be taken out from the bag so that there will be 4 marbles of the same colour?

Analysis: We consider the worst case scenario.

Solution:

R	W	Y	B
1	1	1	1
1	1	1	1
1	1	1	

$$4 + 4 + 3 = 11$$

11 marbles have been taken out.

$$11 + 1 = 12$$

12 marbles must be taken out from the bag so that there will be 4 marbles of the same colour.

2 The height of the shortest student in a group of the same age is 136 cm. The tallest student in that group is 160 cm. How many students must there be before there are 5 students with the same height?
Assume all readings are whole numbers.

Analysis: We first consider each height group having 4 students.

Solution:

$$160 - 136 + 1 = 25$$

There will be 25 different height groups.

In the worst case scenario, there will be 4 students in each height group.

$$25 \times 4 = 100$$

The next student, no matter what his or her height is, will be the 5th student of any height group.

$$100 + 1 = 101$$

There must be **101** students before there are 5 students with the same height.

3 A bag contains some green, white, orange and blue marbles. Each student is to pick two marbles of different colours. How many students will there be before two of them pick marbles of the same colour combination?

Solution:

Possible combinations of colours: gw, go, gb, wo, wb, ob

$$6 + 1 = 7$$

There will be **7** students before two of them pick marbles of the same colour combination.

4 400 marbles are to be given out to a certain number of students. Each student must not get more than 12 marbles. At least how many students will get the same number of marbles?

Analysis: We group the students according to the number of marbles they get, at the same time keeping the number of students in each group to a minimum.

Solution:

1 marble	2 marbles	3 marbles	…	12 marbles
1	1	1	…	1
1	1	1	…	1
⋮				

$$1 + 2 + 3 + \ldots + 12 = 78$$
$$78 \times 5 = 390$$

Each group has 5 students.

$$400 - 390 = 10$$

There are 10 marbles left.

We give the 1st group one marble, 2nd group two marbles and so on.

$$1 + 2 + 3 + 4 = 10$$

The first 4 groups each has

$$5 + 1 = 6 \text{ students.}$$

At least **6** students will get the same number of marbles.

PRACTICE

1 183 students in Russels Elementary School were born in the year of 1998. At least how many students were born in the same month?

2 A deck of 52 cards have four different patterns: hearts, spades, diamonds and clubs. How many cards must you draw so that at least 3 cards have the same pattern?

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3 There are 320 Primary 5 students in a school.
At least how many students from that cohort were born in the same week of that year?
(Take 1 year as 52 weeks. Assume all Primary 5 students were born in the same year.)

4 There are 52 cards in a deck.
Every 13 cards form a same pattern.
Each pattern is labelled

A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K.

At least how many cards must be drawn to ensure there is a card labelled "A"?

5 The highest score in a mathematics test was 100.
The lowest score in the same test for the same class was 81.
At least three students in the class had the same score. What could be the minimum number of students in that class?
(Assume all scores are whole numbers.)

6 600 saga seeds are to be distributed among a number of children. Each child must not get more than 14 saga seeds. At least how many children will get the same number of saga seeds?

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7 443 candidates enter an examination hall.
There are 20 rows of seats in the hall. Each row has 25 seats.
At least how many rows have an equal number of candidates?

8 A bag contains some orange, blue, red and yellow marbles. Each student is allowed to pick two marbles. How many students must there be so that at least two students pick marbles of the same colour?

9 The east wing of a stadium has 1150 spectators.
There are 22 rows of seats in that wing.
Each row has 55 seats.
At least how many rows have an equal number of spectators?

10 A mathematics competition consists of 10 questions.
3 marks are awarded for answering a question correctly. 2 marks will be deducted for a wrong answer. 1 mark will be deducted for not answering a question.
How many students must have taken part in the competition in order to have at least four students of the same score?

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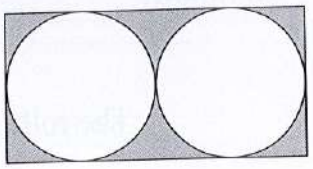
24 Area and Perimeter of Circles

EXAMPLES

1 Two identical circles are embedded in a rectangle. Find the area of the shaded region. (Take $\pi = \frac{22}{7}$)

***Analysis*:** We may assign a value to the radius and work on that.

***Solution*:**



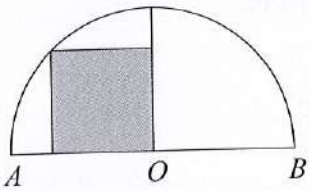
Let the radius be 7 cm.
Width of the rectangle = 14 cm
Length of the rectangle = 28 cm
Area of rectangle = $14 \times 28 = 392$ cm^2

Area of two circles = $2 \times 7 \times 7 \times \frac{22}{7} = 308$ cm^2

Shaded region = $392 - 308 = 84$ cm^2

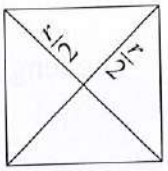
The area of the shaded region is **84 cm^2**.

2 The area of the shaded square is 8 cm^2.
Find the area of the semicircle where O is the centre. (Take $\pi = 3.14$)



***Analysis*:** Assume the square is made up from 4 equal triangles of base $\frac{r}{2}$ and height $\frac{r}{2}$.

***Solution*:**



$$\text{Area} = 8 = \frac{1}{2} \times \frac{r}{2} \times \frac{r}{2} \times 4$$

$$\frac{1}{2}r^2 = 8$$

$$r^2 = 16$$

$$r = 4 \text{ cm}$$

$$\frac{1}{2} \times 4 \times 4 \times 3.14 = 25.12$$

The area of the semicircle is **25.12 cm^2**.

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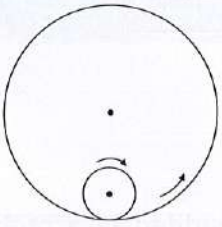
3 A small roller rolls without sliding inside a big circle. The radius of the circle is 4 times that of the roller. How many rounds will the roller have rotated about its own centre by the time it rolls back to its original position?



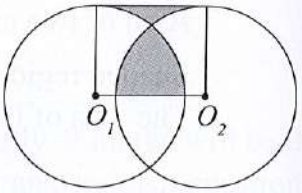
***Analysis*:** Our first intuition suggests $2\pi(4r) \div 2\pi r$ or 4 rounds.
A deeper analysis suggests otherwise.
As the roller rotates about a smaller radius $= 4r - r = 3r$.

***Solution*:**

$$4r - r = 3r$$
$$2 \times \pi \times 3r \div 2\pi r = 3$$

The roller will have rotated about its own centre for **3** rounds.

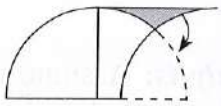
4 The radius of two identical circles is 4 cm.
The two shaded regions have the same area.
Find the length of O_1O_2. (Take $\pi = 3.14$)



***Analysis*:** If the two shaded regions have the same area, the rectangle has the same area as one semicircle.

***Solution*:**

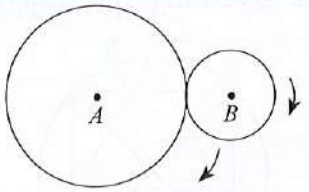
Area of semicircle
$= \frac{1}{2} \times 4 \times 4 \times 3.14$
$= 25.12 \text{ cm}^2$



Area of rectangle $= 4 \times O_1O_2$
$4 \times O_1O_2 = 25.12$
$O_1O_2 = 6.28$

The length of O_1O_2 is **6.28 cm**.

5 The radius of circle A is 3 times that of the radius of circle B. If circle B rolls along the circumference of circle A once without sliding, how many rounds will circle B have rotated about its own centre?



Analysis: Let the radii of circle A and circle B be $3r$ and r respectively. Imagine that the centre of circle B actually travels along a circular path that is $4r$ from the centre of circle A.

Solution:

$$3r + r = 4r$$
$$2 \times \pi \times 4r \div 2\pi r = 4$$

Circle B will have rotated **4** rounds about its own centre.

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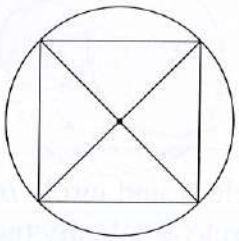
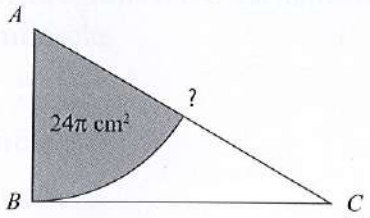
PRACTICE

1 What is the area of the circle given that the area of the square is 32 cm^2? (Take $\pi = 3.14$)



2 Given that the area of a fan is 24π cm^2.
If $\angle A = 60°$ and $\angle B = 90°$, what is the length of AC?



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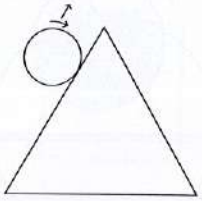
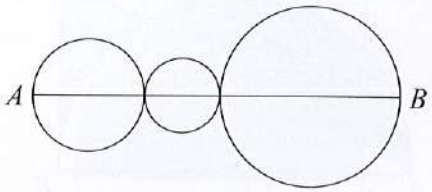
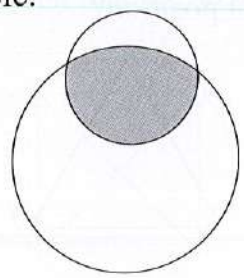
3 Given the perimeter of the triangle is 15 cm and the circumference of the ball is 5 cm.
How many turns will the ball have rolled, without slipping, as it rolls along the triangle and back to its original position?



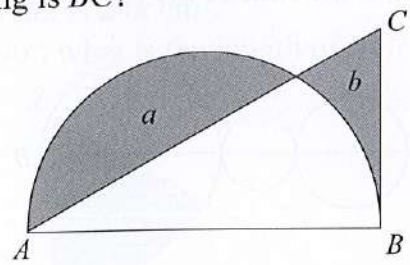
4 Given that $AB = 49$ cm, find the sum of the perimeters of the three circles.
(Take $\pi = \frac{22}{7}$)



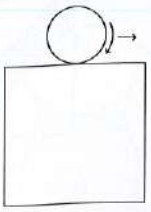
5 The radius of the small circle in the diagram below is 6 cm.
The shaded region is $\frac{2}{3}$ of the small circle and $\frac{1}{6}$ of the big circle.
Find the radius of the big circle.



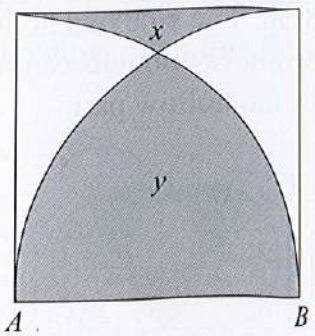
6 A right-angled triangle is placed over a semicircle.
The area of region a is greater than that of region b by 47 cm^2.
If $AB = 20$ cm, how long is BC?



7 Given the perimeter of the square is 12 cm and the circumference of the ball is 3 cm. How many turns will the ball have rolled, without slipping, as it rolls along the square and base to its original position?



8 Two quadrants are drawn using points A and B as centres respectively. The length of AB is 7 cm. Find the difference between areas x and y.



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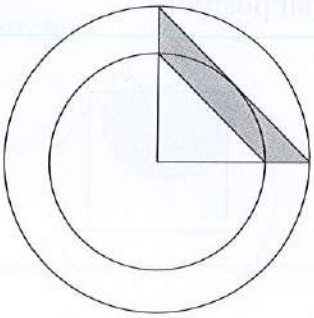
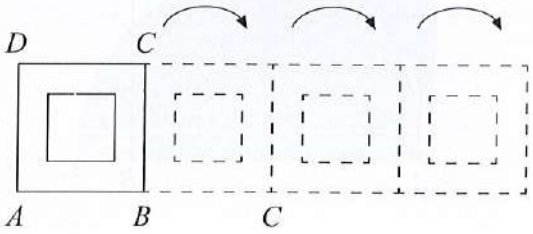
9 The area of the shaded region is 28 cm^2.
Find the area of the circular ring.



10 Square *ABCD* has sides of 8 cm. The side of a hollow part, which is also a square, has sides of 3 cm. The square rotates about its centre along a straight line as shown. Find the distance travelled by the hollow part.



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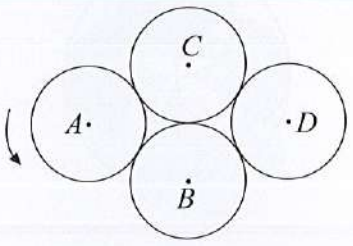
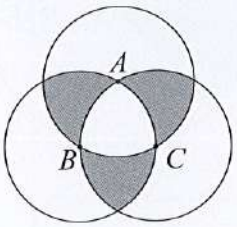
11 Disc A rotates in an anti-clockwise direction around three other identical discs. Find the angle through which disc A has travelled when it returns to the original position without sliding.



12 A, B and C are centres of three identical circles.
Find the area of the shaded region given that the radius is 14 cm.
(Take $\pi = \frac{22}{7}$)



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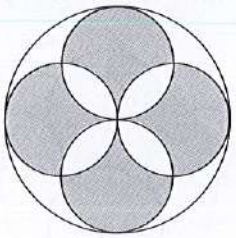
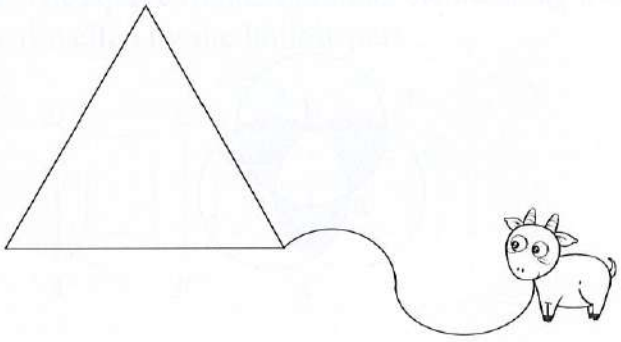
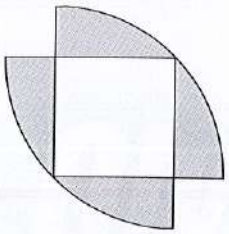
13 Given the radius of the circumscribing circle is 3 cm, find the area of the shaded region.



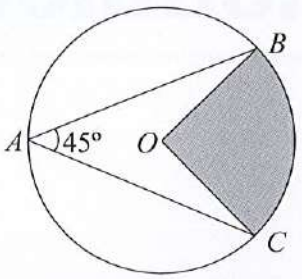
14 A goat is tied using a rope to a triangular fence as shown. Given that each side of the triangle is 17.5 m and the length of the rope is 21 m, find the area of the region in which the goat can move freely.
(Take $\pi = \frac{22}{7}$)



15 Two identical quadrants overlap as shown below. The overlapped region is a square with an area of 25 cm^2. Find the area of the shaded region.
(Take $\pi = 3.14$)



16 Given the radius of the circle is 14 cm, find the area of the fan if "O" is its centre.
(Take $\pi = \frac{22}{7}$)



NOTES

SINGAPORE ASIA education
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Maths Olympiad

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Solutions

Advanced (Pr 5 & 6, suitable for 11-13 years old)

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CHAPTER 1

1. (a) **12, 1122, 111 222, 11 112 222, 1 111 122 222, 111 111 222 222**

(b) **42, 4422, 444 222, 44 442 222, 4 444 422 222, 444 444 222 222**

(c) **45, 5445, 554 445, 55 544 445, 5 555 444 445, 555 554 444 445**

(d) **72, 8712, 887 112, 88 871 112, 8 888 711 112, 888 887 111 112**

2. $12 \div 4 = 3$

$1122 \div 34 = 33$

$111\,222 \div 334 = 333$

$\vdots$

$1\,111\,111\,122\,222\,222 \div 33\,333\,334$

$= \mathbf{33\,333\,333}$

3. $6 \times 7 = 42$

$66 \times 67 = 4422$

$666 \times 667 = 444\,222$

$\vdots$

$\underbrace{666 \ldots 666}_{2008\text{ 6s}} \times \underbrace{666 \ldots 667}_{2007\text{ 6s}}$

$= \underbrace{\mathbf{444 \ldots 444}}_{\mathbf{2008\text{ 4s}}}\underbrace{\mathbf{222 \ldots 222}}_{\mathbf{2008\text{ 2s}}}$

4. $11\,111\,111 \times 99\,999\,999$

$= 11\,111\,111 \times (100\,000\,000 - 1)$

$= 1\,111\,111\,100\,000\,000 - 11\,111\,111$

$= \mathbf{1\,111\,111\,088\,888\,889}$

5. $\underbrace{888 \ldots 888}_{20\text{ 8s}} \times \underbrace{333 \ldots 334}_{19\text{ 3s}} + \underbrace{666 \ldots 666}_{20\text{ 6s}} \times \underbrace{888 \ldots 888}_{20\text{ 8s}}$

$= \underbrace{888 \ldots 888}_{20\text{ 8s}} \times (\underbrace{333 \ldots 334}_{19\text{ 3s}} + \underbrace{666 \ldots 666}_{20\text{ 6s}})$

$= \underbrace{888 \ldots 888}_{20\text{ 8s}} \times \underbrace{1\,000 \ldots 000}_{20\text{ 0s}}$

$= \underbrace{\mathbf{888 \ldots 888}}_{\mathbf{20\text{ 8s}}}\underbrace{\mathbf{000 \ldots 000}}_{\mathbf{20\text{ 0s}}}$

6. $\underbrace{333 \ldots 333}_{2008\text{ 3s}} \times 2 \times \underbrace{444 \ldots 444}_{2008\text{ 4s}} \div \underbrace{666 \ldots 666}_{2008\text{ 6s}}$

$= \underbrace{666 \ldots 666}_{2008\text{ 6s}} \times \underbrace{444 \ldots 444}_{2008\text{ 4s}} \div \underbrace{666 \ldots 666}_{2008\text{ 6s}}$

$= \underbrace{\mathbf{444 \ldots 444}}_{\mathbf{2008\text{ 4s}}}$

7. $(2007\,2007 + 1) \times (2008\,2008 - 1) - 2007\,2007 \times 2008\,2008$

$= 2007\,2007 \times 2008\,2008 + 2008\,2008 - 2007\,2007 - 1 - 2007\,2007 \times 2008\,2008$

$= 2008\,2008 - 2007\,2007 - 1$

$= 10\,001 - 1$

$= \mathbf{10\,000}$

8. $\underbrace{999 \ldots 999}_{2008\text{ 9s}} \times (\underbrace{1\,000 \ldots 000}_{2008\text{ 0s}} - 1) +$

$\underbrace{1\,000 \ldots 000}_{2008\text{ 0s}} + \underbrace{999 \ldots 999}_{2008\text{ 9s}}$

$= \underbrace{999 \ldots 999}_{2008\text{ 9s}}\underbrace{000 \ldots 000}_{2008\text{ 0s}} - \underbrace{999 \ldots 999}_{2008\text{ 9s}} +$

$\underbrace{1\,000 \ldots 000}_{2008\text{ 0s}} + \underbrace{999 \ldots 999}_{2008\text{ 9s}}$

$= \mathbf{1}\underbrace{\mathbf{000 \ldots 000}}_{\mathbf{4016\text{ 0s}}}$

9. Let the two-digit number be mn. It can also be expressed as $10m + n$.

$8(m + n) - (10m + n) = 8$

$8m + 8n - 10m - n = 8$

$7n - 2m = 8$

$7n = 8 + 2m$

$n = \dfrac{8 + 2m}{7}$

When $m = 3$,

$n = \dfrac{8 + 6}{7} = 2$

The two-digit number is **32**.

10. Suppose James was born in the year $1abc$.

$1 + a + b + c = 2008 - (1000 + 100a + 10b + c)$

$= 1008 - 100a - 10b - c$

$1007 = 101a + 11b + 2c$

Take $a = 9$, so that it becomes $19bc$.

$11b + 2c = 1007 - 909$

$= 98$

$b = \dfrac{98 - 2c}{11}$

When $c = 5$,

$b = \dfrac{98 - 2 \times 5}{11} = \dfrac{98 - 10}{11} = \dfrac{88}{11} = 8$

$1 + 9 + 8 + 5 = 23$

as $1985 + 23 = 2008$

James was **23** years old in 2008.

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CHAPTER 2

1. $\square + \square + \varnothing = \bigcirc + \varnothing$

$\square + \square = \bigcirc = \Delta + \Delta + \Delta + \Delta + \Delta + \Delta$

$\bigcirc = \Delta + \Delta + \Delta + \Delta + \Delta + \Delta$

2. Let r and d be the price of a raisin roll and a donut respectively.

$5r + 6d \rightarrow \$14.70$

$1r \rightarrow 3d$

$5r \rightarrow 5 \times 3d = 15d$

$15d + 6d = 21d \rightarrow \$14.70$

$d \rightarrow \$14.70 \div 21 = \$0.70$

$r \rightarrow 3 \times \$0.70 = \$2.10$

A raisin roll is **$2.10**. A donut is **$0.70**.

3. Let T and C be the cost of a table and a chair respectively.

$5T + 12C \rightarrow \$720$

$1T \rightarrow C + \$25$

$5T \rightarrow 5C + \$125$

$5C + \$125 + 12C \rightarrow \$720$

$17C \rightarrow \$720 - \$125 = \$595$

$C \rightarrow \$595 \div 17 = \$35$

$T \rightarrow \$35 + \$25 = \$60$

The cost of a table is **$60**.

4.

US$1 → S$1.45

S$50 → 1000 Thai Baht

S$1 → 20 Thai Baht

S$1.45 → 29 Thai Baht

29 000 Thai Baht → US$1000

29 000 Thai Baht could buy **1000** US dollars.

5. Let f and v be the cost of a football and a volleyball respectively.

$4f + 4v \rightarrow \$240$

$3f + 5v \rightarrow \$234$

$12f + 12v \rightarrow 3 \times \$240 = \$720$

$12f + 20v \rightarrow 4 \times \$234 = \$936$

$8v \rightarrow \$936 - \$720 = \$216$

$v \rightarrow \$216 \div 8 = \$27$

$3f \rightarrow \$234 - 5 \times \$27 = \$99$

$f \rightarrow \$99 \div 3 = \$33$

The cost of a volleyball is **$27**.

The cost of a football is **$33**.

6.

10 rabbits → 2 turkeys

6 turkeys → 2 goats

10 goats → 2 cows

$(5 \div 2) \times 10 = 25$ goats → 5 cows

$(25 \div 2) \times 6 = 75$ turkeys → 25 goats

$(75 \div 2) \times 10 = 375$ rabbits → 75 turkeys

5 cows can be exchanged for **375** rabbits.

7. Let p and a be the cost of a peach and an apple respectively.

$2p + 12a \rightarrow \$14.60$

$6p + 4a \rightarrow \$18.20$

$6p + 36a \rightarrow 3 \times \$14.60 = \$43.80$

$32a \rightarrow \$43.80 - \$18.20 = \$25.60$

$a \rightarrow \$25.60 \div 32 = \$0.80$

$2p \rightarrow \$14.60 - 12 \times \$0.80 = \$5$

$p \rightarrow \$5 \div 2 = \$2.50$

An apple is **$0.80**. A peach is **$2.50**.

8. Let w, b and y be the number of white, blue and yellow balls respectively.

$w + b \rightarrow 160$

$b + y \rightarrow 180$

$w + y \rightarrow 170$

$2w + 2b + 2y \rightarrow 160 + 180 + 170 = 510$

$w + b + y \rightarrow 510 \div 2 = 255$

$y \rightarrow 255 - 160 = 95$

$w \rightarrow 255 - 180 = 75$

$b \rightarrow 255 - 170 = 85$

There are **95** yellow balls, **75** white balls and **85** blue balls.

9. Let f, b and d be the prices of fries, burger and drinks respectively.

$3f + 2b + 1d \rightarrow \$10.95$

$1f + 2b + 3d \rightarrow \$11.25$

$4f + 4b + 4d \rightarrow \$10.95 + \$11.25 = \$22.20$

$f + b + d \rightarrow \$22.20 \div 4 = \$5.55$

The price of a set is **$5.55**.

10. $3A + 7B + 1C \rightarrow \$31.50$

$4A + 10B + 1C \rightarrow \$42$

By comparing,

$1A + 3B \rightarrow \$42 - \$31.50 = \$10.50$

$2A + 6B \rightarrow 2 \times \$10.50 = \$21$

Again by comparing,

$1A + 1B + 1C \rightarrow \$31.50 - \$21$

$= \$10.50$

The price of item A + item B + item C is **$10.50**.

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11. Let c and g be cow and goat respectively.

$5c + 6g \rightarrow 27$

$6c + 5g \rightarrow 28$

A cow feeds on 1 kg more of grass.

Let us assume all to be cows.

$5c + 6c \rightarrow 27 + 6 \times 1 = 33$

$11c \rightarrow 33$

$1c \rightarrow 33 \div 11 = 3$ kg

A cow feeds on **3 kg** of grass each day.

12. Let b and f be the cost of a basketball and a football respectively.

$3b + 5f \rightarrow \$390$

$6b + 3f \rightarrow \$465$

$6b + 10f \rightarrow 2 \times \$390 = \$780$

$7f \rightarrow \$780 - \$465 = \$315$

$f \rightarrow \$315 \div 7 = \$45$

$3b \rightarrow \$390 - 5 \times \$45 = \$165$

$b \rightarrow \$165 \div 3 = \$55$

The cost of a basketball is **\$55** and the cost of a football is **\$45**.

13. Let C and S be Amanda's cycling and swimming speeds respectively.

$3000C + 1000S \rightarrow 30$ min

$2400C + 2000S \rightarrow 48$ min

$6000C + 2000S \rightarrow 2 \times 30$ min $= 60$ min

$3600C \rightarrow 60 - 48 = 12$ min

$C = 3600 \div 12 = 300$ m/min

Amanda could cycle **300 m per minute**.

14. Let p, o and a be the weights of a basket of pears, oranges and apples respectively.

$1p + 2o \rightarrow 13$ kg —— (1)

$2a + 2o \rightarrow 22$ kg —— (2)

$+ \quad 3p + 2a \rightarrow 21$ kg —— (3)

$4a + 4p + 4o \rightarrow 13 + 22 + 21 = 56$ kg

$1a + 1p + 1o \rightarrow 56 \div 4 = 14$ kg

From (2), $1a + 1o \rightarrow 22 \div 2 = 11$ kg

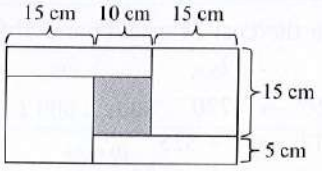
$1p \rightarrow 14 - 11 = 3$ kg

$1o \rightarrow (13 - 3) \div 2 = 5$ kg

$1a \rightarrow (21 - 3 \times 3) \div 2 = 6$ kg

A basket of pears weighs **3 kg**, a basket of oranges weighs **5 kg** and a basket of apples weighs **6 kg**.

15.



10 cm $\times$ 10 cm $= 100$ cm^2

The area of the shaded square is **100 cm²**.

CHAPTER 3

1.

2 1	3 2	3 4
+ 1 2	+ 2 3	+ 4 3
3 3	5 5	7 7
4 5	6 3	7 2
+ 5 4	+ 3 6	+ 2 7
9 9	9 9	9 9

2. **(a)**

$$\begin{array}{r} 578 \\ \times \quad 5 \\ \hline 2890 \end{array}$$

A = **5**
B = **7**
C = **8**

(b)

$$\begin{array}{r} 14 \\ \times 14 \\ \hline 196 \end{array}$$

A = **1**, B = **4**
C = **9**, D = **6**

or

$$\begin{array}{r} 13 \\ \times 13 \\ \hline 169 \end{array}$$

A = **1**, B = **3**
C = **6**, D = **9**

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3. (a)
$$\begin{array}{r} 1\,2\,3 \\ \times\quad 4 \\ \hline 4\,9\,2 \end{array}$$
A = **1**
B = **2**
C = **3**
D = **4**

(b)
$$\begin{array}{r} 2\,5 \\ \times\,2\,5 \\ \hline 6\,2\,5 \end{array}$$
A = **6**
B = **2**
C = **5**

4. (a)
$$\begin{array}{r} 4\,5\,6 \\ \times\quad 4 \\ \hline 1\,8\,2\,4 \end{array}$$
A = **4**
B = **5**
C = **6**
D = **2**

(b)
$$\begin{array}{r} 2\,6 \\ \times\,2\,6 \\ \hline 6\,7\,6 \end{array}$$
A = **2**
B = **6**

5.
$$\begin{array}{r} 9\,4\,8\,0 \\ +\,1\,4\,1\,0 \\ \hline 1\,0\,8\,9\,0 \end{array}$$
A = **0**
B = **8**
C = **4**
D = **9**
F = **1**

6.
$$\begin{array}{r} 1\,2\,3\,4\,5 \\ +\,4\,1\,9\,7\,6 \\ \hline 5\,4\,3\,2\,1 \end{array}$$
A = **1**
B = **2**
C = **3**
D = **4**
E = **5**

7.
$$\begin{array}{r} 3\,5 \\ 1\,3\,5 \\ +\,1\,8\,3\,5 \\ \hline 2\,0\,0\,5 \end{array}$$
A = **1**
B = **8**
C = **3**
D = **5**

8.
$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times\quad\quad 7 \\ \hline 9\,9\,9\,9\,9\,9 \end{array}$$
A = **1**
B = **4**
C = **2**
D = **8**
E = **5**
F = **7**
G = **9**

9.
$$\begin{array}{r} 2\,9\,0 \\ +\,9\,2\,0 \\ \hline 1\,2\,1\,0 \end{array}$$
A = **2**
B = **9**
C = **0**
D = **1**

10.
$$\begin{array}{r} 3\,5 \\ \times\,5\,3 \\ \hline 1\,0\,5 \\ +\,1\,7\,5 \\ \hline 1\,8\,5\,5 \end{array}$$
A = **3**
B = **5**

11.
$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times\quad\quad 3 \\ \hline 4\,2\,8\,5\,7\,1 \end{array}$$
A = **4**
B = **2**
C = **8**
D = **5**
E = **7**

12.
$$\begin{array}{r} 3\,9 \\ \times\,9\,3 \\ \hline 1\,1\,7 \\ +\,3\,5\,1 \\ \hline 3\,6\,2\,7 \end{array}$$
C = **3**
D = **9**

13.
$$\begin{array}{r} 2 \\ 2\,4 \\ 2\,4\,3 \\ +\,2\,4\,3\,6 \\ \hline 2\,7\,0\,5 \end{array}$$
A = **2**
B = **4**
C = **3**
D = **6**

14.
$$\begin{array}{r} 4\,9\,9\,4\,4 \\ +\,4\,9\,9\,4\,4 \\ \hline 9\,9\,8\,8\,8 \end{array}$$
A = **9**
B = **4**
C = **8**

15.
$$\begin{array}{r} 2\,3 \\ 2\,3 \\ 2\,3 \\ +\,2\,3 \\ \hline 9\,2 \end{array}$$
A = **9**
B = **2**
C = **3**

16. Let the original number be $abcd$ and the new number be $dabc$.

$dabc - abcd = 5877$

$1000d + 100a + 10b + c - 1000a - 100b - 10c - d = 5877$

$999d - 900a - 90b - 9c = 5877$

Divide both sides by 9.

$111d - 100a - 10b - c = 653$

When $d = 8$ and $a = 2$,

$688 - 10b - c = 653$

$b = 3$ and $c = 5$

The original combination lock is **2358**.

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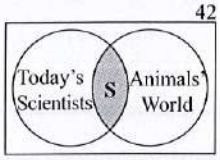
CHAPTER 4

1. $34 + 22 - S = 42$
$56 - S = 42$
$S = 56 - 42$
$= 14$

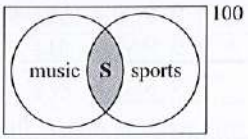


14 students subscribe to both magazines.

2. $8000 + 6000 - 2000 = 12\,000$

The total area of the two pieces of table cloth is **12 000 cm²**.

3. $68 + 75 - S = 100$
$143 - S = 100$
$S = 43$



43 people like both music and sports.

4. $10 \times 10 + 6 \times 6 - 3 \times 3$
$= 100 + 36 - 9$
$= 127$

The area of the figure is **127 cm²**.

5. $100 \div 3 = 33$ R 1
$100 \div 11 = 9$ R 1
$33 + 9 = 42$

Common multiples of 3 and 11: 33, 66, 99.
$42 - 3 = 39$
$100 - 39 = 61$

61 numbers ranging from 1 to 100 are not multiples of 3 and not divisible by 11.

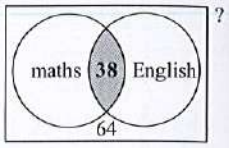
6. $200 \div 5 = 40$
$200 \div 3 = 66$ R 2
$40 + 66 = 106$

Common multiples of 3 and 5: 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180, 195.

$106 - 13 = 93$

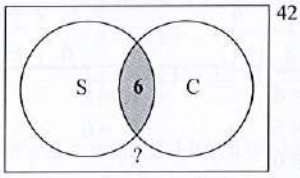
93 numbers ranging from 1 to 200 are multiples of 5 and divisible by 3.

7. $45 + 78 - 38 = 85$
$85 + 64 = 149$



There were **149** Primary 5 students in the cohort.

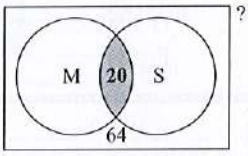
8.



$18 + 14 - 6 = 26$
$42 - 26 = 16$

16 students do not like either activity.

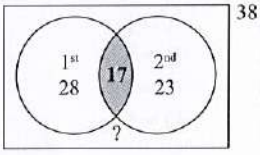
9.



$26 + 32 - 20 = 38$
$38 + 64 = 102$

There were **102** Primary 5 students.

10.



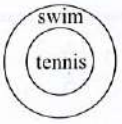
$28 + 23 - 17 = 34$
$38 - 34 = 4$

4 students did not get either question correct.

11. $5050 - 98 - 99 - 100 = \mathbf{4753}$
$4753 - 98 - 99 - 100 = \mathbf{4456}$

The value is **4456**.

12. Scenario 1



56 club members like both sports at the most.

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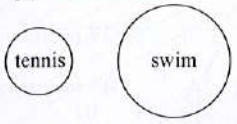
Scenario 2

$56 + 74 = 130$

$130 - 100 = 30$



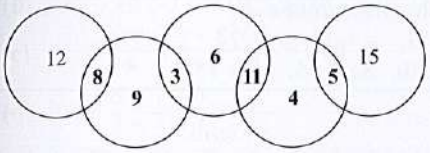
At least **30** of them like both sports.

13. $20 + 26 + 18 - (6 + 8 + 10) = 40$

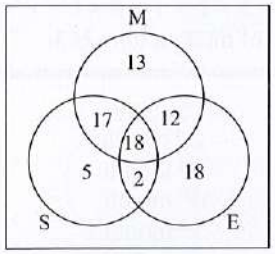
$45 - 40 = 5 \text{ cm}^2$

The area of S_0 is **5 cm²**.

14.



15. Total $(60 + 50 + 42) - (30 + 20 + 35) + 18 = 85$



$13 + 5 + 18 = 36$

36 students have exactly one favourite subject.

16. Numbers that are divisible by 3:

$80 \div 3 = 26$ R 2

Numbers that are divisible by 4: $80 \div 4 = 20$

Numbers that are divisible by 5: $80 \div 5 = 16$

Numbers that are divisible by 3 and 4:

$80 \div 12 = 6$ R 8

Numbers that are divisible by 3 and 5:

$80 \div 15 = 5$ R 5

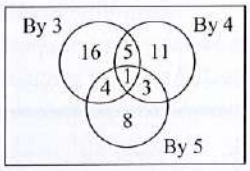
Numbers that are divisible by 4 and 5:

$80 \div 20 = 4$

Numbers that are divisible by 3, 4 and 5:

$80 \div 60 = 1$ R 20

When represented diagrammatically with a Venn diagram, we have



$(26 + 20 + 16) - (6 + 5 + 4) + 1 = 48$

48 numbers from 1 to 80 are divisible by 3, 4 and 5.

CHAPTER 5

1. **(a)** 3, 6, **9**, **15**, 24, **39**, ...

(b) 10, 15, **25**, **40**, 65, **105**, ...

(c) 2, 2, **4**, **6**, **10**, **16**, ...

(d) 3, 5, **8**, **13**, 21, **34**, ...

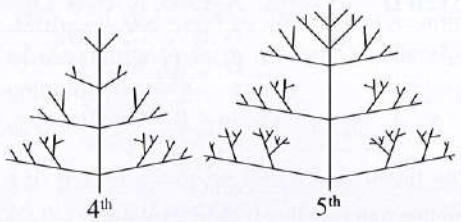
2. **(a)** **4**, **8**, 12, 20, **32**, **52**, ...

(b) **5**, **12**, 17, 29, **46**, **75**, ...

(c) **20**, 20, **40**, 60, **100**, **160**, ...

(d) **17**, 18, **35**, 53, **88**, 141, ...

3.



4. $a, 3, a + 3, a + 6, 2a + 9, 3a + 15, ...$

$3a + 15 = 60$

$3a = 45$

$a = 15$

The value of a is **15**.

5. $3, b, 3 + b, 3 + 2b, 6 + 3b, 9 + 5b, ...$

$9 + 5b = 84$

$5b = 75$

$b = 15$

The value of the sixth term is **15**.

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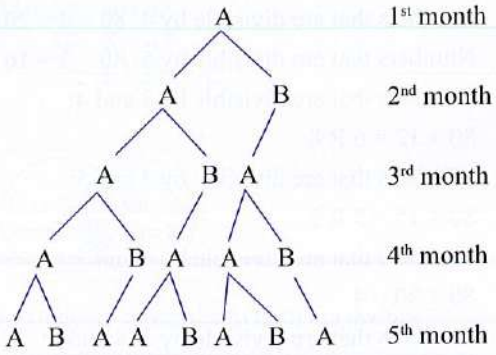
6. Suppose A denotes a pair of adult guinea pigs and B denotes a pair of baby guinea pigs.



We have the Fibonacci numbers as below.

1, 2, 3, 5, 8, 13, 21, 34, 55, 89,144, 233

↑ 8th month ↑ 12th month

There will be 34 pairs of guinea pigs in the **8th** month.
There will be **233** pairs of guinea pigs in a year.

---

7. 1 matchstick: 1 way
2 matchsticks: 2 ways: 1–1 or 2
3 matchsticks: 3 ways: 1–1–1, 2–1, 1–2
4 matchsticks: 5 ways: 1–1–1–1, 2–2, 1–2–1, 1–1–2, 2–1–1
5 matchsticks: 8 ways: 1–1–1–1–1, 1–1–1–2, 1–1–2–1, 1–2–1–1, 2–1–1–1, 1–2–2, 2–2–1, 2–1–2

1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, ...
I can draw the matchsticks in **233** ways.

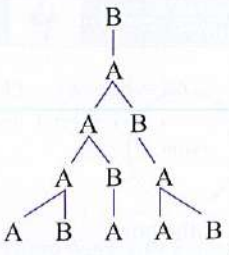
---

8. To go to 1st step: 1 way
To go to 2nd step: 2 ways: 1–1, 2
To go to 3rd step: 3 ways: 1–1–1, 1–2, 2–1
To go to 4th step: 5 ways: 1–1–1–1, 1–1–2, 1–2–1, 2–1–1, 2–2
To go to 5th step: 8 ways: 1–1–1–1–1, 1–1–1–2, 1–1–2–1, 1–2–1–1, 2–1–1–1, 1–2–2, 2–2–1, 2–1–2

1, 2, 3, 5, 8, 13, 21, 34, ...
↑ 8th step

The person can reach the eighth step in **34** ways.

9.



1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89,144, ...
↑ 12th month

There would be **144** pairs of rabbits in 12 months.

---

10. Let the first term be a.
a, 12, $a+12$, $a+24$, $2a+36$, $3a+60$, $5a+96$, $8a+156$
↑ 8th term

$8a + 156 = 180$
$8a = 180 - 156$
$8a = 24$
$a = 3$

The value of the first term is **3**.

11.

a	1st month
b	2nd month
$a+b$	3rd month
$a+2b$	4th month
$2a+3b$	5th month
$3a+5b$	6th month
$5a+8b$	7th month
$8a+13b$	8th month
$13a+21b$	9th month

$13a + 21b = 280$
$21b = 280 - 13a$
$b = \dfrac{280 - 13a}{21}$

$a = 1$, $b \neq$ whole number
$a = 2$, $b \neq$ whole number
⋮
$a = 7$, $b = 9$

The values of a and b were **7** and **9** respectively.

---

12. To cell A: 1 way
To cell B: 2 ways: A–B, B
To cell C: 3 ways: A–C, A–B–C, B–C
To cell D: 5 ways: A–C–D, A–C–B–D, A–B–C–D, B–D, B–C–D, A–B–D
⋮

1, 2, 3, 5, 8, 13, 21, ...
↑ cell G

The bee can reach cell G in **21** ways.

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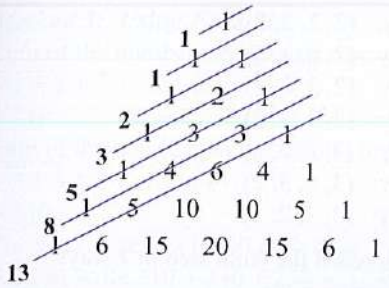
13.



14. Side of A = 16

$16 \times 16 = 256$

The area of A is **256 cm²**.

CHAPTER 6

1. (a) ${}^5P_4 = 5 \times 4 \times 3 \times 2 = \mathbf{120}$

(b) ${}^{18}P_3 = 18 \times 17 \times 16 = \mathbf{4896}$

(c) ${}^{10}C_4 = \dfrac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = \mathbf{210}$

(d) ${}^8C_3 = \dfrac{8 \times 7 \times 6}{3 \times 2 \times 1} = \mathbf{56}$

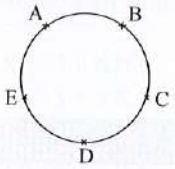
(e) $2 \times {}^5P_2 - {}^3P_2 = 2 \times 5 \times 4 - 3 \times 2$

$= 40 - 6 = \mathbf{34}$

(f) ${}^7C_3 \times {}^6C_2 = \dfrac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \dfrac{6 \times 5}{2 \times 1}$

$= 35 \times 15 = \mathbf{525}$

2.



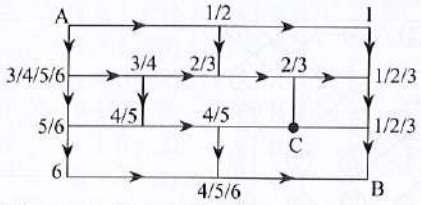
Scenario 1: Using AB as the base, 3 triangles can be formed.

Scenario 2: Using either A or B as one of the vertices, 6 triangles can be formed.

$3 + 6 = 9$

9 triangles can be formed.

3.



There are **6** ways for the worker to reach site B.

4. *Analysis:* We need to choose three points out of ten points to form a triangle. Therefore, the combination is ${}^{10}C_3$.

${}^{10}C_3 = \dfrac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$

120 triangles can be formed by using any three points as their vertices.

5. There are 4 ways to colour region A, 3 ways to colour region B, 2 ways each to colour region C, D and E.

$4 \times 3 \times 2 \times 2 \times 2 = 96$

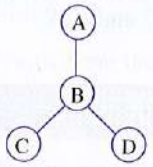
Each region of the figure can be coloured using 4 different colours in **96** ways.

6. *Analysis:* We need 2 lines to form an angle. There are 12 lines including Line A and Line B.

${}^{12}C_2 = \dfrac{12 \times 11}{2 \times 1} = \dfrac{132}{2} = 66$

66 acute angles can be formed from the figure if $\angle BOA$ is less than 90°.

7.



There are 3 ways to colour A, 2 ways to colour B, 2 ways to colour C and 1 way to colour D.

$3 \times 2 \times 2 \times 1 = 12$

There are **12** ways to colour the circles using 3 different colours.

8. Scenario 1: When A and C are of the same colour, there are

3 ways to colour A and C,
2 ways to colour B,
2 ways to colour D.

$3 \times 2 \times 2 = 12$

Scenario 2: When A and C are of different colours, there are

3 ways to colour A,
2 ways to colour B,
1 way to colour C,
1 way to colour D.

$3 \times 2 \times 1 \times 1 = 6$

$12 + 6 = 18$

There are **18** ways to colour the four circles if no two circles connected by a line can have the same colour.

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9. ${}^6C_2 \times {}^8C_1 + {}^8C_2 \times {}^6C_1$

$= \frac{6 \times 5}{2 \times 1} \times 8 + \frac{8 \times 7}{2 \times 1} \times 6$

$= 120 + 168$

$= 288$

288 triangles can be formed using any of these 3 points as their vertices.

10. ***Method 1:***

Step 1: There are 3 odd digits that we can choose to place in the ones place, therefore ${}^3C_1 = 3$.

Step 2: We choose from 4 remaining digits to be placed in the hundreds and tens places. ${}^4P_2 = 4 \times 3 = 12$.

$3 \times 12 = 36$

Method 2:

Step 1: □ □ □

3 ways to fill the ones place

Step 2: □ □ □

3 ways to fill the tens place

4 ways to fill the hundreds place

$4 \times 3 \times 3 = 36$

36 three-digit odd numbers can be formed using the digits 1, 3, 6, 7 and 8.

11. 1st way: (2, 2, 2, 2, 2)
2nd way: (2, 2, 3, 3)
3rd way: (2, 3, 2, 3)
4th way: (2, 3, 3, 2)
5th way: (3, 2, 2, 3)
6th way: (3, 2, 3, 2)
7th way: (3, 3, 2, 2)

He can reach the tenth step in **7** ways.

12. Let us first look at the possible combinations, and the number of ways to arrange each combination.

Possible combinations	Number of ways to arrange
1 + 1 + 1 + 7 = 10	4
1 + 1 + 2 + 6 = 10	12
1 + 1 + 3 + 5 = 10	12
1 + 1 + 4 + 4 = 10	6
1 + 2 + 2 + 5 = 10	12
1 + 2 + 3 + 4 = 10	24
2 + 2 + 2 + 4 = 10	4
2 + 2 + 3 + 3 = 10	6

$4 + 12 + 12 + 6 + 12 + 24 + 4 + 6 = 80$

Mrs Reed has **80** times at the most to try to open her safe.

CHAPTER 7

1. *Analysis:* It suffices to look at the first number of each row.

1 3 9 19 33, …

2 6 10 14

We notice that the difference between two numbers increases by 4 each time.

1 3 9 19 33 51 73 99

2 6 10 14 18 22 26

129 163 201 243 289 339 393

30 34 38 42 46 50 54

The first number in the 15th row is **393**.

2. ***Method 1: Inductive Reasoning***

If we look at a small sample,

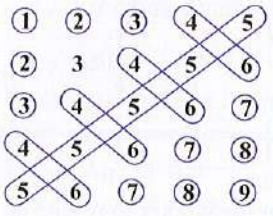
For 1 2
2 3 the sum is $1 + 2 + 2 + 3 = 8$.

For 1 2 3
2 3 4
3 4 5 the sum is $1 + 2 + 3 + 2 + 3 + 4 + 3 + 4 + 5 = 27$.

We can make a conjecture that
the sum of a 2×2 pattern is $2 \times 2 \times 2 = 8$,
the sum of a 3×3 pattern is $3 \times 3 \times 3 = 27$,
the sum of a 4×4 pattern is $4 \times 4 \times 4 = 64$, and so on.

$20 \times 20 \times 20 = 8000$

Method 2: Deduction



We realise there is one pair of (1 + 9), 2 pairs of (2 + 8), 3 pairs of (3 + 7), …

There are 10 pairs of 10.

$10 \times 10 + 5 \times 5 = 125 = 5 \times 5 \times 5$

$20 \times 20 \times 20 = 8000$

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Method 3: Using Formula

Sum of the numbers in the first row
$= 1 + 2 + 3 + \ldots + 20$
$= 210$

Sum of the numbers in the second row
$= 2 + 3 + 2 + \ldots + 21$
$= 230$

The sum increases by 20 each time.

We can write $210 + 230 + 250 + \ldots + 590$

$$(210 + 590) \times 10 = 8000$$

The sum of the number pattern is **8000**.

3. *Analysis:* Each group has 8 numbers.

We can use the concept of remainder to find the position of 2008.

$$8 \times 2 = 16$$

A	B	C	D	E
	R8	R6	R4	R2
R10	R12	R14	R0	

$$2008 \div 16 = 125 \text{ R } 8$$
$$125 \times 2 = 250$$
$$250 + 1 = 251$$

Number 2008 appears in **Row 251** and **Column B**.

4. *Analysis:* We make a list to find out how the pattern evolves.

$(1 + 9 + 8) \div 3 = 6$ R 0
$(9 + 8 + 0) \div 3 = 5$ R 2
$(8 + 0 + 2) \div 3 = 3$ R 1
$(0 + 2 + 1) \div 3 = 1$ R 0
$(2 + 1 + 0) \div 3 = 1$ R 0
$(1 + 0 + 0) \div 3 = 0$ R 1
$(0 + 0 + 1) \div 3 = 0$ R 1
$(0 + 1 + 1) \div 3 = 0$ R 2
$(1 + 1 + 2) \div 3 = 1$ R 1
$(1 + 2 + 1) \div 3 = 1$ R 1
$(2 + 1 + 1) \div 3 = 1$ R 1
$(1 + 1 + 1) \div 3 = 1$ R 0
$(1 + 1 + 0) \div 3 = 0$ R 2
$(1 + 0 + 2) \div 3 = 1$ R 0
$(0 + 2 + 0) \div 3 = 0$ R 2
$(2 + 0 + 2) \div 3 = 1$ R 1
$(0 + 2 + 1) \div 3 = 1$ R 0

The pattern recurs every 13 numbers.

1, 9, 8, 0, 2, 1, 0, 0, 1, 1, 2,
1, 1, 1, 0, 2, 0, 2, 1, 0, 0, 1, 1, 2

↑ ↑ ↑
R1 R2 R3

The first 3 numbers are not part of the recurrence.

$$2008 - 3 = 2005$$
$$2005 \div 13 = 154 \text{ R } 3$$

The 2008th number is **1**.

5. *Analysis:* It becomes laborious if we attempt to find the 2008th number.

Instead, we observe how the pattern of the remainders develop.

numbers	1	2	4	7	11	16	22	29	37	46	56
remainders	1	2	4	2	1	1	2	4	2	1	1

The remainders recur at

1, 2, 4, 2, 1, 1, 2, 4, 2, 1

↑ ↑ ↑ ↑ ↑
R1 R2 R3 R4 R0

It suffices to find the remainder when 2008 is divided by 5.

$$2008 \div 5 = 401 \text{ R } 3$$
$$\text{R3} \rightarrow 4$$

The remainder is **4** when the 2008th number in the pattern is divided by 5.

6. Transform the pattern.

Clearly,
1st line → 1 number
2nd line → 2 numbers
⋮

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

Make a guess.

$$\frac{(1 + 63) \times 63}{2} = 2016$$

Count backwards from the RHS.

∴ Column 9 or 2016 is the last number of row 63.

7.

1st → 2
2nd → 3 } 2
3rd → 5 } 2
4th → 7 } 3
5th → 10 } 3
6th → 13 } 4
7th → 17

8th → 21 } 4
9th → 26 } 5
10th → 31 } 5
11th → 37 } 6
12th → 43 } 6
13th → 50 } 7
14th → 57 } 7

The pattern is shown above.

15th : $57 + 8 = 65$
16th : $65 + 8 = 73$
17th : $73 + 9 = 82$
18th : $82 + 9 = 91$
19th : $91 + 10 = 101$
20th : $101 + 10 = 111$
21st : $111 + 11 = 122$
22nd : $122 + 11 = 133$

The number at 22nd corner is **133**.

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8. *Analysis:* We simplify the problem to 3 columns of households.

$2008 - 10 = 1998$

$1998 \div 3 = 666$ households each column

$666 + 4 - 1 = 669$ at the second corner

672 is at the third corner.

The house marked with "A" is **673**.

9. *Analysis:* If we assume the number at the centre of a cross to be n, the rest of the numbers are:

$n - 8, n - 1, n + 1, n + 8$

The sum of the five numbers is 625.

$n + n - 8 + n - 1 + n + 1 + n + 8 = 625$

$5n = 625$

$n = 125$

The five numbers are **117**, **124**, **125**, **126**, **133**.

10. *Analysis:* Suppose the first number of the 9 numbers is m, we will have

m	$m + 1$	$m + 2$
$m + 7$	$m + 8$	$m + 9$
$m + 14$	$m + 15$	$m + 16$

where the sum must be 12 573.

$m + m + 1 + m + 2 + m + 7 + m + 8 + m + 9 + m + 14 + m + 15 + m + 16 = 12\,573$

$9m + 72 = 12\,573$

$9m = 12\,501$

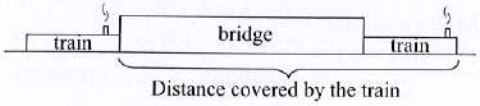
$m = 1389$

The 3-by-3 number block is

1389	**1390**	**1391**
1396	**1397**	**1398**
1403	**1404**	**1405**.

CHAPTER 8

1. *Analysis:* The distance covered is the length of the train + the length of the bridge.



$1625 - 1000 = 625$ m

$75 - 50 = 25$ s

$625 \div 25 = 25$ m/s

It was travelling at **25 m/s**.

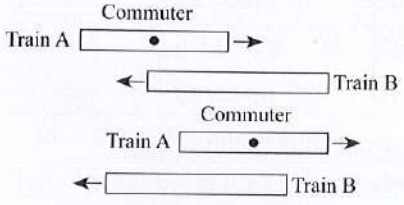
$25 \times 50 - 1000 = 250$ m

or

$25 \times 75 - 1625 = 250$ m

The train was **250 m** long.

2. *Analysis:* Think of the combined speed.



Speed of Train A in m/s

$= 54 \times 1000 \div 3600 = 15$ m/s

Speed of Train B in m/s

$= 72 \times 1000 \div 3600 = 20$ m/s

Combined speed $= 15 + 20 = 35$ m/s

$35 \times 8 = 280$ m

Train B was **280 m**.

3. ***Method 1: Write Equations***

Let d be the distance.

Total time taken $= \frac{d}{108} + \frac{d}{72}$

$= \frac{2d + 3d}{216}$

$= \frac{5d}{216}$

Total distance covered $= d + d = 2d$

Average speed $= \frac{\text{distance}}{\text{time}}$

$= 2d \div \frac{5d}{216}$

$= 2\cancel{d} \times \frac{216}{5\cancel{d}}$

$= 86\frac{2}{5}$ km/h

Method 2: Solve by Assuming

Assume the distance to be 108 km.

Total time taken $= \frac{108}{108} + \frac{108}{72}$

$= 1 + 1\frac{1}{2}$

$= 2\frac{1}{2}$ h

Total distance covered $= 108 \times 2$

$= 216$ km

Average speed $= 216 \div 2\frac{1}{2}$

$= 216 \times \frac{2}{5}$

$= 86\frac{2}{5}$ km/h

His average speed for the round-trip is **$86\frac{2}{5}$ km/h**.

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4. ***Method 1: Write Equations***

Suppose he needs m hours to reach there, the distance becomes $90m$.

The distance is also $120(m-2)$.

$90m = 120(m-2)$

$90m = 120m - 240$

$30m = 240$

$m = 8$ h

$90 \times 8 = 720$ km or $120 \times (8-2) = 720$ km

$720 \div 7.2 = 100$ km/h

Method 2: Solve by Reasoning

At 2 pm, he still has 180 km to travel as

$90 \times 2 = 180$ km.

Difference in speed $= 120 - 90 = 30$ km/h

$180 \div 30 = 6$ h (at speed of 120 km/h)

$6 + 2 = 8$ h (at speed of 90 km/h)

$6 \times 120 = 720$ km or $8 \times 90 = 720$ km

$720 \div 7.2 = 100$ km/h

Mr Cartland should be driving at a speed of **100 km/h** if he wants to arrive at 3.12 pm.

5. Let the speed of the escalator be n steps per second.

$20 \times 3 + 20n = 25 \times 2 + 25n$

$60 + 20n = 50 + 25n$

$5n = 10$

$n = 2$

$20 \times 3 + 20 \times 2 = 100$

or

$25 \times 2 \times 25 \times 2 = 100$

The escalator had **100** steps when it was stationary.

6. If he had taken a ride from the beginning,

$25 \times 2.5 = 62.5$ km

$62.5 - 40 = 22.5$

he could have saved a walking distance of 22.5 km.

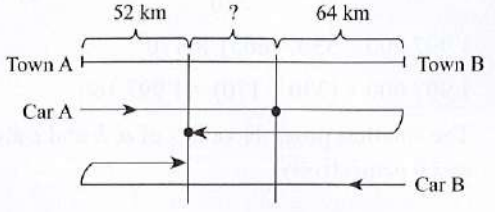
$22.5 \div (25 - 5) = 1.125$ h

1 h $\rightarrow$ 5 km

1.125 h $\rightarrow 5 \times 1.125 = 5.625$ km

The backpacker had walked **5.625 km** before he took a ride.

7. *Analysis:* The total distance covered by the two cars is 3 times the distance between Town A and Town B.



Distance covered by Car B $= 64 \times 3$

$= 192$ km

Distance between Towns A and B

$= 192 - 52 = 140$ km

$140 - 64 - 52 = 24$ km

The distance between the two points where they passed each other was **24 km**.

8. When Barton was 90 m, Dan was 85 m.

Their ratio is $85 : 90 = 17 : 18$.

$17 : 18 = n : 100$

$\frac{17}{18} = \frac{n}{100}$

Cross multiply,

$n = \frac{1700}{18}$

$n = 94\frac{4}{9}$ m

$100 - 94\frac{4}{9} = 5\frac{5}{9}$ m

Dan was $\mathbf{5\frac{5}{9}}$ **m** away from the finishing line when Barton completed the race..

9. ***Method 1: Solve by Reasoning***

Assume each side of the triangle to be 300 cm.

Total time taken $= \frac{300}{30} + \frac{300}{50} + \frac{300}{60}$

$= 10 + 6 + 5$

$= 21$ min

Total distance $= 3 \times 300 = 900$ cm

Average speed $= \frac{900}{21} = 42\frac{6}{7}$ cm/min

Method 2: Write Equations

Let the length of each side of the triangle be l.

Total time taken $= \frac{l}{30} + \frac{l}{50} + \frac{l}{60}$

$= \frac{10l + 6l + 5l}{300}$

$= \frac{21l}{300}$

Total distance covered $= 3l$

Average speed $= 3l \div \frac{21l}{300}$

$= 3l \times \frac{300}{21l}$

$= 42$ cm/min

The average speed for the beetle to complete the whole journey is $\mathbf{42\frac{6}{7}}$ **cm/min**.

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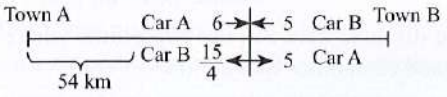
10. Car A's reduced speed $= 6 \times (1 - \frac{1}{6})$

$= 5$ units

Car B's reduced speed $= 5 \times (1 - \frac{1}{4})$

$= \frac{15}{4}$ units



Car A travelled $5 \div 5 = 1$ h to Town B from where they met.

$\frac{15}{4} \times 1 \text{ h} = \frac{15}{4}$

1 unit of the journey $= 54 \div (6 - \frac{15}{4})$

$= 54 \times \frac{4}{9}$

$= 24$ km

Distance between Town A and B

$= 24 \times (6 + 5)$

$= 264$ km

The distance between Town A and Town B is **264 km**.

11. The combined speeds $= 6 + 5 = 11$ m/s

$\frac{400}{11} = 36\frac{4}{11}$

They will meet each other every $36\frac{4}{11}$ s.

$15 \times 60 = 900$ s

$900 \div \frac{400}{11} = 900 \times \frac{400}{11} = \frac{99}{4} = 24\frac{3}{4}$

They pass each other **24** times in 15 minutes.

12. Time ratio $= 6 : 5$

Speed ratio $= 5 : 6$

$15 \div (\frac{6}{5} - 1) = 75$ km/h at first.

If the speed is reduced by 15 km/h,

$\frac{60}{75} = \frac{4}{5}$

Time ratio $= 5 : 4$

105 min $= 1.75$ h

$1.75 \div (\frac{5}{4} - 1) = 7$ h

$75 \times 7 = 525$ km

The distance travelled by the car is **525 km**.

CHAPTER 9

1. For 9 | 1 7 3 a, $a = 7$

For 11 | 1 7 3 b, $b = 8$

For 6 | 1 7 3 c, $c = 4$

$7 + 8 + 4 = 19$

The value of $a + b + c$ is **19**.

2. *Analysis:* If 1997 is divisible by 23 and 19, then it must be divisible by the product of 23 and 19.

Solution:

It suffices to find a multiple of 437 that is more than 1997.

$23 \times 19 = 437$

$437 \times 5 = 2185$

$2185 - 1997 = 188$

The smallest possible value of this number is **188**.

3. *Analysis:* Since 72 | □ 71.5 and $72 = 8 \times 9$, we can have 8 | □ 71.5 and 9 | □ 71.5 □.

Solution:

For a number to be divisible by 8, the last 3 digits must be divisible by 8.

8 | 15 □, □ = 2

□ + 7 + 1 + 5 + 2 = 15 + □

9 | 15 + □, □ = 3

The correct amount of money he had collected was **$371.52**.

4. 2860 = 2 × 1430; 1430 = 2 × 715; 715 = 5 × 143; 143 = 11 × 13

We are looking for pairs of prime factors.

$2860 \times m = 2 \times 2 \times 5 \times 11 \times 13 \times m$

$m = 5 \times 11 \times 13 = 715$

The smallest possible value of m is **715**.

5. 210 = 2 × 105; 105 = 7 × 15; 15 = 5 × 3

$210 = 2 \times 3 \times 5 \times 7 = 6 \times 5 \times 7$

$6 + 5 + 7 = 18$

The age of each boy is **5**, **6** and **7**.

6. Common multiple $= 2 \times 3 \times 5 \times 11$

$= 330$

$1\,997\,000 \div 330 = 6051$ R 170

$1\,997\,000 + (330 - 170) = 1\,997\,160$

The smallest possible values of a, b and c are **1**, **6** and **0** respectively.

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7. *Analysis:* First, we find all the multiples of 2, 5, 7 up to 2008 or smaller. Next, we find the common multiples of two of the numbers. Lastly, we find the common multiple of the three numbers.

Solution:

Number of multiples of 2 = 2008 ÷ 2
= 1004

Number of multiples of 5 = 2008 ÷ 5
= 401 R 3

Number of multiples of 7 = 2008 ÷ 7
= 286 R 6

Number of common multiples of
2 and 5 = 2008 ÷ (2 × 5) = 200 R 8
2 and 7 = 2008 ÷ (2 × 7) = 143 R 6
5 and 7 = 2008 ÷ (5 × 7) = 57 R 13

Number of common multiples of 2, 5 and 7 = 2008 ÷ (2 × 5 × 7) = 28 R 48

Think of Venn Diagram,
1004 + 401 + 286 − (200 + 143 + 57) + 28
= 1319

1319 numbers from 1 to 2008 are divisible by either 2, 5 or 7.

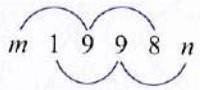
8. *Analysis:* We work on the basis properties that govern the divisibility of 9 and 11.

Solution:

For divisibility by 9,
$m + 1 + 9 + 9 + 8 + n$
$= 27 + m + n$ is divisible by 9.

That leads us to three possibilities,
$m + n = 0$, $m + n = 9$, $m + n = 18$ —— (1)

For divisibility by 11,



$m + 9 + 8 = m + 17$
$n + 9 + 1 = n + 10$

The difference of $(m + 17)$ and $(n + 10)$ must be 0 or 11.

Suppose the difference is 0,
$m + 17 = n + 10$
$n - m = 7$ —— (2)

Only $m + n = 9$ from (1) satisfies $n - m = 7$.

When $m = 1$ and $n = 8$,
(1) → $1 + 8 = 9$
(2) → $8 - 1 = 7$

The values of m and n are **1** and **8** respectively.

9. After 1st round: 6, 12, 18, 24, 30, 36, 42, ...

After 2nd round: 36, 72, 108, 144, 180, 216, ...

After 3rd round: 216, 432, 648, 864, 1080, 1296, 1512, 1728, 1944

The original number of the 3rd electronic part from the left is **648**.

10. For this number to be minimum, we begin with 1500*cd*.

Condition 1: *cd* must be divisible by 4.
$1 + 5 + 0 + 0 + c + d = 6 + c + d$

Condition 2: $(6 + c + d)$ is a multiple of 9, or $c + d = 12$.

$c = 4$, $d = 8$

The number is **150048**.

11. In Prime Factorisation from Intermediate, we learn
$1001 = 7 \times 11 \times 13$.
Now, there are 1004 pairs of $7ab7ab$ when $7ab$ is written 2008 times.

$7ab \times 1001 = 7ab \times 11 \times 91$

Therefore, for any value of a or b, this condition will always be satisfied, and it will be a multiple of 91.

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CHAPTER 10

1. Exchange the red marbles in the first group with the blue marbles in the second group.

1st	R	R	R
2nd	B	B	B
3rd	R	R	B

The fraction of blue marbles in all the three groups of marbles is $\frac{4}{9}$.

2. *Analysis:* Make a table to allow us to work backwards.

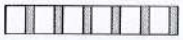
Solution:

$156 \div 3 = 52$

A	B	C
52	52	52
24	24	108
10	94	52
87	45	24

A got **87** marbles, B got **45** marbles and C got **24** marbles at first.

3. Jonathan

Samuel 

$6 \times \$48 = \$288$

$\$288 + \$712 = \$1000$

$\$1000 \div 10 = \$100$

$4 \times \$100 = \$400$

$\$100 - \$48 = \$52$

$6 \times \$52 = \$312$

Jonathan and Samuel saved **\$400** and **\$312** respectively.

4. *Analysis:* For each number, it suffices to count the number of small numbers from its right and move accordingly.

	10	3	7	4	6	5	8	9	1	2
Number of times to move	9	2	5	2	3	2	2	2	0	0

$9 + 2 + 5 + 2 + 3 + 2 + 2 + 2 + 0 + 0 = 27$

I will need to adjust the numbers **27** times.

5. Every 2 hours, it would come across one coach.

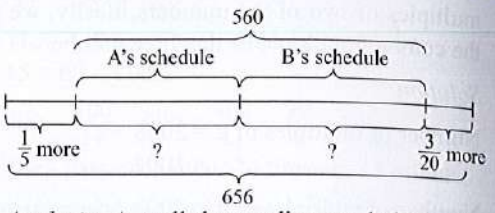
$48 \div 2 = 24$

There would be one coach just before setting off.

$24 + 1 = 25$

An eastbound coach would come across **25** coaches before its arrival at the destination.

6.



Analysis: A well-drawn diagram helps us to see the mathematical problem better.

Solution:

Take 560 spare parts as '1 whole'.

$(1 + \frac{3}{20}) \times 560 = 644$

$656 - 644 = 12$

$\frac{1}{5} - \frac{3}{20} = 4 - \frac{3}{20} = \frac{1}{20}$

$1\text{ u} = 12$

$20\text{ u} = 12 \times 20 = 240$

$560 - 240 = 320$

Machines A and B were supposed to produce **240** and **320** spare parts at first respectively.

7.

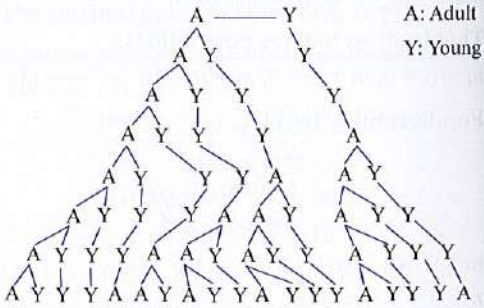
	Win	Loss
C scored 3 goals:	1 against team A	
	2 against team B	
A scored 6 goals:	2 against team B	
	4 against team C	
B scored 2 goals:	both team C	

Results: **A vs B = 2 : 0**

B vs C = 2 : 2

A vs C = 4 : 1

8.



There will be **19** cows in the eighth year.

9. *Analysis:* We work backwards with an even number 8.

$\boxed{2} + \boxed{2} + \boxed{2} + 2 = 8$

8

$\boxed{2} + \boxed{2} + \boxed{2} + 2 = 14$

14

$\boxed{2} + \boxed{2} + \boxed{2} + 2 = 23$

Jeremy had **23** marbles at first.

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1. Number of small triangles in
Figure 1: $1 = 1 \times 1$
Figure 2: $4 = 2 \times 2$
Figure 3: $9 = 3 \times 3$
Figure 4: $16 = 4 \times 4$
⋮
Figure 8: $64 = 8 \times 8$
Figure 8 has **64** small triangles.

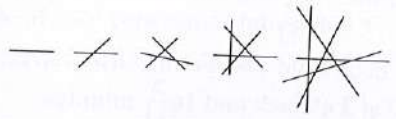
2. $9 \times 9 = 81$ → 1 odd digit
$99 \times 99 = 9801$ → 2 odd digits
$999 \times 999 = 998\,001$ → 3 odd digits
⋮
There are **2008** odd digits in the product of $\underbrace{999...999}_{\text{2008 9s}}$ and $\underbrace{999...999}_{\text{2008 9s}}$.

3. $1^3 + 2^3 = 1 + 8 = 9$
$= (1 + 2)^2$
$1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36$
$= (1 + 2 + 3)^2$
$1^3 + 2^3 + 3^3 + 4^3 = 1 + 8 + 27 + 64$
$= (1 + 2 + 3 + 4)^2$
⋮
$A = (1 + 2 + 3 + ... + 11 + 12)^2$
$= 78^2$
$= 6084$
A is **6084**.

4. *Analysis:* From observation,
(a) the denominator of each term is the term number for the row;
(b) numerator + denominator − 1
= number of terms in the row

Solution:
$2008 + 1998 - 1 = 4005$
$\frac{2008}{1998}$ is in **row 4005**.
The term number from the left is **1998**.

5. ***Method 1:***
Analysis: Make an observation and record the finding.



Solution:
We attempt to establish a relationship between number of lines and number of intersections.
1 line: 0 intersection
2 lines: 1 intersection $= 0 + 1$
3 lines: 3 intersections $= 0 + 1 + 2$
4 lines: 6 intersections $= 0 + 1 + 2 + 3$
5 lines: 10 intersections $= 0 + 1 + 2 + 3 + 4$
⋮
20 lines $= 0 + 1 + 2 + 3 + 4 + ... + 19$
$= (1 + 19) \times 19 \div 2$
$= 190$ intersections

Method 2:
We need 2 lines to form an intersection.
$^{20}C_2 = \frac{20 \times 19}{2 \times 1}$
$= 190$ intersections
These lines can form **190** intersections at the most.

6.

Number of rectangles	1	2	3	4	5
Number of regions	2	10	26	50	82
Increase in number of regions	0	8	16	24	32

Five rectangles can divide a plane into **82** regions.

7. *Analysis:* $1 + 2 + 3 + ... + 62 = 1953$
So the last term is $\frac{62}{62}$.

Solution:
$A = 1 + 1.5 + 2 + 2.5 + 3.0 + ... + \frac{1953}{62}$
$= (1 + 31.5) \times 62 \div 2$
$= 1007.5$
The sum of the first 1953 terms of the series is **1007.5**.

8. The law that governs this sequence: From the 4th term onwards, the 4th term is the sum of the 1st and 2nd terms.
The 5th term is the sum of the 2nd and 3rd terms, ...

1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, 16, 21, 28,
37, 49, 65, 86, 114, 151, ...

The 16th term is **49**. The 20th term is **151**.

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9. *Analysis:* Every denominator is $n^2 + 1$, where n is the term number.
The numerator is 1 throughout the series.
Solution:
$1^2 + 1 = 2$
$2^2 + 1 = 5$
$3^2 + 1 = 10$
$4^2 + 1 = 17$
⋮

$$1008^{\text{th}} \text{ term} = \frac{1}{1008 \times 1008 + 1}$$
$$= \frac{1}{1\,016\,065}$$

10. *Analysis:*
(a) The 1st figure has 10 sides, 2nd figure has 16 sides, 3rd figure has 22 sides, and so on.
Each side is $30 \div 10 = 3$ cm.
(b) The number of squares in each figure is a square number.

Number of squares	4	9	16	25	36	49	64
Number of sides of the figure	10	16	22	28	34	40	46
Perimeter	30	48	66	84	102	120	138

The perimeter of the figure formed by 64 squares is **138 cm**.

11. (a) From the 2nd pattern onwards, we have
$2 \times 2 - 1 = 3$, or $2^2 - 1^2 = 4 - 1 = 3$
$2 \times 3 - 1 = 5$, or $3^2 - 2^2 = 9 - 4 = 5$
$2 \times 4 - 1 = 7$, or $4^2 - 3^2 = 16 - 9 = 7$
⋮
The pattern is $\mathbf{2n - 1 = n^2 - (n-1)^2}$.

(b) $(2 \times 2008 - 1) + (2 \times 2006 - 1)$
$+ (2 \times 2004 - 1) + \ldots + (2 \times 2 - 1)$
$= 4015 + 4011 + 4007 + \ldots + 3$
We have a sequence whose common difference is 4.
number of terms $= (4015 - 3) \div 4 + 1$
$= 1003 + 1 = 1004$
Sum of the pattern
$= (4015 + 3) \times 1004 \div 2 = \mathbf{2\,017\,036}$

12. If one side of the equilateral triangle has $(n - 1)$ coins, then one side of the square will have $(n - 7)$ coins.
$3(n - 1) = 4(n - 7)$
$3n - 3 = 4n - 28$
$n = 25$
$3n - 3 = 3 \times 25 - 3$
$= 72$
There are **72** fifty-cent coins altogether.

CHAPTER 12

1. When the minute hand has travelled 90°,
$= 120° + 30° \times \frac{15}{60}$
$= 127.5°$
the hour hand has travelled 127.5°.
$127.5° - 90° = 37.5°$
The angle formed is **37.5°**.

2. The hour hand has travelled
$= 270° + 30° \times \frac{10}{60}$
$= 275°$
The minute hand has travelled
$= 360° \times \frac{10}{60}$
$= 60°$
$275° - 60° = 215°$
The angle formed is **215°**.

3. Angle formed by the hour hand and an imaginary hand pointing at 12
$= 210° + 30° \times \frac{15}{60}$
$= 217.5°$
Angle formed by the minute hand and an imaginary hand pointing at 12
$= 360° \times \frac{15}{60}$
$= 90°$
$217.5° - 90° = 127.5°$
The angle formed is **127.5°**.

4. *Analysis:* When the minute hand completes one cycle, the hour hand completes only $\frac{1}{12}$ of a cycle.
Difference in speed $= 1 - \frac{1}{12} = \frac{11}{12}$
Distance to catch up = 15 out of 60 scales
$15 \div \frac{11}{12} = 15 \times \frac{12}{11}$
$= 16\frac{4}{11}$ min
The hour hand will lie directly over the minute hand at **3 o'clock and $16\frac{4}{11}$ minutes**.

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5. *Analysis:* The clock will show 10 o'clock after the minute hand has completed 12 cycles.

$2008 \div 12 = 167$ R 4

10.00 pm + 4 hours = 2.00 am

The clock will show **2.00 am**.

6. *Analysis:* When the minute hand completes one cycle, the hour hand completes $\frac{1}{12}$ of a cycle.

Difference in speed $= 1 - \frac{1}{12}$

$= \frac{11}{12}$

1st scenario is when the minute hand is ahead.
Distance to catch up = 15 – 5 = 10 scales

Time taken $= 10 \div \frac{11}{12} = 10 \times \frac{12}{11}$

$= 10\frac{10}{11}$ min

2nd scenario is when the hour hand is ahead.
Distance to catch up = 40 scales

$40 \div \frac{11}{12} = 40 \times \frac{12}{11} = 43\frac{7}{11}$

The right angles are formed at **11 o'clock and $10\frac{10}{11}$ minutes** and at **11 o'clock and $43\frac{7}{11}$ minutes**.

7. *Analysis:* In an hour, the minute hand travels 360° and the hour hand travels 30°.
In a minute, the minute hand travels 6° and the hour hand travels 0.5°.

At 5 o'clock, the angle formed is $= 30° \times 5 = 150°$
Difference in speed per min = 6 – 0.5

$5.5° \times 10 = 55°$

$150° - 55° = 95°$

The angle formed is **95°**.

8. *Analysis:* minute hand to travel 1 h → 360°

1 min → 6°

hour hand to travel 1 h → 30°

1 min → 0.5°

Difference in speed per minute = 6° – 0.5°

= 5.5°

$\frac{1}{x} = \frac{5.5}{210}$ if x is the time required by minute hand to travel 210°

$x = 38\frac{2}{11}$

The hour hand and the minute hand make an angle of 150° between 2 o'clock and 3 o'clock at **2 o'clock and $38\frac{2}{11}$ minute**.

9. The actual time to travel 12 h 10 min = 730 min
His watch travelled 720 min.
The actual time travels 60 min in 1 hour.
The watch travelled n min in one hour.

$\frac{720}{730} = \frac{n}{60}$

$n = \frac{4320}{73} = 59\frac{13}{73}$ min

60 min $- 59\frac{13}{73}$ min $= \frac{60}{73}$ min

His watch was $\frac{60}{73}$ **min** slower than the actual time in an hour.

10. ***Method 1: Think Algebra***

$6 \times 60 = 360$

The clock travels 360 min in 6 hours.
Say the actual time travels $n \times 60$ minutes,
speed of the clock = 59 min in an hour
speed of the actual time = 60 min in an hour

$\frac{60}{59} = \frac{n \times 60}{360}$

$n = 6\frac{6}{59}$ h

$= 6$ h $+ \frac{6}{59} \times 60$

$= 6$ h and $6\frac{6}{59}$ min

Method 2: Solve Using Ratio

The ratio of speeds = 59 : 60

1 min $\div \frac{59}{60} = \frac{60}{59} = 1\frac{1}{59}$ min faster in an hour

In 6 hours, $6 \times 1\frac{1}{59} = 6\frac{6}{59}$ min

The actual time is **6 o'clock and $6\frac{6}{59}$ minutes**.

11. *Analysis:* Use a table to record all numbers.

Alarm clock	Watch	Actual
3600 s	3620 s	
3570 s	t s	3600 s

3600 : 3570 = 3620 : t

$\frac{3600}{3570} = \frac{3620}{t}$

Cross multiply,

$3600t = 3620 \times 3570$

$t = \frac{3620 \times 3570}{3600}$

Number of seconds in 24 hours using the watch

$= 24 \times \frac{3620 \times 3570}{3600} = 86\,156$ s

Number of seconds in one day using actual time

$= 3600 \times 24 = 86\,400$ s

$86\,400 - 86\,156 = 244$

His watch deviated **244 s** from the actual time in a day.

CHAPTER 13

1. If $x = 1, y = 2, z = 3$

$2 \cdot 1 + 6 \cdot 2 + 2 \cdot 3 = 2 + 12 + 6$

$= 20$ (rejected)

If $x = 3, y = 6, z = 9$

$2 \cdot 3 + 6 \cdot 6 + 2 \cdot 9 = 6 + 36 + 18$

$= 60$

The values of x, y and z are **3**, **6** and **9** respectively.

2. From (1),

$A + C + E = 120$ — (4)

From (2),

$A + B + D = 84$ — (5)

From (3),

$B + C + D + E = 132$ — (6)

(4) + (5),

$2A + B + C + D + E = 204$ — (7)

Compare (6) and (7)

$2A = 72$

$A =$ **36**

3. Let the first number be m, then the 2nd number is $m + 1$, ⋯, and the 11th number is $m + 10$.

$m + m + 1 + m + 2 + \cdots + m + 10 = 187$

$11m = 187 - 55$

$= 132$

$m = 12$

$m + 10 = 22$,

so the next 15 numbers start from 23.

$23 + 24 + 25 + \cdots + 36 + 37$

$= (23 + 37) \times 7 + 30$

$=$ **450**

4. Let the weight of the saline water now be x.

The original salt content

$= 50\,000 \times 30\%$

$= 15\,000$ g

$15\,000 = 40\% \cdot x$

$x =$ **37.5 kg**

5. A wrong answer loses $8 + 5 = 13$ marks.

Let a, b and c be correct answers, wrong answers and unattempted questions respectively.

$a + b + c = 30$ — (1)

$8a - 13b - 5c = 49$ — (2)

(1) × 5,

$5a + 5b + 5c = 150$ — (3)

(2) + (3),

$13a - 8b = 199$

$13a = 199 + 8b$

$a = \dfrac{199 + 8b}{13}$

When $b = 6$,

$a = 19$,

$c = 5$

He did not attempt **5 questions**.

6. ***Method 1***

$(3 + 2) = 5$ u

5 u $\rightarrow 120$

1 u $\rightarrow 24$

$3 \times 24 = 72$ female guests

Method 2

Let f be the number of female guests, so $(120 - f)$ is the number of gentlemen.

$\dfrac{f}{3}$ is the number of bottles of wine consumed by ladies.

$\dfrac{(120 - f)}{2}$ is the number of bottles of wine consumed by gentlemen.

$$\frac{f}{3} + \frac{(120 - f)}{2} = 48$$

Multiply both sides by 6 to eliminate the denominator.

$2f + 360 - 3f = 288$

$f = 360 - 288$

$= 72$

There are **72 female guests.**

7. Let the original number of days be x. The rate is $\frac{1}{x}$.

The rate for the new schedule is $\frac{1}{x}(1 + 40\%)$.

The new schedule needs $(x - 20)$ days, so the rate is $\dfrac{1}{x - 20}$.

We have

$\frac{1}{x}(1 + 40\%)$ — (1)

$\dfrac{1}{x - 20}$ — (2)

Equating, $\frac{1}{x}(1 \cdot 4) = \dfrac{1}{x - 20}$

Cross multiply,

$1 \cdot 4x - 28 = x$

$0.4x = 28$

$x = 70$ days

It was scheduled to take **70 days**.

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8. Let c be the number of crickets and s be the number of spiders.

$$6c + 8s = 52$$
$$6c = 52 - 8s$$
$$c = \frac{52 - 8s}{6}$$

When $s = 1$,

$$c = \frac{52 - 8}{6}$$
$$= \frac{44}{6} \text{ (rejected)}$$

When $s = 2$,

$$c = \frac{52 - 16}{6}$$
$$= \frac{36}{6}$$
$$= 6$$

When $s = 3$,

$$c = \frac{52 - 24}{6}$$
$$= \frac{28}{6} \text{ (rejected)}$$

When $s = 4$,

$$c = \frac{52 - 32}{6}$$
$$= \frac{20}{6} \text{ (rejected)}$$

When $s = 5$,

$$c = \frac{52 - 40}{6}$$
$$= \frac{12}{6}$$
$$= 2$$

There are 2 sets of answers:
6 crickets and 2 spiders or **2 crickets and 5 spiders**.

9. *Analysis*: The scores show that

(i) each did not get different cards in each round, otherwise all would have the same score;

(ii) each did not get the same cards in each round, otherwise all scores would be multiples of 3.

We have

$2x + y = 18$ — (1)
$2y + z = 25$ — (2)
$2z + x = 23$ — (3)

where x, y and z are the numbers on each of the cards.

(1) × 2, $4x + 2y = 36$ — (4)
(4) − (2), $4x - z = 11$ — (5)
(5) × 2, $8x - 2z = 22$ — (6)
(6) + (3), $9x = 45$

$$x = 5$$
$$y = 8$$
$$z = 9$$

10. Let f, p and b be the prices of fan, printer and bookshelf respectively.

$2f + p + b = 700$ — (1)
$f + 2p + b = 800$ — (2)
$f + p + 2b = 900$ — (3)

(1) + (2) + (3),

$4f + 4p + 4b = 2400$
$f + p + b = 600$

from (3) $\quad b = 300$
from (2) $\quad p = 200$
from (1) $\quad f = 100$

11. Let the total number of balls be b.

Scenario 1:

$\frac{1}{3} \cdot (b - 1)$ is the number of red balls after one red ball is removed.

Scenario 2:

$\frac{2}{5} \cdot (b - 2)$ is the number of red balls after two black balls are removed.

Scenario 3:

$\frac{2}{5} \cdot (b - 2) - 1$ is the number of red balls after one red ball is removed regardless of the black balls.

Equating Scenario 1 and Scenario 3,

$$\frac{1}{3} \cdot (b - 1) = \frac{2}{5} \cdot (b - 2) - 1$$
$$\frac{1}{3}b - \frac{1}{3} = \frac{2}{5}b - \frac{4}{5} - 1$$
$$\frac{2}{5}b - \frac{1}{3}b = \frac{4}{5} + 1 - \frac{1}{3}$$
$$\frac{(6 - 5)}{15}b = 12 + 15 - \frac{5}{15}$$
$$b = 15 \times \frac{22}{15}$$
$$= \mathbf{22}$$

12. Let e be the escalator's speed, a be Alan's speed and b be Bryan's speed.

We have $\frac{55}{a}(a + e) = n$ — (1)
$\frac{60}{2a}(2a + e) = n$ — (2)

where n is the number of steps.

(1) = (2), $\frac{55}{a}(a + e) = \frac{60}{2a}(2a + e)$

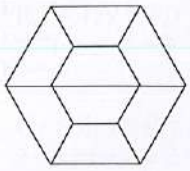
$$55a + 55e = 60a + 30e$$
$$60a - 55a = 55e - 30e$$
$$5a = 25e$$
$$a = 5e$$

Substitute into (1),

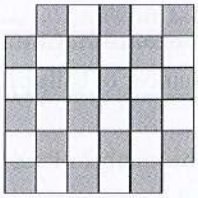
$$n = \frac{55}{5e}(5e + e) = \frac{55}{5e} \cdot 6e$$
$$= \mathbf{66 \text{ steps}}$$

CHAPTER 14

1.



2. Suppose we could colour the area as shown.



There are 18 black tiles and 16 white tiles. We conclude that cannot cover the whole area.

3. $4 : 9 = 8 : 18$
$= 16 : 36$
The sides of A and B are 4 and 6 respectively.
When $c = 8$, area of $c = 64$.
Area of reactangle
$= (8 + 6) \times (4 + 6)$
$= 140 \text{ cm}^2$
Area of shaded region
$= 140 - 64 - 16 - 36$
$= 24 \text{ cm}^2$
When $c = 9$, area of $c = 81$.
Area of rectangle
$= (9 + 6) \times (4 + 6)$
$= 150 \text{ cm}^2$
Area of shaded region
$= 150 - 81 - 16 - 36$
$= \mathbf{17 \text{ cm}^2}$

4. Area of ΔABC = Area of ΔACD
If follows,

$$\frac{1}{2} \times BC \times 9 = \frac{1}{2} \times CD \times 7$$

$$BC = \frac{7}{9} \times CD$$

Now, $BC + CD = 80 \div 2 = 40$ cm

$$CD \times \frac{7}{9} + CD = CD \times (\frac{7}{9} + 1)$$

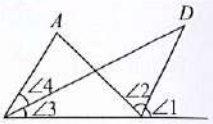
Equate it to 40, $CD \times \frac{16}{9} = 40$

$$CD = 22.5$$

Area of $ABCD = 2 \times \frac{1}{2} \times 22.5 \times 7$

$$= \mathbf{157.5 \text{ cm}^2}$$

5. The exterior angle is the sum of 2 opposite interior angles.



$\angle 1 + \angle 2$
$= \angle A + \angle 3 + \angle 4$

We know
$\angle 1 = \angle 2, \angle 3 = \angle 4$
$2\angle 1 = \angle A + 2\angle 3$

$$\angle 1 = \frac{1}{2}\angle A + \angle 3 \quad \text{—(1)}$$

$$\angle 1 = \angle D + \angle 3 \quad \text{—(2)}$$

Equating, we get

$$\frac{1}{2}\angle A + \angle 3 = \angle D + \angle 3$$

$$\frac{1}{2} \times 76° = \angle D$$

$$\mathbf{\angle D = 38°}$$

6. *Analysis*: In Fig. 3, the shaded regions each has 2 layers. The unshaded region has 4 layers.

$$10 \div 2 = 5 \text{ cm}^2$$

One shaded region is 5 cm^2. Every region is 4 layered now.

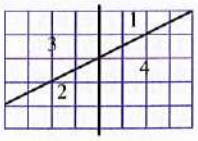
Area of Fig. 3 is $\frac{4}{11}$ of the area of the rectangle.

$$5 \div (\frac{4}{11} - \frac{1}{4}) = 5 \div (\frac{16 - 11}{44})$$

$$= 44 \text{ cm}^2$$

The area is **44 cm^2**.

7.



$1 + 2 + 3 + 4 = 10$
$8 \times 5 = 40 \text{ cm}^2$
$1 \text{ u} \rightarrow 40 \div 10 = 4 \text{ cm}^2$

Area of region $1 = \frac{1}{2} \times 4 \times 2$
$= 4 \text{ cm}^2$

Area of region $2 = (1 + 3) \times 4 \div 2$
$= 8 \text{ cm}^2$

Area of region $3 = (2 + 4) \times 4 \div 2$
$= 12 \text{ cm}^2$

Area of region $4 = (3 + 5) \times 4 \div 2$
$= 16 \text{ cm}^2$

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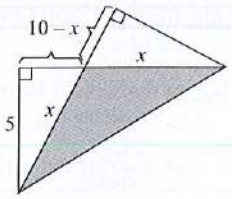
8.



Use Pythagoras Theorem,

$x^2 = 5^2 + (10 - x)^2$

$x^2 = 25 + 100 - 20x + x^2$

$x = 6\frac{1}{4}$ cm

Area of shaded region

$= \frac{1}{2} \times 6\frac{1}{4} \times 5$

$= \frac{125}{8}$

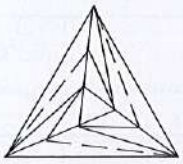
$= \mathbf{15\frac{5}{8}\ cm^2}$

9. The solid lines divide part of the Δ into 7 equal parts.

Each Δ with a dotted side has 2 equal parts.

$6 \times 2 = 12$

$7 + 12 = 19$



The ratio is **1 : 19**.

10. $\angle BFC = \angle B + \angle BEF$
$\angle BEF = \angle A + \angle C$
} Exterior angle is sum of 2 opposing interior angles

$\angle BFC = \angle B + \angle 1 + \angle C$ given, $AB = BD$, $AC = CE$

$\angle B = 180° - 2\angle A$

$\angle C = 180° - 2\angle A$

$105° = 180° - 2\angle A + 180° - 2\angle A + \angle A$

$= 360° - 3\angle A$

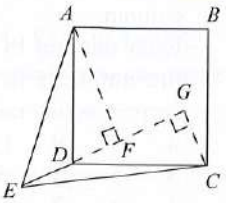
$3\angle A = 255°$

$\angle A = \mathbf{85°}$

11. $DE \times AF \div 2 = 18\ cm^2$

but $DE = 4$ cm

$AF = \frac{18 \times 2}{4} = 9$ cm



Again,

$DE \times CG \div 2 = 10\ cm^2$

$CG = \frac{10 \times 2}{4} = 5$ cm

Use Pythagoras Theorem, since $AD = CD$,

$DG = AF$

$(CD)^2 = (CG)^2 + (DG)^2$

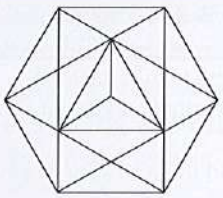
$= 5^2 + 9^2$

$= 25 + 81$

$= 106\ cm^2$

The area of $ABCD$ is **106 cm²**.

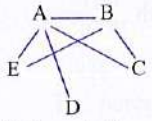
12.



CHAPTER 15

1. **A D E C F B**

2.



Edward played **2** games so far.

3. From the left,

$2000 \div 3 = 666$ R 2

"2" is the last number.

From the right,

$2000 \div 4 = 500$ R 0

"4" is the last number.

1 2 3 1 2 3 1 2 3 ... 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2

4 3 2 1 4 3 2 1 4 ... 1 4 3 2 1 4 3 2 1 4 3 2 1 4 3 2 1

$2000 - 2 = 1998$

Beginning from the right, 1 in every 12 soldiers calls out "3" twice.

$1998 \div 12 = 166$ R 6

$166 + 1 = 167$

167 soldiers call out "3" in both scenarios.

4. If David was the thief, he and Ellen lied.
Fred and George told the truth.

If Ellen stole the jewel, she lied and the rest told the truth.

If Fred did it, he, David and George lied and Ellen told the truth.

Fred was the thief.

5. $31 \div 7 = 4$ R 3

4 days of the week happened 4 times in the month of May. They were Thursday, Friday, Saturday and Sunday.

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3 days of the week happened 5 times and they were Monday, Tuesday and Wednesday.
The first day of the month was Monday, so was the 15th of that month in that year.
15th May was a **Monday** in that year.

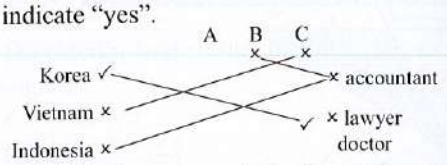
6. Each number in the 1st row is the sum of the numbers in the 2nd and 3rd rows in each corresponding column.
Each number in the last row is the product of the numbers in the 2nd and 3rd rows in each corresponding column.

4	9	9	11	**6**	12	15	8
3	5	**2**	3	4	**5**	9	**3**
1	4	**7**	8	2	**7**	6	**5**
3	20	14	**24**	8	35	**54**	15

7. **(a)** $18 \div 3 = 6$, so $a = \mathbf{6}$.
(b) Fibonacci numbers in an anti-clockwise direction. $a = 2 + 4 = \mathbf{6}$
(c) $a = 4 \times 4 = \mathbf{16}$ (square number)
(d) $a = 6 \times 6 \times 6 + 1 = \mathbf{217}$

8. If "today" was Thursday according to B, then C and D were wrong. Both A and B were right.
If "today" was Monday according to C, then B and D were wrong. Both A and C told the truth.
If "today" was Wednesday, that means only D told the truth.
So, "today" was supposed to be **Wednesday**.

9. We will put a "×" to indicate "no" and a "✓" to indicate "yes".



From the clues given, Chloe is not the lawyer and neither is she born in Vietnam. Since the Korean is a lawyer, **Chloe** must be an **Indonesian doctor**. **Alice** is the **Vietnamese accountant**. **Bryan** is a **Korean lawyer**.

10. $5 \times 10 = 50$ marks if all are correct.
(1) Compare Leon's answers with Natalie's answers. Leon gets questions 5 and 8 correct.
(2) Jolene gets questions 5 and 8 wrong.
(3) It follows that among questions 1, 3, 7 and 9, there is only one wrong answer.
(4) That implies Jolene answered questions 2, 4, 6 and 10 correctly.
(5) Wilfred has 3 right answers from questions 1, 3, 7 and 9. He also answered questions 2 and 6 correctly. Therefore, he gets $15 + 10 = 25$ marks.
Wilfred's score is **25**.

11.

	Conduct	Language	Maths	Aesthetics	PE	Total
Annabelle	3	5	5	5	5	23
Shola	5	1	1	1	1	9
Ernest	1	3	3	3	3	13

Ernest scores the least in this category.

CHAPTER 16

1. cost price $= \frac{\$6500}{130} \times 100 = \$5000$
$\frac{5000}{5000} \times 100\% = 100\%$
The percentage of profit is **100%** if the profit is \$5000.

2. Ratio of profits
$= (100\% - 60\%) : (100\% - 70\%)$
$= 40 : 30$
$= 4 : 3$
Profit from selling mineral water
$= 1200 \div \frac{4}{3}$
$= 1200 \times \frac{3}{4}$
$= \$900$
The profit gained from selling the mineral water was **\$900**.

3. profit in the first month $= (\$4000 \times 20\%) \times 50$
$= \$40\,000$
selling price in the second month
$= (100\% + 20\%) \times 75\%$
$= 120\% \times 75\%$
$= 90\%$
loss in the second month
$= \$4000 \times (100\% - 90\%) \times (80 - 50) = \$12\,000$
Total profit $= \$40\,000 - \$12\,000 = \$28\,000$
A profit of **\$28 000** was made altogether.

4. Let the cost price be C.
marked price $= C \times (1 + 0.2) = 1.2C$
selling price $= 1.2C \times 0.8 = 0.96C$
loss $= C - 0.96C = 0.04C$
$0.04C = \$40$
$C = \$1000$
The cost price of the product was **\$1000**.

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5. total profit = \$525 + 8 × \$175
= \$1925

profit from each racket = \$175 – \$150
= \$25

Number of rackets = \$1925 ÷ \$25
= 77

The shop purchased **77** tennis rackets at first.

6. *Analysis:* We can simplify the problem by giving a price to the question.

Solution:

(1) Suppose the cost price and selling price of the book are \$8 and \$10 respectively.
Last year's profit = \$10 – \$8
= \$2 per book

The cost price increases to 8 × 110% = \$8.80

This year's profit = \$10 – \$8.80
= \$1.20 per book

$$\text{Percentage drop} = \frac{2 - 1.20}{2} \times 100\%$$
$$= \frac{0.80}{2} \times 100\%$$
$$= 40\%$$

(2) Suppose the number of books sold last year was 1000, so this year it will be 1800.

Last year's profit = 1000 × \$2
= \$2000

This year's profit = 1800 × \$1.2
= \$2160

Increment in profit %

$$= \frac{2160 - 2000}{2000} \times 100\%$$
$$= \frac{160}{2000} \times 100\%$$
$$= 8\%$$

The percentage increment in the profit is **8%**.

7. Cost price of the 1st set = \$2040 ÷ (1 + 0.2)
= \$1700

Cost price of the 2nd set = \$2040 ÷ (1 – 0.2)
= \$2550

profit amount = \$2040 – \$1700 = \$340
loss amount = \$2550 – \$2040 = \$510
\$510 – \$340 = \$170

No, the shopowner did not profit from the sales of the two television sets.

8. Interest for 3 years = \$800 × 0.08 × 3 = \$192
The amount of simple interest earned in 3 years is **\$192**.

Suppose it takes n years to have an interest of \$512.

$512 = \$800 \times 0.08 \times n$
$512 = 64n$
$n = 512 \div 64 = 8$

His interest will accumulate to \$512 in **8** years.

9. Interest at the end of 1st year
= \$500 × 3% × 1
= \$15

Total amount at the end of 1st year
= \$500 + \$15
= \$515

Interest at the end of 2nd year
= \$515 × 3% × 1
= \$15.45

Total amount at the end of 2nd year
= \$515 + \$15.45
= \$530.45

She will get **\$530.45** at the end of 2 years.

10. *Analysis:* If the amount of barley borrowed was 1 kg, the repayable amount would be $(1 + \frac{1}{3})$ kg at the end of 4 months.

4 months later = $n \times (1 + \frac{1}{3})$
$= \frac{4}{3}n$ kg

8 months later = $\frac{4}{3}n \times (1 + \frac{1}{3})$
$= \frac{16}{9}n$ kg

1 year later = $\frac{16}{9}n \times (1 + \frac{1}{3})$
$= \frac{64}{27}n$ kg

$\frac{64}{27}n = 2\frac{10}{27}n > 2n$

The amount of barley received from the repayment with interest compounded every 4 months would be greater than the repayment with interest at the end of the first year.

CHAPTER 17

1. There is a multiplier k, such that

$$\frac{1}{11k} - \frac{1}{13k} = \frac{1}{286} \text{ (since A : B = 11 : 13)}$$

$$= \frac{13 - 11}{143k}$$

$$= \frac{2}{143k}$$

$k = 4$

$A + B = 11 \times 4 + 13 \times 4 = 44 + 52 = \mathbf{96}$

---

2. Make a guess,

$1 : 4 = 7 : 28$

$7 + 4 : 28 + 4 = 21 : 32$ (rejected)

Next, we try

$1 : 4 = 8 : 32$

$8 + 4 : 32 + 4 = 12 : 36 = 1 : 3$

Suppose x years ago, the ratio is 1 : 9.

$$\frac{8 - x}{32 - x} = \frac{1}{9}$$

Cross multiply,

$72 - 9x = 32 - x$

$8x = 40$

$x = \mathbf{5}$ **years ago**

---

3. The initial sugar content = $900 \times 0.06 = 54$ g

We want to increase it to 10%.

Let the extra amount be s.

$$\frac{54 + s}{900 + s} = \frac{10}{100} = \frac{1}{10}$$

Cross multiply,

$540 + 10s = 900 + s$

$9s = 360$

$s = \mathbf{40}$ **g**

---

4. Think in "cut and paste", the right-hand region has 6 circles while the left-hand region has 4 circles.

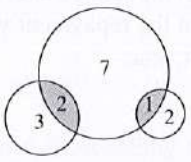
$4 : 6 = 2 : 3$

The ratio of areas is **2 : 3**.

---

5. We assign values to both the shaded and unshaded regions of circles A, B and C until all the conditions stated in the questions are satisfied.

A : B : C = 3 : 5 : 10



The ratio of areas of circles A to B to C is **3 : 5 : 10**.

---

6. $\frac{a + 23}{b + 32} = \frac{2}{3}$

$3(a + 23) = 2(b + 32)$

Since $a + b = 100$,

$b = 100 - a$

$3a + 69 = 2(100 - a + 32) = 2(132 - a) = 264 - 2a$

$3a + 2a = 264 - 69$

$5a = 195$

$a = 39$

$b = 100 - 39 = 61$

The fraction is $\mathbf{\frac{39}{61}}$.

---

7. For every 1 white face, there are 3 black faces.

For every 1 black face, there are 5 white faces.

5 : 3

10 : 6

⋮

20 : 12

There are **20 white faces**.

---

8. ***Method 1*** Using LCM

Analysis: The lowest common multiple of 2, 3 and 4 is 12. We can begin with 60 mℓ of red wine.

After 1st round:

30 mℓ red wine + 30 mℓ white wine

After 2nd round:

20 mℓ red wine + 40 mℓ white wine

After 3rd round:

15 mℓ red wine + 45 mℓ white wine

The ratio is 15 : 45 = **1 : 3**.

Method 2 Use Reasoning

We consider instead on the presence of white wine in each round.

After 1st round:

$\frac{1}{2}$

After 2nd round:

$\frac{1}{2} \times (1 - \frac{1}{3}) + \frac{1}{3} = \frac{2}{3}$

After 3rd round:

$\frac{2}{3} \times (1 - \frac{1}{4}) + \frac{1}{4} = \frac{3}{4}$

The amount of white wine is $\frac{3}{4}$, the amount of red wine is $\frac{1}{4}$.

The ratio is **1 : 3**.

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9. Let the initial heights of the 2 candles be m and n.

Amount of candle burnt

$= m \times \frac{120}{180} = \frac{2m}{3}$

Amount of candle burnt

$= n \times \frac{120}{300} = \frac{2n}{5}$

Remaining heights: $m - \frac{2m}{3} = \frac{m}{3}$

$n - \frac{2n}{5} = \frac{3n}{5}$

The height are the same after 2 hours.

$\frac{m}{3} : \frac{3n}{5} = 1 : 1$

$\frac{\frac{m}{3}}{\frac{3n}{5}} = \frac{1}{1}$

Cross multiply, $\frac{m}{3} = \frac{3n}{5}$

$\frac{m}{n} = \frac{9}{5}$

$m : n = 9 : 5$

The ratio of height was **9 : 5** at first.

10. Ratio of speed = 3 : 1

$90 \div 3 = 30$ steps

Jane would have needed only 30 steps if she was as fast as Carl.

Actual time used by Carl
$= 45 \div 30$
$= 1.5$ times

$(90 - 45) \div (1.5 + 1) = 18$

$18 \times 1.5 + 45 = 72$ steps

or $90 - 18 = 72$ steps

The escalator has **72 steps**.

11. If the total number is 100, there are 60 males and 40 females.

No. of golfers $= \frac{10}{10 + 8 + 7} \times 100$

$= 40$

No. of male golfers = 30
No. of female golfers = 10

No. of tennis players $= \frac{8}{25} \times 100$

$= 32$

No. of male players = 20

No. of female players = 12

No. of swimmers $= \frac{7}{25} \times 100$

$= 28$

No. of male swimmers $= 60 - 30 - 20$

$= 10$

No. of female swimmers $= 28 - 10$

$= 18$

$10 : 18 = 5 : 9$

The ratio is **5 : 9**.

12. ***Method 1*** Use a table

Vanilla	Orange	% of Vanilla
36	4	90%
44	7	86.3%
52	10	83.9%
60	13	82.1%
68	16	80.9%
76	19	80%

Method 2 Write an inequality

$(40 + 11n) \times 80\% \leqslant 36 + 8n$

where $\leqslant$ means smaller or equal to.

$(40 + 11n) \times \frac{4}{5} \leqslant 36 + 8n$

$160 + 44n \leqslant 180 + 40n$

$4n \leqslant 20$

$n \leqslant 5$

$36 + 8 \times 5 = 76$ sweets

$\frac{76}{m} = \frac{80}{100}$

$m = 95$

The maximum number of sweets is **95**.

1. Mr Kaufman's working rate $= \frac{1}{24}$

$$\frac{1}{24} \times 16 = \frac{2}{3}$$

He can complete $\frac{2}{3}$ of the harvest.

$$1 - \frac{2}{3} = \frac{1}{3}$$

His son does $\frac{1}{3}$ of the harvest.

His son's working rate $= \frac{1}{3} \div 6 = \frac{1}{18}$

$$1 \div \frac{1}{18} = 18$$

His son will take **18 hours** to harvest the crops all by himself.

2. *Analysis:* Together, they will complete $\frac{1}{12}$ of the project in a day.
We treat the question as if John and Peter work together for 3 days.
Peter works alone for $(8 - 3) = 5$ days.
John and Peter work together for 3 days, they will complete $\frac{3}{12}$ of the project.

$$\frac{5}{12} - \frac{3}{12} = \frac{1}{6}$$

Peter will complete $\frac{1}{6}$ of the project alone in 5 days.

$$5 \times 6 = 30$$

Peter can complete the project in 30 days.

$$\frac{1}{12} - \frac{1}{30} = \frac{1}{20}$$

John completes $\frac{1}{20}$ of the project in a day.

$$1 \div \frac{1}{20} = 20$$

John can complete the project in **20 days**.

3. *Analysis:* Treat the painting task as "1".
The combined working rate of

Randy and Steven $= \frac{1}{6}$

Steve and Tim $= \frac{1}{9}$

Randy and Tim $= \frac{1}{12}$

$$\frac{1}{6} + \frac{1}{9} + \frac{1}{12} = \frac{13}{36}$$

$\frac{13}{36}$ is twice the working rate for the three boys.

$$\frac{13}{36} \times \frac{1}{2} = \frac{13}{72}$$

$\frac{13}{72}$ is the combined working rate for the three boys.

$$1 \div \frac{13}{72} = 5\frac{7}{13}$$

It will take the three boys $\mathbf{5\frac{7}{13}}$ days to do the painting together.

4. *Analysis:* Treat the task as "1".
The combined rate of

Hoses A and B $= \frac{1}{9}$

Hoses A and C $= \frac{1}{12}$

Hoses B and C $= \frac{1}{18}$

$$\frac{1}{9} + \frac{1}{12} + \frac{1}{18} = \frac{1}{4}$$

$$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

$$1 \div \frac{1}{8} = 8$$

It will take **8 hours** to fill the pool if all three hoses are turned on at the same time.

5. ***Method 1:***

Working rate of Paul $= \frac{1}{6 + 12 \div 3} = \frac{1}{10}$

Working rate of Mary $= \frac{1}{8 \times 3 + 6} = \frac{1}{30}$

$$(1 - \frac{1}{10} \times 3) = \frac{7}{10}$$

$$\frac{7}{10} \div \frac{1}{30} = \frac{7}{10} \times \frac{30}{1} = 21$$

Method 2:
Let x and y be the working rate of Paul and Mary respectively.

$$6x + 12y = 1 \quad \text{—— (1)}$$
$$8x + 6y = 1 \quad \text{—— (2)}$$

(2) × 2

$$16x + 12y = 2 \quad \text{—— (3)}$$

(3) − (1)

$$10x = 1$$
$$x = \frac{1}{10}$$
$$y = \frac{1 - \left(8 \times \frac{1}{10}\right)}{6}$$
$$y = \frac{1}{30}$$

If Paul works on it for 3 hours before handing to Mary who will work for m hours,

$$3 \times \frac{1}{10} + m \times \frac{1}{30} = 1$$
$$\frac{m}{30} = 1 - \frac{3}{10} = \frac{7}{10}$$
$$m = 30 \times \frac{7}{10}$$
$$m = 21$$

Mary alone needs **21 hours** to complete the project if Paul has already worked on it for 3 hours.

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6. *Analysis:* Let the heights of the two candles be m and n respectively.

Step 1: After 2 hours, heights of

1^{st} candle $= m \times \frac{120}{240} = \frac{m}{2}$

2^{nd} candle $= n \times \frac{120}{300} = \frac{2n}{5}$

Step 2: Remaining heights of

1^{st} candle $= m - \frac{m}{2} = \frac{m}{2}$

2^{nd} candle $= n - \frac{2n}{5} = \frac{3n}{5}$

Step 3: $\frac{m}{2} : \frac{3n}{5} = 1 : 1$

$$\frac{\frac{m}{2}}{\frac{3n}{5}} = \frac{1}{1}$$

Cross multiply,

$$\frac{m}{2} = \frac{3n}{5}$$

$$\frac{m}{n} = \frac{6}{5}$$

$$m : n = 6 : 5$$

The ratio of the heights of the two candles at first was **6 : 5**.

7. Step 1: Length of 1^{st} candle after n minutes $= \frac{n}{240}$

Length of 2^{nd} candle after n minutes $= \frac{n}{120}$

Step 2: Remaining length after n minutes

1^{st} candle $= 1 - \frac{n}{240}$

2^{nd} candle $= 1 - \frac{n}{120}$

Step 3: We seek a ratio of 1 : 2 after burning for n minutes

$$(1 - \frac{n}{120}) : (1 - \frac{n}{240}) = 1 : 2$$

$$\frac{(1 - \frac{n}{120})}{(1 - \frac{n}{240})} = \frac{1}{2}$$

Cross multiply,

$$2 \times (1 - \frac{n}{120}) = (1 - \frac{n}{240})$$

$$2 - \frac{n}{60} = 1 - \frac{n}{240}$$

$$\frac{n}{60} - \frac{n}{240} = 2 - 1$$

$$\frac{4n - n}{240} = 1$$

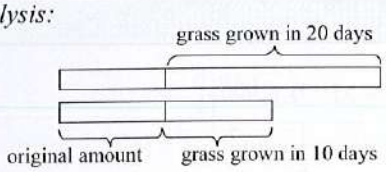
$$3n = 240$$

$$n = 240 \div 3 = 80$$

80 min = 1 h 20 min

It will take **1 h 20 min** for the ratio of the lengths of the candles to be 1 : 2 if both candles are lit at the same time.

8. *Analysis:*



$100 \times 20 = 2000$

$150 \times 10 = 1500$

Difference in the total amount
$= 2000 - 1500$
$= 500$ shares

Growth rate $= 500 \div (20 - 10)$
$= 50$ shares each day

Original amount $= 2000 - 50 \times 20$
$= 1000$ shares

or

$1500 - 50 \times 10 = 1000$ shares

$1000 \div (250 - 50) = 5$

The piece of pasture is able to feed 250 cows for **5** days.

9. $120 \times 12 - 150 \times 8 = 240$ shares

Growth rate $= 240 \div (12 - 8)$
$= 60$ shares each week

Original amount $= 1440 - 60 \times 12 = 720$

or

$1200 - 60 \times 8 = 720$

$720 \div (m - 60) = 6$

$$\frac{720}{(m - 60)} = 6$$

$$720 = 6m - 360$$

$$6m = 1080$$

$$m = 1080 \div 6$$

$$= 180$$

The pasture can sustain **180** cows for 6 weeks.

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CHAPTER 19

1. $= \dfrac{17 + 21 - 9 + \dfrac{12 + 9 - 6}{18}}{79\frac{1}{2} - 49\frac{2}{3}}$

$= \dfrac{29 + \dfrac{15}{18}}{29 + \dfrac{5}{6}}$

$= \dfrac{15}{18} \times \dfrac{6}{5}$

$= \mathbf{1}$

2. $= (1 + 2 + 3 + ... + 11) + \dfrac{1}{2} + \dfrac{1}{6} + \dfrac{1}{12} + ...$

$+ \dfrac{1}{132} + \dfrac{1}{156}$

$= \dfrac{(1 + 11) \times 11}{2} + \dfrac{1}{1 \times 2} + \dfrac{1}{2 \times 3} + \dfrac{1}{3 \times 4} + ...$

$+ \dfrac{1}{11 \times 12} + \dfrac{1}{12 \times 13}$

$= 66 + (1 - \dfrac{1}{2} + \dfrac{1}{2} - \dfrac{1}{3} + \dfrac{1}{3} - \dfrac{1}{4} + ...$

$+ \dfrac{1}{11} - \dfrac{1}{12} + \dfrac{1}{12} - \dfrac{1}{13})$

$= 66 + (1 - \dfrac{1}{13})$

$= \mathbf{66\frac{12}{13}}$

3. Make use of the formula

$a^2 - b^2 = (a - b)(a + b)$

Let $a = 123456$

$A = \dfrac{2000}{a^2 - (a + 1)(a - 1)}$

$= \dfrac{2000}{a^2 - (a^2 - 1)}$

$= \dfrac{2000}{a^2 - a^2 + 1}$

$= \mathbf{2000}$

4. Write each denominator as

$\text{Sum} = \dfrac{(1^{st} \text{ no.} + \text{last no.}) \times \text{how many no. s}}{2}$

We have,

$\dfrac{1}{\dfrac{(1 + 2) \times 2}{2}} + \dfrac{1}{\dfrac{(1 + 3) \times 3}{2}} + \dfrac{1}{\dfrac{(1 + 4) \times 4}{2}} + ...$

$+ \dfrac{1}{\dfrac{(1 + 100) \times 100}{2}}$

$= \dfrac{2}{2 \times 3} + \dfrac{2}{3 \times 4} + \dfrac{2}{4 \times 5} + ... + \dfrac{2}{100 \times 101}$

$= 2 \times (\dfrac{1}{2 \times 3} + \dfrac{1}{3 \times 4} + \dfrac{1}{4 \times 5} + ... + \dfrac{1}{100 \times 101})$

$= 2 \times (\dfrac{1}{2} - \dfrac{1}{3} + \dfrac{1}{3} - \dfrac{1}{4} + \dfrac{1}{4} - \dfrac{1}{5} + ... + \dfrac{1}{100} - \dfrac{1}{101})$

$= 2 \times (\dfrac{1}{2} - \dfrac{1}{101})$

$= 2 \times (\dfrac{101 - 2}{202}) = \mathbf{\dfrac{99}{101}}$

5. "3" is common to all.

$= \dfrac{1}{3} \times (\dfrac{1}{1 \times 2 \times 3} + \dfrac{1}{2 \times 3 \times 4} + \dfrac{1}{3 \times 4 \times 5} + ...$

$+ \dfrac{1}{98 \times 99 \times 100})$

Use the formula $\dfrac{1}{n(n + 1)(n + 2)}$

$= \dfrac{1}{2} \cdot [\dfrac{1}{n(n + 1)} - \dfrac{1}{(n + 1)(n + 2)}]$

$= \dfrac{1}{3} \times \dfrac{1}{2} \times (\dfrac{1}{1 \times 2} - \dfrac{1}{2 \times 3} + \dfrac{1}{2 \times 3} - \dfrac{1}{3 \times 4} + \dfrac{1}{3 \times 4}$

$- \dfrac{1}{4 \times 5} + ... + \dfrac{1}{98 \times 99} - \dfrac{1}{99 \times 100})$

Eliminate the opposite pairs accordingly.

$- \dfrac{1}{2 \times 3} + \dfrac{1}{2 \times 3} = 0$

$- \dfrac{1}{3 \times 4} + \dfrac{1}{3 \times 4} = 0$

$\vdots$

$- \dfrac{1}{98 \times 99} + \dfrac{1}{98 \times 99} = 0$

$= \dfrac{1}{6} \times (\dfrac{1}{2} - \dfrac{1}{9900})$

$= \dfrac{1}{6} \times \dfrac{4950 - 1}{9900}$

$= \dfrac{1}{6} \times \dfrac{4949}{9900} = \mathbf{\dfrac{4949}{59\,400}}$

6. $\dfrac{1}{1 * 2} = \dfrac{1}{\dfrac{1 \times 2}{1 + 2}} = \dfrac{1 + 2}{1 \times 2}$

$\dfrac{1}{2 * 3} = \dfrac{1}{\dfrac{2 \times 3}{2 + 3}} = \dfrac{2 + 3}{2 \times 3}$

$\vdots$

$\dfrac{1}{2000 * 2001} = \dfrac{1}{\dfrac{2000 \times 2001}{2000 + 2001}} = \dfrac{2000 + 2001}{2000 \times 2001}$

$= \dfrac{1 + 2}{1 \times 2} - \dfrac{2 + 3}{2 \times 3} + \dfrac{3 + 4}{3 \times 4} - ... - \dfrac{2000 + 2001}{2000 \times 2001}$

$= \dfrac{1 + 2}{1 \times 2} - \dfrac{2 + 3}{2 \times 3} + \dfrac{3 + 4}{3 \times 4} - ... - \dfrac{2000 + 2001}{2000 \times 2001}$

$= (1 + \dfrac{1}{2}) - (\dfrac{1}{2} + \dfrac{1}{3}) + (\dfrac{1}{3} + \dfrac{1}{4}) - ... - (\dfrac{1}{2000} + \dfrac{1}{2001})$

$= 1 - \dfrac{1}{2001}$

$= \mathbf{\dfrac{2000}{2001}}$

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7. $A = \frac{1+2}{1\times 2} - \frac{2+3}{2\times 3} + \frac{3+4}{3\times 4} - \frac{4+5}{4\times 5} + \ldots + \frac{99+100}{99\times 100}$

It is because $1 + \frac{1}{2} = \frac{3}{2}$,

$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$,

$\vdots$

$\frac{1}{99} + \frac{1}{100} = \frac{199}{9900}$

We can write,

$A = (1 + \frac{1}{2}) - (\frac{1}{2} + \frac{1}{3}) + (\frac{1}{3} + \frac{1}{4}) - (\frac{1}{4} + \frac{1}{5}) + \ldots$

$+ (\frac{1}{99} + \frac{1}{100})$

$= 1 + \frac{1}{100}$

$= \mathbf{\frac{101}{100}}$

8. $3 + \cfrac{1}{4 + \cfrac{1}{b + \frac{1}{6}}} = \frac{421}{130} = 3 + \frac{31}{130}$

$4 + \cfrac{1}{b + \frac{1}{6}} = \frac{130}{31} = 4 + \frac{6}{31}$

$b + \frac{1}{6} = \frac{31}{6}$

$b = \frac{31}{6} - \frac{1}{6}$

$= \frac{30}{6}$

$= \mathbf{5}$

9. $= 5 \times (1 - \frac{1}{5^2}) \times (1 - \frac{1}{6^2}) \times \ldots \times (1 - \frac{1}{1000^2})$

$\times 1000$

Use the formula $a^2 - b^2 = (a - b)(a + b)$

$= 5 \times (1 - \frac{1}{5})(1 + \frac{1}{5})(1 - \frac{1}{6})(1 + \frac{1}{6}) \times \ldots$

$\times (1 - \frac{1}{1000})(1 + \frac{1}{1000}) \times 1000$

$= 5 \times \frac{4}{5} \times \frac{6}{5} \times \frac{5}{6} \times \frac{7}{6} \times \ldots \times \frac{999}{1000} \times \frac{1001}{1000} \times 1000$

$= 4 \times 1001$

$= \mathbf{4004}$

10. $= \cfrac{\frac{1}{2\times 3}}{\frac{1}{3\times 4}} \times \cfrac{\frac{1}{4\times 5}}{\frac{1}{5\times 6}} \times \ldots \times \cfrac{\frac{1}{1999\times 2000}}{\frac{1}{2000\times 2001}}$

$= \frac{1}{2\times 3} \times 3 \times 4 \times \frac{1}{4\times 5} \times 5 \times 6 \times \ldots \times \frac{1}{1999\times 2000}$

$\times 2000 \times 2001$

$= \frac{2001}{2}$

$= \mathbf{1000.5}$

11. 1st: $1 - \frac{139}{365} = \frac{226}{365}$

2nd: $1 - \cfrac{1}{\frac{226}{365}} = 1 - \frac{365}{226}$

$= -\frac{139}{226}$

3rd: $1 - \cfrac{1}{(-\frac{139}{226})} = 1 + \frac{226}{139}$

$= 365$

4th: $1 - \cfrac{1}{\frac{365}{139}} = 1 - \frac{139}{365}$

$= \frac{226}{365}$

The pattern recurs at $\frac{226}{365}, -\frac{139}{226}, \frac{365}{139}, \frac{226}{365}, \ldots$

$2000 \div 3 = 666 \times 3 + 2$

The expression has a value of $\mathbf{-\frac{139}{226}}$.

12. Let $A = 1 + \frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{12}}$

Then,

$\frac{1}{3}A = \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \ldots + \frac{1}{3^{13}}$

$A - \frac{1}{3}A = \frac{2}{3}A = (1 + \frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{12}}) -$

$(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \ldots + \frac{1}{3^{13}})$

$\frac{2}{3}A = 1 + \frac{1}{3} + \frac{1}{3^2} + \ldots + \frac{1}{3^{12}} - \frac{1}{3} - \frac{1}{3^2} - \frac{1}{3^3} - \ldots - \frac{1}{3^{13}}$

$= 1 - \frac{1}{3^{13}}$

$A = \mathbf{\frac{3}{2} \times \left(1 - \frac{1}{3^{13}}\right)}$

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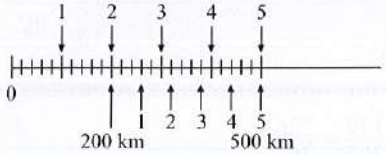
CHAPTER 20

1. ***Method 1*** Solve by Drawing

Ratio of distance = 1000 : 600
= 5 : 3

Ratio of speed = 5 : 3

The truck has a headstart of 200 km.



The car caught up at 500 km from Town A.

Method 2 Solve by Reasoning

Ratio of speed = 5 : 3

Difference in speed = 5 – 3
= 2 units

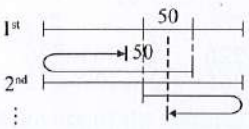
Distance to catch up = 20 km

$200 \div 2 = 100$ units of time

$100 \times 5 = 500$ km

The car caught up at **500 km** from Town A.

2. The combined speed = (300 – 50) ÷ 2
= 125 km/h



The total distance is $(4 \times 300 + 250)$

$(4 \times 300 + 250) \div 125 = 11.6$ hours

They are 50 km apart for the fifth time at **1936 h**.

3. We find a common multiple.

$8 \times 7 = 56$

$56 \times 2 = 1120$

Let the distance be 1120 m.

Time taken by Nick

$$= \frac{1120}{2} \times \frac{1}{70} + \frac{1120}{2} \times \frac{1}{80}$$

$= 8 + 7$

= 15 min

Time taken by Mark

$$= \frac{2 \times 1120}{80 + 70}$$

= 14.93 min

$14.93 < 15$

Mark arrived first.

4. Lenny has walked

$40 \times 2 = 80$ steps

Marie has walked

$20 \times 3 = 60$ steps

The escalator moves at

$(80 - 60) \div (3 - 2) = 20$ steps/min

$(40 + 20) \times 2 = 120$ steps

or $(20 + 20) \times 3 = 120$ steps

The escalator has **120 steps**.

5. Let the distance be d.

The messenger has covered

$(d - 2 + 2)$ km

The platoon of soldiers have marched

$(d - 2 - 2)$ km

The time taken by both is the same.

$$\left(\frac{d-2+2}{10.8_3}\right) = \left(\frac{d-2-2}{7.2_2}\right)$$

$2d = 3d - 12$

$d =$ **12 km**

6. Let Jessie's speed be S m/min.

Mark's speed would be $3S$ m/min.

The speed of the buses is V m/min.

The intervals of buses in terms of distance:

$10 \cdot (V - S)$ — (1)

Also, $20 \cdot (V - 3S)$ — (2)

Equate (1) and (2),

$10 \cdot (V - S) = 20 \cdot (V - 3S)$

$10V - 10S = 20V - 60S$

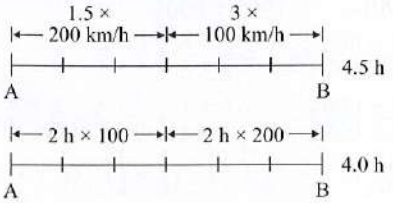
$V = 5S$

The intervals at which the buses left the terminal $= \dfrac{(V - S) \cdot 10}{V}$

$$= \frac{4S \cdot 10}{5S}$$

= **8 min**

7. ***Method 1*** Solve by Drawing



$1.5 \times 200 + 3 \times 100$

$= 300 + 300$

= 600 km

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Method 2 Write Equations

Going for the mission,

$$S = \frac{1}{2} \times t \times 200 + \frac{1}{2} \times t \times 100$$
$$= 100t + 50t$$
$$= 150t$$

$$t = \frac{S}{150} \quad \text{—— (1) where } S = \text{distance},\ t = \text{time}$$

Return trip,

$$\frac{S}{2 \times 200} + \frac{S}{2 \times 100} = t + 0.5$$
$$\frac{3S}{400} = t + 0.5$$
$$t = \frac{3S}{400} - 0.5 \quad \text{—— (2)}$$

Sub (1) into (2)

$$\frac{S}{150} = \frac{3S}{400} - 0.5$$
$$0.5 = \frac{95 - 85}{1200}$$
$$S = \mathbf{600\ km}$$

8. Let the speed of the trains be S.

$$\text{Mary's speed} = \frac{9 \times 1000}{3600}$$
$$= 150 \text{ m/min}$$

The distance completed by her and a train from behind
$= 8 \cdot (S - 150)$ —— (1)
The distance completed by her and an oncoming train
$= 5 \cdot (S + 150)$ —— (2)
Equating (1) & (2),
$8 \cdot (S - 150) = 5 \cdot (S + 150)$
$8S - 1200 = 5S + 750$
$3S = 1950$
$S = 650$ m/min
$= 650 \times \frac{3600}{1000}$
$= \mathbf{39\ km/h}$

9. Mary and Nikki met every

$600 \div (8 - 6) = 300$ seconds

Mary and Polly met every

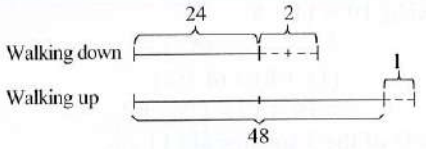
$600 \div (8 + 4) = 50$ seconds

Nikki and Polly met every

$600 \div (6 + 4) = 60$ seconds

The common multiple is **300 seconds**.

10. Ratio of speed = 4 : 1
$48 \div 4 = 12$ steps
Sam will have taken 12 steps if he is as fast as Ted.
Actual time used by Sam $= 24 \div 12$
$= 2$ times



$(48 - 24) \div (1 + 2) = 8$
$8 \times 2 + 24 = 40$ steps
or $48 - 8 \times 1 = 40$ steps

The escalator has **40 steps** when it is stationary.

11. Ratio of distance travelled by Nick and Peter,
$n : p = 30 : (10 + 30)$
$= 3 : 4$

Ratio of distance by Mark and Peter,
$m : p = 100 : (100 + 10 + 30)$
$= 100 : 140$
$= 5 : 7$

$m : n : p = 20 : 21 : 28$

Time taken for Mark to catch up with Nick
$= 30 \div (21 - 20) \times 20$
$= \mathbf{600\ minutes\ later}$

12. Let the distance from A to B be x and that from B to C be y.

$$\frac{x}{200} + \frac{y}{500} = 11.5 \quad \text{—— (1)}$$
$$\frac{x}{500} + \frac{y}{200} = 13 \quad \text{—— (2)}$$

(1) × 1000,
$5x + 2y = 11500$ —— (3)

(2) × 1000,
$2x + 5y = 13000$ —— (4)

(3) + (4),
$7x + 7y = 24500$
$x + y = 3500$
$5x + 5y = 17500$ —— (5)

Compare to (3),
$3y = 6000$
$y = \mathbf{2000\ m}$

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CHAPTER 21

1. The side C : B = 1 : 2
= 2 : 4
= 4 : 8
Likewise, 19 = 11 + 8
⋮
$(11 + 8) \times (4 + 8)$
$= 19 \times 12 = 228 \text{ cm}^2$
The side of the 3 squares are 11 cm, 8 cm and 4 cm respectively.
Area of the shaded region
$= 228 - 11^2 - 8^2 - 4^2$
$= 228 - 121 - 64 - 16$
$= \mathbf{27 \text{ cm}^2}$

2. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3 + 2 + 1}{6}$
$= 1$

$\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$

$\frac{1}{5} + \frac{1}{10} = \frac{3}{10}$

Approximating,

$1 + \frac{3}{8} + \frac{3}{10} + \frac{1}{7} + \frac{1}{9} < 1 + \frac{3}{8} + \frac{3}{10} + \frac{1}{5} + \frac{1}{8}$
$= 1 + \frac{1}{2} + \frac{1}{2}$
$= 2$

The integer part is **1**.

3. Possible lowest value, by setting all to 9.8, is
$9.8 \times 8 = 78.4$

Possible highest value, by setting all to 10, is
$10 \times 8 = 80$

$\because \quad 78.4 < A < 80$

The integer part of A is **79**.

4. First, prime factorise 1001.
$1001 = 7 \times 11 \times 13$
The expression A can be simplified.

$A = \frac{1001}{2} + \frac{1001}{3} + \frac{1001}{5} + 11 \times 13 + 7 \times 13 + 7 \times 11$

$= 500\frac{1}{2} + 333\frac{2}{3} + 200\frac{1}{5} + 143 + 91 + 77$

Clearly,

$1 < \frac{1}{2} + \frac{2}{3} + \frac{1}{5} < 2$

$500 + 333 + 200 + 143 + 91 + 77 + 1 = 1345$

The integer part of A is **1345**.

5. The maximum approximation,

$M < 1 + 1 + 1 + \ldots + 1 = 20$

The minimum approximation,

$M > \frac{9}{10} + \frac{9}{10} + \frac{9}{10} + \ldots + \frac{9}{10} = 18$

$18 < M < 20$

The integer part of M is **19**.

6. Total length $= 1 + 2 + 3 + \ldots + 9 = 45$ cm

$45 \div 4 = 11.25 < 12$

The longest side is 11 cm.

When the side is 11 cm,
$9 + 2 = 8 + 3 = 7 + 4 = 5 + 6$, there is one way.

When the side is 10 cm,
$9 + 1 = 8 + 2 = 7 + 3 = 6 + 4$, there is one way.

When the side is 9 cm,
$8 + 1 = 7 + 2 = 6 + 3 = 5 + 4 = 9$

${}^5C_4 = \frac{5 \times 4 \times 3 \times 2}{1 \times 2 \times 3 \times 4} = 5$ ways.

When the side is 8 cm,
$7 + 1 = 6 + 2 = 5 + 3 = 8$, there is one way.

When the side is 7 cm,
$6 + 1 = 5 + 2 = 4 + 3 = 7$, there is one way.

$1 + 1 + 5 + 1 + 1 = 9$

There are **9 such squares**.

7. Compare each to $\frac{1}{72}$.

$\frac{1}{73} < \frac{1}{72}, \frac{1}{74} < \frac{1}{72}, \cdots, \frac{1}{80} < \frac{1}{72}$

$\frac{1}{72} + \frac{1}{73} + \frac{1}{74} + \ldots + \frac{1}{80} < \frac{1}{72} \times 9 = \frac{1}{8}$

Next, compare each to $\frac{1}{80}$.

$\frac{1}{80} < \frac{1}{72}, \frac{1}{80} < \frac{1}{73}, \cdots, \frac{1}{80} < \frac{1}{79}$

$9 \times \frac{1}{80} < \frac{1}{72} + \frac{1}{73} + \frac{1}{74} + \ldots + \frac{1}{80}$

$\frac{9}{80} < \frac{1}{72} + \frac{1}{73} + \frac{1}{74} + \ldots + \frac{1}{80} < \frac{1}{8}$

Let the original expression be s.

$\frac{80}{9} > s > 8$

$8.88 > s > 8$

The integer part is **8**.

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8. Observe the pattern,

$$1-\frac{1}{2}=\frac{1}{2}$$

$$1-\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$$

$$1-\frac{1}{2}-\frac{1}{4}-\frac{1}{8}=\frac{1}{8}$$

$$\vdots$$

$$1-\frac{1}{2}-\frac{1}{4}-\frac{1}{8}-\frac{1}{16}-\ldots-\frac{1}{512}=\frac{1}{512}$$

But $\frac{1}{512}=0.002$.

Clearly, $A=\frac{1}{512}<0.002$.

9. $\frac{41245}{123456}-\frac{1}{3}=\frac{41245-41152}{123456}$

$$=\frac{93}{123456}$$

$$\frac{78287}{234567}-\frac{1}{3}=\frac{78282-78189}{234567}$$

$$=\frac{93}{234567}$$

Clearly, $\frac{93}{123456}>\frac{93}{234567}$

$$\therefore \frac{41245}{123456}>\frac{78282}{234567}$$

10. Let the left-hand-side, LHS, be A.

$$\frac{1}{1\times 2}=1-\frac{1}{2}$$

$$\frac{1}{1\times 2}+\frac{1}{2\times 3}=1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}$$

$$=\frac{2}{3}$$

$$\frac{1}{1\times 2}+\frac{1}{2\times 3}+\frac{1}{3\times 4}$$

$$=1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}$$

$$=1-\frac{1}{4}$$

$$=\frac{3}{4}$$

$$\vdots$$

$$A=1-\frac{1}{n\times 1}$$

$$1-\frac{1}{n\times 1}>\frac{1949}{2000}$$

Add $\frac{1}{n\times 1}$ to both sides,

$$1>\frac{1949}{2000}+\frac{1}{n\times 1}$$

Subtract $\frac{1949}{2000}$ from both sides,

$$\frac{51}{2000}>\frac{1}{n\times 1}$$

$$51n+51>2000$$

$$51n>1949$$

$$n>\frac{1949}{51}$$

$$n>38\frac{11}{51}$$

The smallest value of n is **39**.

11. (a) $\frac{342}{113}=3\frac{2}{113}$

$$\frac{755}{241}=3\frac{2}{251}$$

$$\because 3\frac{2}{113}>3\frac{2}{251}$$

$$\because \frac{113}{342}<\frac{255}{755}$$

(b) $\frac{1784}{3456}-\frac{1}{2}$

$$=\frac{1784-1728}{3456}$$

$$=\frac{56}{3456}$$

$$\frac{673}{1234}-\frac{1}{2}$$

$$=\frac{673-617}{1234}$$

$$=\frac{56}{1234}$$

$$\because \frac{56}{1234}>\frac{56}{3456}$$

$$\because \frac{673}{1234}<\frac{1784}{3456}$$

12. Let $B=\frac{2}{3}\times\frac{4}{5}\times\frac{6}{7}\times\ldots\times\frac{98}{99}\times\frac{100}{100}$

$$\frac{1}{2}<\frac{2}{3},\frac{3}{4}<\frac{4}{5},\frac{5}{6}<\frac{6}{7},\ldots,\frac{97}{98}<\frac{98}{99},\frac{99}{100}<\frac{100}{100}$$

$$\therefore A<B$$

$$A\times B=\frac{1}{2}\times\frac{2}{3}\times\frac{3}{4}\times\frac{4}{5}\times\ldots\times\frac{98}{99}\times\frac{99}{100}$$

$$=\frac{1}{100}$$

$$=\frac{1}{10}\times\frac{1}{10}$$

$$A^2<\frac{1}{100}=\frac{1}{10}\times\frac{1}{10}$$

$$\therefore A<\frac{1}{10}$$

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CHAPTER 22

1. **(a)**
$$\begin{array}{r} 1011 \\ +\ 1111 \\ \hline \mathbf{11010} \end{array}$$

(b)
$$\begin{array}{r} 11011 \\ +\ 10011 \\ \hline \mathbf{101110} \end{array}$$

(c)
$$\begin{array}{r} 111111 \\ +\ 11011 \\ \hline \mathbf{1011010} \end{array}$$

(d)
$$\begin{array}{r} 11111 \\ -\ 10011 \\ \hline \mathbf{1100} \end{array}$$

2. **(a)**
$4 \overline{)1237}$
$4 \overline{)309}$ — R1
$4 \overline{)77}$ — R1
$4 \overline{)19}$ — R1
$4 \overline{)4}$ — R3
1 — R0

$(1237)_{10} = \mathbf{(103111)_4}$

(b)
$8 \overline{)9653}$
$8 \overline{)1206}$ — R5
$8 \overline{)150}$ — R6
$8 \overline{)18}$ — R6
2 — R2

$(9653)_{10} = \mathbf{(22665)_8}$

3. **(a)** $(894)_{10} = \mathbf{8 \times 10^2 + 9 \times 10^1 + 4 \times 10^0}$

(b) $(17653)_{10} = \mathbf{1 \times 10^4 + 7 \times 10^3 + 6 \times 10^2 + 5 \times 10^1 + 3 \times 10^0}$

(c) $(4321)_5 = \mathbf{4 \times 5^3 + 3 \times 5^2 + 2 \times 5^1 + 1 \times 5^0}$

(d) $(6544)_7 = \mathbf{6 \times 7^3 + 5 \times 7^2 + 4 \times 7^1 + 4 \times 7^0}$

(e) $(8888)_9 = \mathbf{8 \times 9^3 + 8 \times 9^2 + 8 \times 9^1 + 8 \times 9^0}$

(f) $(7546)_8 = \mathbf{7 \times 8^3 + 5 \times 8^2 + 4 \times 8^1 + 6 \times 8^0}$

4. **(a)** $(10111)_2 = 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
$= 16 + 4 + 2 + 1 = \mathbf{(23)_{10}}$

(b) $(100100)_2 = 1 \times 2^5 + 1 \times 2^2$
$= 32 + 4 = \mathbf{(36)_{10}}$

(c) $(123123)_4 = 1 \times 4^5 + 2 \times 4^4 + 3 \times 4^3 + 1 \times 4^2 + 2 \times 4^1 + 3 \times 4^0$
$= 1024 + 512 + 192 + 16 + 8 + 3$
$= \mathbf{(1755)_{10}}$

(d) $(443322)_5 = 4 \times 5^5 + 4 \times 5^4 + 3 \times 5^3 + 3 \times 5^2 + 2 \times 5^1 + 2 \times 5^0$
$= 12500 + 2500 + 375 + 75 + 10 + 2$
$= \mathbf{(15462)_{10}}$

5.

Base 10	2	4	6	8	10
Base 3	**2**	11	**20**	22	**101**

6. **(a)** $(1202)_3 = 1 \times 3^3 + 2 \times 3^2 + 2 \times 3^0 \quad (3^0 = 1)$
$= 27 + 18 + 2$
$= \mathbf{(47)_{10}}$

(b) $(4321)_5$
$= 4 \times 5^3 + 3 \times 5^2 + 2 \times 5^1 + 1 \times 5^0 \quad (5^0 = 1)$
$= 500 + 75 + 10 + 1$
$= \mathbf{(586)_{10}}$

7.

	1	2	3	4	5
1	**1**	2	3	4	5
2	**2**	4	10	12	14
3	**3**	**10**	13	20	23
4	**4**	**12**	**20**	24	32
5	**5**	**14**	**23**	**32**	41

8.
$$\begin{array}{r} 4321 \\ +\ 3124 \\ \hline 13000 \end{array}$$

This computation is based in **base 5**.

9.

	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	**2**	4	6	10	12	14	16
3	**3**	**6**	11	14	17	22	25
4	**4**	**10**	**14**	20	24	30	34
5	**5**	**12**	**17**	**24**	31	36	43
6	**6**	**14**	**22**	**30**	**36**	44	52
7	**7**	**16**	**25**	**24**	**43**	**52**	61

10. $(122)_4 = 1 \times 4^2 + 2 \times 4^1 + 2 \times 4^0$
$= 16 + 8 + 2$
$= (26)_{10}$

$5 \overline{)26}$
$5 \overline{)5}$ — R1
1 — R0

abc = 101
The number represented by abc is **101**.

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11. $(724)_8 = 7 \times 8^2 + 2 \times 8^1 + 4 \times 8^0$
$= 448 + 16 + 4$
$= (468)_{10}$

$9 \overline{)4\ 6\ 8}$
$9 \overline{)5\ 2}$ — R0
5 — R7

abc = 570
The value represented by abc is **570**.

12. $(abc)_3 = a \times 3^2 + b \times 3^1 + c \times 3^0 = 9a + 3b + c$
$(cba)_4 = c \times 4^2 + b \times 4^1 + a \times 4^0 = 16c + 4b + a$
Equating, $9a + 3b + c = 16c + 4b + a$
$8a = b + 15c$
We are bound by base 3 and base 4.
$a = 2, b = 1, c = 1$
The number is **211**.

CHAPTER 23

1. $183 \div 12 = 15$ R 3
$15 + 1 = 16$
At least **16** students were born in the same month.

2.

♥	♠	♦	♣
1	1	1	1
1	1	1	1

$4 + 4 = 8$
$8 + 1 = 9$
I must draw **9** cards so that at least 3 cards have the same pattern.

3. $320 \div 52 = 6$ R 8
$6 + 1 = 7$
At least **7** students from that cohort were born in the same week of that year.

4. There are 4 cards labelled "A", one from each pattern.
In the worst case scenario,
$52 - 4 = 48$ cards drawn without an "A".
$48 + 1 = 49$
At least **49** cards must be drawn to ensure there is a card labelled "A".

5. $100 - 81 = 19$
$19 + 1 = 20$
There are 20 different scores between 81 and 100.
$20 \times 2 = 40$
$40 + 1 = 41$
The minimum number of students in that class could be **41**.

6. *Analysis:* We want to keep the number of children having the same number of saga seeds to a minimum.

1 seed	2 seeds	3 seeds	⋯	14 seeds
1	1	1	⋯	1
1	1	1	⋯	1
⋮				

$1 + 2 + 3 + \ldots\ldots + 14 = 15 \times 7 = 105$
$600 \div 105 = 5$ R 75
$5 + 1 = 6$
At least **6** children will get the same number of saga seeds.

7. *Analysis:* We suppose there are 3 rows of 25 candidates each, 3 rows of 24 candidates each and so on.
For the first 18 rows,
$3 \times (25 + 24 + 23 + 22 + 21 + 20) = 3 \times 135$
$= 405$
$443 - 405 = 38$
$38 \div 2 = 19$
There are 2 rows of 19 candidates each.
At least **3** rows have an equal number of candidates.

8. Possible colour combinations =
oo, ob, or, oy, bb,
br, by, rr, ry, yy
$10 + 1 = 11$
There must be **11** students so that at least two students pick marbles of the same colour.

9. *Analysis:* We assume there are 4 rows of 55 spectators each, 4 rows of 54 spectators each and so on.
For the first 20 rows,
$4 \times (55 + 54 + 53 + 52 + 51)$
$= 4 \times 265$
$= 1060$
$\frac{1150 - 1060}{2} = 45$
There are 2 rows of 45 spectators each.
At least **4** rows have an equal number of spectators.

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10. *Analysis:* If all questions are answered correctly, a candidate should get 30 marks. The range of marks is therefore between 0 and 30. Yet certain scores are not possible.

1 w → 25	1 na → 26	1 w, 1 na → 21
2 w → 20	2 na → 22	2 w, 1 na → 16
3 w → 15	3 na → 18	1 w, 2 na → 17
4 w → 10	4 na → 14	2 w, 2 na → 12
	5 na → 10	
5 w → ⑤	6 na → ⑥	3 w, 1 na → 11
6 w → 0	7 na → ②	3 w, 2 na → 7
		3 w, 3 na → 3
		1 w, 3 na → 13
		2 w, 3 na → 8
		4 w, 1 na → ⑥
		4 w, 2 na → ②
		1 w, 4 na → 9
		5 w, r na → 1
		1 w, 5 na → ⑤

The circled numbers represent repeats.

19, 23, 24, 27, 28 and 29 are not possible.

$$30 - 6 = 24$$

There are 24 different marks.

$$24 \times 3 + 1 = 73$$

73 students must have taken part in the competition.

CHAPTER 24

1. *Analysis:* Treat the square as 4 identical triangles.

$32 \div 4 = 8$ cm²

$\frac{1}{2} \times r \times r = 8$

$r \times r = 16$

$r = 4$

$4 \times 4 \times 3.14 = 50.24$

The area of the circle is **50.24 cm²**.

2. An equilateral triangle can be formed.

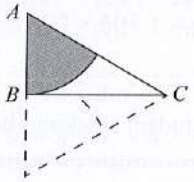
Area of fan $= \frac{1}{3} \times \frac{1}{2} \times \pi \times r \times r = 24\pi$

$r \times r = 144$

$r = 12$

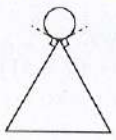
$AC = 2 \times 12$

$= 24$ cm



The length of AC is **24 cm**.

3.



$360° - 90° - 90° - 60° = 120°$

The ball rolls 120° at each corner.

$120° \times 3 = 360°$ for 3 corners

$= 1$ round

$3 + 1 =$ **4 rounds**

4. Let the sum of three diameters be $d_1 + d_2 + d_3$.

$d_1 + d_2 + d_3 = AB = 49$ cm

Sum of perimeters $= \pi d_1 + \pi d_2 + \pi d_3$

$= \pi(d_1 + d_2 + d_3)$

$= \frac{22}{7} \times 49 = 154$ cm

The sum of the perimeters of the three circles is **154 cm**.

5. Area of small circle $= 6 \times 6 \times \pi = 36\pi$ cm²

Area of shaded region $= 36\pi \times \frac{2}{3} = 24\pi$ cm²

Area of big circle $= 24\pi \div \frac{1}{6} = 144\pi$

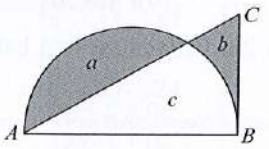
If the radius is r,

$r \times r \times \pi = 144\pi$

$r = 12$ cm

The radius of the big circle is **12 cm**.

6.



$a - b = 47$

$(a + c) - (b + c) = 47$

$\frac{1}{2} \times 10 \times 10 \times 3.14 - \frac{1}{2} \times 20 \times h = 47$

$10h = 157 - 47 = 110$

$h = 110 \div 10 = 11$

BC is **11 cm**.

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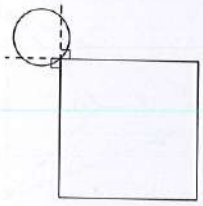
7.



$12 \div 4 = 3$ cm

The ball rolls 90° at each corner.

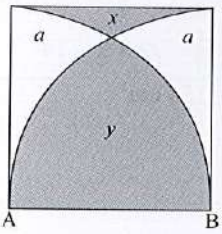
$90° \times 4 = 360°$ for 4 corners

$= 1$ round

$4 + 1 =$ **5 rounds**

8. *Analysis:* We are looking for $(y - x)$.

The two areas marked a are identical.



$y - x = (y + a) - (x + a)$

$= \frac{1}{4} \times 7 \times 7 \times \pi - (7 \times 7 - \frac{1}{4} \times 7 \times 7 \times \pi)$

$= \frac{49\pi}{4} - 49 + \frac{49\pi}{4}$

$= \frac{49\pi}{2} - 49$

$= \frac{49}{2} \times \frac{22}{7} - 49$

$= 77 - 49$

$= 28$

The difference between areas x and y is **28 cm²**.

9. Let the 2 radii be r_1 and r_2.

Area of the shaded region

$= \frac{1}{2} r_1 \times r_1 - \frac{1}{2} \times r_2 \times r_2$

$= \frac{1}{2} \times (r_1^2 - r_2^2)$

$\frac{1}{2} \times (r_1^2 - r_2^2) = 28$

$r_1^2 - r_2^2 = 56$ cm²

The area of the ring $= \pi r_1^2 - \pi r_2^2$

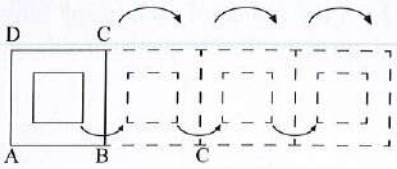
$= \pi(r_1^2 - r_2^2)$

$= \frac{22}{7} \times 56$

$= 176$ cm²

The area of the circular ring is **176 cm²**.

10. *Analysis:* The hollow part travels the distance shown by the 3 arrows.



$8 - 3 = 5$

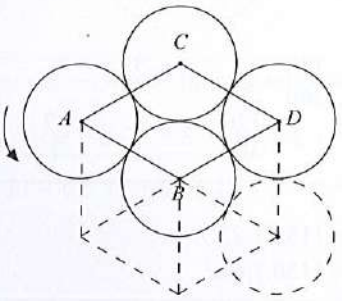
$5 \div 2 = 2.5$

$2.5 \times 6 = 15$ cm

The distance travelled by the hollow part is **15 cm**.

11. *Analysis:* $\angle ABC = 60°$ as ΔABC is an equilateral triangle.

$\angle ABD = 120°$



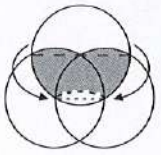
Disc A has travelled 180° to the position indicated.

To return to the original position,

$180° \times 3 = 540°$

The angle through which disc A has travelled when it returns to the original position without sliding is **540°**.

12. *Analysis:* The resultant shaded region is a semicircle.

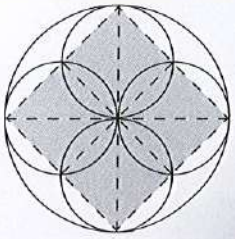


Area of semicircle $= 14 \times 14 \times \frac{22}{7} \times \frac{1}{2}$

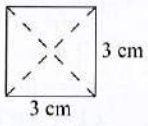
$= 308$ cm²

The area of the shaded region is **308 cm²**.

13.

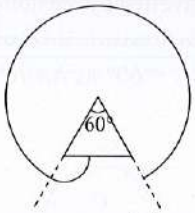


Area of originally shaded region
= Area of square whose diagonal is
$3 \times 2 = 6$ cm, as half of each 'petal' moves out.
There are altogether 2 squares of side 3 cm.



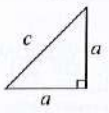
$3^2 + 3^2 = \mathbf{18\ cm^2}$

14.



$$\text{Area} = \frac{300}{360} \times 21 \times 21 \times \frac{22}{7} + 2 \times \frac{120}{360} \times 3.5 \times 3.5 \times \frac{22}{7}$$
$$= \frac{5}{6} \times 3 \times 21 \times 22 + \frac{2}{3} \times 3.5 \times 11$$
$$= 1155 + 25.7$$
$$= \mathbf{1180.7\ cm^2}$$

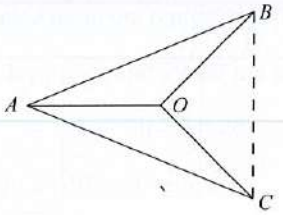
15. Recall Pythagoras' Theorem, given a right-angled triangle,



$a^2 + a^2 = c^2$
But c^2 = radius of quadrant
$= 2a^2$
$= 2 \times 25$
$= 50\ \text{cm}^2$

Area of shaded region
$$= \left(\frac{1}{4} \times \pi \times c^2 - 25\right) \times 2$$
$$= \left(\frac{1}{4} \times 3.14 \times 50 - 25\right) \times 2$$
$$= \mathbf{28.5\ cm^2}$$

16.



Join A and O.
$OA = OB = OC$
$\angle BAO = \angle CAO = 45 \div 2 = 22.5°$
$\Rightarrow \angle AOB = 180° - 22.5 \times 2 = 135°$
Likewise,
$\angle AOC = 135°$
$\angle BOC = 360° - 135° - 135° = 90°$

$$\text{Area of fan} = \frac{90}{360} \times \frac{22}{7} \times 14 \times 14$$
$$= \frac{1}{4} \times 22 \times 2 \times 14$$
$$= \mathbf{154\ cm^2}$$

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